EUROPEAN HEALTH AND DIGITAL EXECUTIVE AGENCY (HADEA)

HADEA.B – Digital, Industry and Space
B.03 – Industry

AMENDMENT Reference No AMD-820789-26

Grant Agreement number: 820789 — Innovative manufacturing processes and in-line monitoring techniques for the OLED and thin film and organic photovoltaic industries (CIGS and OPV) (OLEDSOLAR)

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

1. Change of Annex 1 (description of the action)

Annex 1 is changed and replaced by the Annex 1 attached to this Amendment.

2. Change of the action's duration

The duration of the action in **Article 3** is changed to 42 months.

3. Change of the reporting periods

The reporting periods are changed.

This implies the **following changes** to the Grant Agreement:

- The reporting periods in **Article 20.2** are replaced by:

  - RP1: from month 1 to month 18
  - RP2: from month 19 to month 42

- The option for information on cumulative expenditure incurred in **Article 20.5** becomes applicable.

4 . Changes of Annex 2 (estimated budget)

Annex 2 is changed and replaced by the Annex 2 attached to this Amendment.

5. Change of the beneficiary short name

The short name is changed:

- for ARMOR SOLAR POWER FILMS GMBH (OPVIUS): to ARMOR SOLAR POWER FILMS GMBH (ASPF)

- for SILICON AUSTRIA LABS GMBH (CTR): to SILICON AUSTRIA LABS GMBH (SAL)

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator | For the Agency

Enclosures:

Annex 1
Annex 2

EUROPEAN COMMISSION
European Health and Digital Executive Agency

The Director

ANNEX 1 (part A)

Research and Innovation action

NUMBER — 820789 — OLEDSOLAR

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1.1. The project summary

Project Number [1]	820789	Project Acronym [2]	OLEDSOLAR

One form per project

General information	
Project title [3]	Innovative manufacturing processes and in-line monitoring techniques for the OLED and thin film and organic photovoltaic industries (CIGS and OPV)
Starting date [4]	01/10/2018
Duration in months [5]	42
Call (part) identifier [6]	H2020-NMBP-FOF-2018
Topic	DT-FOF-03-2018 Innovative manufacturing of opto-electrical parts (RIA)
Fixed EC Keywords	Coating and films, Materials engineering, Electrical and electronic engineering, Robotics and automatic control
Free keywords	OLEDs, CIGs, Solar cells, Spatial atomic layer deposition, R2R, Inkjet deposition, Inline metrology, High yield processes, Automation, Product handling, Recycling, Non-destructive Testing (NDT)

Abstract [7]

Opto-electronic devices are opening exciting new applications everyday. With new display options using pliable substrates such as plastic and flexible glass, OLEDs manufacturers are bringing a wide range of new applications in lighting (e.g. energy efficient lighting) and different type of displays. Similarly, with the emergence of thin-film technologies in the solar cells market, new applications ranging from installations on curved surfaces to building-integrated PV has become possible. However, to meet the industry requirements for mass production, including low cost, manufacturing volumes and efficiency, many challenges still need to be addressed. These challenges for OPV, OLED and CIGs are scale-up from laboratory to mass production, selection of efficient manufacturing processes, employing inspection, control and measurement techniques to improve yield, quality and time-to-market.
OLEDSOLAR aims to tackle these challenges by developinginnovative manufacturing processes for critical steps in the production of opto-electronic devices including OLEDs, OPVs and CIGs solar cells. The proposed activities include reconfigurable high yield (>10% improvement) processes to be scaled up, tested at pilot lines and implemented in production line for validation. A complete system of inspection, quality control, functional testing and measurements using advance system and sensors will be optimised in the project for efficient manufacturing of opto-electronics parts. Recycling and re-use strategies will be developed allowing resource efficiency and reduction of high value product wastes. Automation and advance processing software will be developed for overall control and monitoring of roll-to-roll (R2R) and sheet-to-sheet (S2S) manufacturing process.
During 36 months, a multidisciplinary team of leading RTOs and industries in this field will dedicate their resources and effort to perform proposed activities in 8 WPs and guarantee the maximum impact of OLEDSOLAR project.

1.2. List of Beneficiaries

Project Number [1]	820789	Project Acronym [2]	OLEDSOLAR

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	TEKNOLOGIAN TUTKIMUSKESKUS VTT OY	VTT	Finland		
2	FRAUNHOFER GESELLSCHAFT ZUR FOERDERUNG DER ANGEWANDTEN FORSCHUNG E.V.	Fraunhofer	Germany		
3	NEDERLANDSE ORGANISATIE VOOR TOEGEPAST NATUURWETENSCHAPPELIJK ONDERZOEK TNO	TNO	Netherlands		
4	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain		
5	ARMOR SOLAR POWER FILMS GMBH	ASPF	Germany		
6	TWI LIMITED	TWI	United Kingdom		
7	CSEM CENTRE SUISSE D'ELECTRONIQUE ET DE MICROTECHNIQUE SA - RECHERCHE ET DEVELOPPEMENT	CSEM	Switzerland		
8	SOLIBRO HI-TECH GMBH	Solibro	Germany		27/05/2020
9	MEYER BURGER (NETHERLANDS) BV	MBNL	Netherlands		28/02/2019
10	BRUNEL UNIVERSITY LONDON	BUL	United Kingdom		
11	ACONDICIONAMIENTO TARRASENSE ASSOCIACION	LEITAT	Spain		
12	COATEMA COATING MACHINERY GMBH	Coatema	Germany		
13	DUPONT TEIJIN FILMS UK LTD	DTF	United Kingdom		
14	SILICON AUSTRIA LABS GMBH	SAL	Austria		
15	ADVANCED ENERGY TECHNOLOGIES AE EREUNAS & ANAPTYXIS YLIKON & PROIONTONANANEOSIMON PIGON ENERGEIAS & SYNAFON SYMVOULEFTIKON Y PIRESION	Advent	Greece		
16	INURU GMBH	INURU	Germany		
17	SMIT THERMAL SOLUTIONS BV	SMIT	Netherlands	01/03/2019	

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	[illegible][11]	Start month[12]	End month[13]
WP1	Identification of opto-electrical components manufacturing and inspection requirements at production scale	5 - ASPF	[illegible]	1	4
WP2	Development of innovative manufacturing processes for OLEDs and solar cells	3 - TNO	[illegible]	4	34
WP3	Development of in-line inspection and quality control system	6 - TWI	[illegible]	5	34
WP4	Development of AI driven production process automation and quality control system	4 - IRIS	[illegible]	8	32
WP5	Development of processes for materials recycling, repair and re-use approaches for OLEDs and solar cells	11 - LEITAT	[illegible]	13	36
WP6	Integration, Testing, demonstration and performance validation	17 - SMIT	[illegible]	26	42
WP7	Dissemination, communication and Exploitation	1 - VTT	[illegible]	3	42
WP8	Project Management and Coordination	1 - VTT	[illegible]	1	42
		Total	[illegible]		

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Requirements and specifications of OLEDSOLAR solution	WP1	5 - ASPF	Report	Confidential, only for members of the consortium (including the Commission Services)	4
D2.1	SALD AND PESALD Deposition process	WP2	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	11
D2.2	Report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools	WP2	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Test bench for PE deposition	WP2	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D2.4	Final report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools	WP2	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D2.5	Web handling procedure and how to wind damage free barrier film	WP2	2 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D2.6	Patterning techniques	WP2	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D2.7	Packaging techniques	WP2	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	21

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.8	Testing and trial of manufacturing processes	WP2	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D3.1	Optical inspection system	WP3	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D3.2	Inline NDT techniques and processes	WP3	6 - TWI	Report	Confidential, only for members of the consortium (including the Commission Services)	19
D3.3	Report on Functional Testing	WP3	2 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	21
D3.4	Data processing, machine learning and defect detection software	WP3	10 - BUL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D3.5	Raman spectroscopy and photo-luminescence techniques	WP3	14 - SAL	Report	Confidential, only for members of the consortium (including the Commission Services)	19
D3.6	Large area printing quality measurement and registration accuracy	WP3	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	23
D3.7	Report on aging and performance testing	WP3	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	31
D3.8	Pilot line trials of inline inspection	WP3	1 - VTT	Report	Confidential, only for members	34

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					of the consortium (including the Commission Services)	
D4.1	Process control system requirements	WP4	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D4.2	Sensor System Design	WP4	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D4.3	Intelligent decision management and control software design	WP4	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	13
D4.4	Technical report on the sensor system and process control platform	WP4	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	23
D4.5	Process control system user and installation manual	WP4	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D5.1	Report on potential existing pre-treatment technologies and recycling	WP5	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.2	Report on potential recycling routes and waste reuse along the manufacturing process	WP5	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D6.1	Demonstration of integrated OLEDSOLAR system	WP6	17 - SMIT	Demonstrator	Confidential, only for members of the consortium (including the	39

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D6.2	Finalisation of OLEDSOLAR and testing performance result report	WP6	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.3	Environmental and economic performance of OLEDSOLAR system	WP6	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.4	Report on standardisation	WP6	6 - TWI	Report	Confidential, only for members of the consortium (including the Commission Services)	35
D6.5	Report on scalability and performance validation of OLEDSOLAR processes and solution	WP6	12 - Coatema	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D7.1	Dissemination toolkit	WP7	1 - VTT	Report	Public	7
D7.2	Project website	WP7	1 - VTT	Demonstrator	Public	4
D7.3	Plan for the Dissemination and Exploitation of Results	WP7	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	41
D7.4	Open access summary report	WP7	1 - VTT	Report	Public	42
D7.5	Exploitation workshop	WP7	1 - VTT	Demonstrator	Public	24
D7.6	Business Plan	WP7	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D8.1	Project Manual	WP8	1 - VTT	Report	Confidential, only for members of the consortium (including the	3

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D8.2	Project Innovation and Risk management report	WP8	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	41
D8.3	Report on cumulative expenditure	WP8	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	39

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	5 - ASPF
Work package title	Identification of opto-electrical components manufacturing and inspection requirements at production scale		
Start month	1	**End month**	4

Objectives

To develop specification and requirements of manufacturing, inspection, quality control, automation and recycling.

Description of work and role of partners

WP1 - Identification of opto-electrical components manufacturing and inspection requirements at production scale [Months: 1-4]
ASPF, VTT, Fraunhofer, TNO, IRIS, TWI, Solibro, BUL, LEITAT, Coatema, DTF, SAL, Advent, INURU, SMIT
Description of work
T1.1 Identification of OLEDs manufacturing requirements (Lead: INURU) In this task OLEDs manufacturing requirements will be analysed for rigid displays and flexible displays application. The need of process optimisation in S2S manufacturing as well as implementation in roll to roll manufacturing process will be established.
T1.2 Identification of solar cells manufacturing requirements (Lead: OPVIUS, partner: Solibro) In this task manufacturing process factors for yield improvement, improvement of throughput, and reduction of waste will be analysed for CIGS and OPVs. Different existing market solutions will be analysed and potential and benefits of proposed solution will be compared. The end users' product and production requirements will be specified as well. The requirements/specifications in general are identified which can then serve as a benchmark for the assessment of the current production process in T1.3.
T1.3 Assessment of current production processes and process tools adaptability as per production requirements (Lead: Smit TS, Partner: Coatema, OPVIUS, Solibro, INURU) In this tasks assessment of current production processes of OLEDs, OPVs and CIGs will be analysed and adaptability of tools as per change of demand or change of product will be studied.
T1.4 Establishing requirements for inspection, quality and process control (Lead: SOLIBRO, Partners: OPVIUS, Smit TS, DTF, INURU, CTR, DTF, FhG, Coatema) In this task metrology, in-line inspection and measurement techniques requirements will be established for OLEDs, OPVs, and CIGs manufacturing in S2S and roll to roll manufacturing process.
T1.5 Analysis of material challenges for manufacturing process reconfiguration and recyclability (Lead: Advent, Partners: LEITAT, VTT, Smit TS, DTF) Material challenges will be analysed from manufacturing reconfigurability, material transport and handling and recycling issues in production lines. Advent will investigate highly efficient organic materials in order to tweak the material properties towards a simultaneous improvement of synthesis scalability. This will be accomplished by the synergistic application of continuous flow synthesis and traditional organic synthesis reactors. Continuous flow methods are employed for the controlled polymerization of the state of the art polymeric semiconductors including optimization and upscaling experiments. Furthermore, eco-friendly ink formulations will be developed for the efficient printing and/or coating of large area processing.

Participation per Partner

Partner number and short name	**WP1 effort**
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	

Partner number and short name	WP1 effort
6 - TWI	
8 - Solibro	
10 - BUL	
11 - LEITAT	
12 - Coatema	
13 - DTF	
14 - SAL	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Requirements and specifications of OLEDSOLAR solution	5 - ASPF	Report	Confidential, only for members of the consortium (including the Commission Services)	4

Description of deliverables

D1.1 Requirements and specifications of OLEDSOLAR solution (Month 4)

D1.1 : Requirements and specifications of OLEDSOLAR solution [4]
Requirements and specifications of OLEDSOLAR solution

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Establishing Requirements and specifications of OLEDSOLAR	5 - ASPF	4	Establishing Requirements and specifications of OLEDSOLAR

Work package number [9]	WP2	Lead beneficiary [10]	3 - TNO
Work package title	Development of innovative manufacturing processes for OLEDs and solar cells		
Start month	4	End month	34

Objectives

To develop and optimize innovative deposition, packaging and patterning techniques for high volume manufacturing with regard to product quality and yield and industrial scaled up implementation.This concerns spatial atomic layer (inkjet, screen and gravure) printing, slot-die coating and lamination, PECVD, evaporation and sputtering deposition techniques for adding barrier, front window, packaging and encapsulation functionality.
The resulting techniques will be used at pilot scale for demonstration and validation activities in and in conjunction with production lines of end user partners in WP6.

Description of work and role of partners

WP2 - Development of innovative manufacturing processes for OLEDs and solar cells [Months: 4-34]
TNO, VTT, Fraunhofer, IRIS, ASPF, CSEM, Solibro, Coatema, DTF, Advent, INURU, SMIT
T2.1 Developing adaptive material handling system for R2R and S2S production line (Lead: Coatema, Partner: DTF, VTT) In this task substrate material handling system will be developed and tested in order to increase the registration accuracy and the substrate dimensional stability. Intra equipment automation and control via central decision management software will be developed in this task as well.
T2.2 Optimisation of scalable manufacturing processes for solar cells and OLEDs (Lead: VTT, Partner: ASPF, Inuru, Solibro) This task is based on the requirements of the industrial production lines at ASPF (OPV), Solibro (CIGS) and Inuru (OLED) and focuses on optimizing those printing/coating processes (OPV/OLED) and studying if e.g. inkjet printing processes can be utilized in CIGS production. ASPF uses a R2R slot die coating process, combined with laser structuring and screen printing to prepare free-form OPV devices. The OLED process at Inuru is based on inkjet printing. Solibro has a glass-glass process for producing CIGS modules. The bottle-necks and development needs of those processes will be clarified, and solutions will be searched together with VTT. For OPVs, VTT can offer the R2R pilot lines for testing e.g. screen printing of Ag electrodes with several different screen types to optimize the printing quality/ resolution, together with high registration accuracy (50 µm) to minimize waste and number of misaligned components. ASPF will also focus on developing a post-patterning process, i.e. performing all laser steps after all layers have been coated. The developed OPV process will be validated in T2.7/WP6 with ASPF for automation of the selected process steps and feasibility in OPV production.
For OLEDs and CIGSs, VTT can offer S2S scale inkjet printing facilities with industrial scale print heads. The development needs will be collected from Inuru and Solibro and solutions are looked for in the S2S test setup.The OLED printing will be finally validated at Inuru's production lines in T2.7/WP6 and VTT's devices can be used to demonstrate the possible utilization of inkjet in CIGS modules.
T2.3 Optimisation of R2R and S2S atomic layer deposition techniques and processes in R2R and S2S production process (Lead: TNO, Partner: Smit TS, VTT, DTF, FhG; M3 – M28) In this task spatial atomic layer deposition technique will be optimised, tested and scaled up. Carrier application is that of Al2O3 barrier coatings on PET, supplied by DTF. In order to upscale the process for mass production, the lowest requirements will be identified in terms of temperature and speed deposition as well as defect presence in correlation with the barrier properties on lab-scale and R2R tool (in static mode) at TNO. While reducing the thermal budget of the s-ALD processes generally lower the quality barrier properties, the use of atmospheric plasma enhancement (PE) technology will be explored to provide even better performance in terms of WVTR barrier properties. Test will be carried out on lab scale and S2S tool with PE deposition at TNO to determine effect on thermal budget and allowable deposition speed. Investigated will be by TNO, Smit TS the integration options for in-line inspection of pre-deposition and post deposition coated foil such as particle detection, ellipsometry, Raman, PL for automatic adjustment of process settings (for enhanced process control) Being the barrier (and potential other coatings) applied for flexible and rolled applications, the barrier needs to withstand mechanical stress without failures. Developed will be by Smit TS the use of organic pre-coating and post-coating respectively for planarization and mechanical protection purposes as well as use of PET foil supplied with PSC layer from DTF with as aim a WVTR better than 0.5 10-4 g /m2 d. The above will be used to prepare a Customer Requirement Specification (CRS) which will detail the intended layer, the parameter ranges to enable PE s-ALD and the allocation slots for in-line characterization in all relevant disciplines (mechanical, electrical, pneumatic and software) for a R2R-ALD tool.

Smit TS will build a test bench which will be a loop-coater with three segments (set of deposition slots) of a R2R-ALD drum. The test bench will make possible coating of foil in R2R mode with and without PE for analysis of layer functionality and product compatibility (half-fabricates from partners). The tool is meant to be compatible with an existing R2R sALD pilot production system to allow for end-users to select such a system for a (future) pilot line and suitable for installing and/or interacting with in-line inspection of coatings such as ellipsometry, Raman, PL for automatic adjustment of process settings (for enhanced process control), and to be available for demonstration in WP6. In addition, Fraunhofer will develop their R2R evaporation process for OLEDs and provides also OLED samples/rolls for testing the different encapsulation processes (WP2: inkjet/ALD/lamination and their combinations) and to develop the R2R electrical characterization setup (WP3).
T2.4 Development of inkjet-based deposition technique for pre-encapsulation of solar cells and flexible OLEDs (Lead: CSEM, Partner: VTT, Smit TS, INURU) Inkjet based deposition techniques for pre-encapsulation will be developed and tested for R2R and S2S production of OLEDs, OPVs and CIGSs by CSEM where pilot testing will be done in S2S setup (cleanroom environment, in air). R2R Scale up strategy for TRL9 system will be done by Smit TS prior commercialisation. Adhesives that require a nitrogen environment, may be tested with the automated syringe dispensing and UV curing platform (S2S, up to 5-inch square substrates) at CSEM. VTT will provide access to industrial scale print heads in S2S scale from MBNL, which will allow transferring the process to pilot scale. The inkjet-based encapsulation will be compared to other encapsulation techniques as such (for low WVTR requirements) and as a pre-encapsulation technique (for high WVTR requirements) where inkjet is combined with ALD (T2.3) or barrier film lamination (T2.5).
T2.5 Development of reconfigurable and scalable lamination, joining and packaging process for OLEDs and solar cells (Lead: VTT, Partner: CSEM, ASPF, Solibro) Different lamination, packaging and coating technologies will be optimised in this task by VTT for efficient, high quality devices fabrication. Support from CSEM by testing new materials for the lamination process, validating the quality of the encapsulation using accelerated life time and mechanical stress testing, and performing WVTR measurements. Feasibility studies (CSEM) of the fabrication of optical micro/nano structures for in/out coupling directly on the barrier or substrates. Materials for R2R lamination test in air at VTT will be selected based on S2S tests at CSEM. ASPF will work on validating further automatization of its encapsulation (via lamination) and finishing processes, using materials that have shown the best potential at VTT and CSEM. Different methods and materials for joining/contacts will be tested and produced at VTT and their effect on lifetime and mechanical stability will be tested at CSEM.
T2.6 Testing and validation of on production pilot line for roll-to-roll fabricated barrier films (LEAD: Fraunhofer, Partner: TNO, Smit TS) Validation of defect free winding and web handling for barrier films with and without protective liner films.
• Validation of the drying kinetics for sufficient dry and suitable pre-treatment (plasma process and cleaning) steps for conditioning the substrates. Especially, an efficient substrate drying process is needed for a roll-to-roll manufacturing process.
• Determination of substrate surface quality regarding defect level and surface topography of roll materials.
• Substrate packaging and shipment concepts (air-tight packaging to minimize moisture ingress during transportation especially for roll materials)
T2.7 Systems level Testing and validation on production pilot lines (Lead: VTT, Partner: Fraunhofer, CSEM, Smit TS, TNO, ASPF, SOLIBRO, Advent) In this task developed handling and printing techniques in T2.2, T2.4 and T2.5 will be fitted in the production pilot lines of OLEDs, OPVs and CIGSs at VTT (R2R and S2S), TNO (R2R and S2S), Fraunhofer and CSEM (S2S), SOLIBRO (S2S) and ASPF (R2R) and Smit TS (R2R) pilot lines. The S2S environment (single and multi-nozzle inkjet printers, slot-die coating system, screen printing, gravure printing, laminator) at CSEM, will be used for testing new processes developed in T2.2, T2.4 and T2.5. DTF and VTT will investigate the fabrication process of optical hard coats on PCS for improved encapsulation film performance and perform initial trials related to adhesive-free encapsulation trials.

Participation per Partner

Partner number and short name	WP2 effort
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	

Partner number and short name	WP2 effort
7 - CSEM	
8 - Solibro	
12 - Coatema	
13 - DTF	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	SALD AND PESALD Deposition process	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	11
D2.2	Report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Test bench for PE deposition	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D2.4	Final report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools	3 - TNO	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D2.5	Web handling procedure and how to wind damage free barrier film	2 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D2.6	Patterning techniques	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D2.7	Packaging techniques	1 - VTT	Report	Confidential, only for members of the	21

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D2.8	Testing and trial of manufacturing processes	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D2.1 SALD and PESALD Deposition process
D2.2 Report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools
D2.3 Test bench for PE deposition
D2.4 Final report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools
D2.5 Web handling procedure and how to wind damage free barrier film
D2.6 Patterning techniques
D2.7 Packaging techniques
D2.8 Testing and trial of manufacturing processes

D2.1 : SALD AND PESALD Deposition process [11]

SALD AND PESALD Deposition process

D2.2 : Report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools [24]

Report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools

D2.3 : Test bench for PE deposition [34]

Test bench for PE deposition

D2.4 : Final report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools [34]

Final report on PE s-ALD technology for implementation of R2R-ALD and S2S-ALD production tools

D2.5 : Web handling procedure and how to wind damage free barrier film [28]

Web handling procedure and how to wind damage free ALD barrier film

D2.6 : Patterning techniques [16]

Patterning techniques

D2.7 : Packaging techniques [21]

Patterning techniques

D2.8 : Testing and trial of manufacturing processes [34]

Testing and trial of manufacturing processes

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS2	Validated manufacturing process and tools	3 - TNO	22	Validated manufacturing process and tools

Work package number [9]	WP3	Lead beneficiary [10]	6 - TWI
Work package title	Development of in-line inspection and quality control system		
Start month	5	End month	34

Objectives

To develop in-line inspection, monitoring and quality control system.

Description of work and role of partners

WP3 - Development of in-line inspection and quality control system [Months: 5-34]
TWI, VTT, Fraunhofer, TNO, IRIS, ASPF, CSEM, Solibro, BUL, LEITAT, Coatema, DTF, SAL, Advent, INURU, SMIT
T3.1 Development of in-line optical inspection system for component level testing at production scale (Lead: IRIS, Partner: CSEM, Fraunhofer, DTF) Different optical systems, based on both spectral reflectance and spectroscopic imaging ellipsometry will be delivered for high speed inline monitoring of, on the one hand, the coating application processes (range 1-50 microns dry) and, on the other hand, to assess the quality of organic deposits (range expected 50-200 nm dry) on a range of substrates. The selected optical systems, described below, will be assembled and pre-calibrated in the lab scale on a relevant set of samples for obtaining respectively coatings and deposition in non-continuous modes. It is envisaged that at least 3 systems will be placed along the film width to allow getting an idea of the variation of the thickness. The thickness of the samples will be tested by TEM and the coatings through other conventional gages to calibrate the optical systems. The measurement system will be evaluated for Repeatability and Reproducibility using ANOVA gauge R&R system. CSEM supports with: 1) Evaluate the potential of OCT (optical coherence tomography) as technique for inline inspection and position it as a benchmark to the inspection methods developed in the project 2) Off-line inspection for referencing and calibration of the optical inline systems by SEM/ AFM/OCT, 3) Testing of inspection tools in S2S fabrication environment.
T3.2 Implementation and optimisation of NDT testing for zero-defect testing at device level for OLEDs and solar cells (Lead: TWI, Partner: VTT) In this task, advanced NDT techniques will be developed to be deployed in a production line with S2S or roll to roll environment. These techniques include shearography, thermography and other optical techniques. Digital Shearography is an optical measurement technique for NDT and quality control which has found wide industrial application especially for composite materials testing. It detects sub-surface defects (such as voids, dis-bonds, delamination) within an object by identifying defect-induced deformation anomalies as depicted by shearography fringe patterns. When there is no defect in the material, the fringe pattern is smooth or uniform. Otherwise, the fringe pattern will be distorted. The commonly used means of stressing are vacuum pressure (exerted by a vacuum hood or a vacuum chamber), thermal heating and vibration excitation. The application of shearography in this project will be focused on pinhole detection in CIGS solar panels and OLED sheet. Since the pinholes are very small in size (less than 1mm), efforts will be made to find appropriate stressing methods such that the thin plate or film under inspection can have sufficient deformation for the pinholes to be revealed.
For thermography, dark and illuminated lock-in thermography (LIT) will be optimised and implemented by TWI, and finally tested at VTT's R2R electrical measurement setup line.
The lock-in thermography is a non-destructive characterization method, which allows for high-resolution, noise reduced, thermographic imaging for failure analysis based on imaging and localization of electrical defects. Analysis of hot spot, electrical shunt defects and thermal discontinuities. For instance, any disruption of the p-n junction due to cracks and scratches that occur during the processing of a solar cell may produce failures, which may lead to unwanted leakage current (i.e. process-induced shunts). Due to the heat dissipation, these shunts can be detected thermographically. Different modes of LIT to be implemented in OLEDSOLAR:
1. Dark LIT methods:
o Pulsed voltage is applied to the solar cell without illuminating it. It allows the detection of all types of shunts (e.g. linear and non-linear shunts);
2. Illuminated LIT methods:
o Performed by illuminating the solar cell with pulsed light. With this technique, different common sources of shunting in standard industrial-type solar cells can be detected (e.g. points like shunts, cracks, poor edge isolation, contaminations). Spatial and time resolution enables use for inline characterization.
T3.3 Development of in-line Raman spectroscopy and luminescence testing processes and in-line prototype devices (Lead: CTR, Partner: Solibro) This task will first establish the feasibility of non-destructive photonic in-line testing techniques for the applications and needs defined in T1.4. In a second step, suitable in-line system demonstrators

for deployment in relevant manufacturing settings will be developed. This will involve in particular two fundamental techniques, (process) Raman spectroscopy and luminescence imaging. Raman spectroscopy: As a fully spectroscopic technique, Raman spectroscopy is capable of non-invasive, contact-free qualitative and quantitative analysis of a wide range of substance of direct relevance to the project. As a first step, the basic feasibility of the process analysis tasks defined in T1.4 will be investigated under controlled laboratory conditions using a high-end multi-purpose Raman spectrometer. This will involve establishing whether and how e.g. thin film thicknesses and composition (different materials, doping levels etc.) of one or more components within OLED, CIGS and OPV manufacturing can be reliably detected. In the second phase, selected of these approaches will be refined to be applicable in the process, i.e. be robust enough for in-line use (exploiting e.g. self-referencing etc.) and transferred to suitable designed and adapted test units for testing within T3.7.

Luminescence: Techniques to permit in-line characterization during various stages of solar cell manufacturing provide a convenient means for optimizing yield, cost, and efficiency. In particular photo-luminescence measurements are widely used for material characterization. In this project photoluminescence excitation spectroscopy (PLE), in which the steady-state photoluminescence is monitored as a function of the wavelength of the incident illumination, will be implemented. The method is also of high interest to CIGS deposition process control to ensure the optical & electrical quality of the absorber layer, get a faster feedback during the introduction of new processes & materials and improve the yield. The use of spectral and/or transient photoluminescence has shown promise on laboratory scale to supply valuable information about the recombination properties of the absorber and its interfaces enabling a forward calculation of the Voc. The proposed inline PL systems will be used to investigate the correlation of the absorber quality with its properties under field conditions. The optimum PL characterization method and position within the process chain should be identified, which enables the best prediction of the field properties based on inline measurement.

TNO, SAL, FhG-IWS and Brunel will work together on using above given optical measuring techniques as well as the HSI technique for in-line quality control of plates with CIGS PV cell layer stacks, i.e. to identify (areas with) a low quality absorber layer that would lead to poor cell performance. TNO will produce series of semi-fabricates on glass and/or steel plates on their batch-production pilot line(s) at Solliance for which the performance and properties will be varied (by changing Cu, In and Ga ratios and layers' thickness). The plates will be sent to SAL for Raman and luminescent measurements to FhG-IWS for (VNIR and laser) HSI measurements) and/or finished into working devices and their performance determined at TNO.

Brunel will train predictive models to link the luminescent image data with the cell performance parameters (particularly Voc) and possibly also with shunts between back and front electrodes. Similarly FHG-IWS will use the hyper spectral image data sets for training and validating models for predicting the Raman and luminescent measurement and the cell performance.

T3.4 Qualification of thin film barrier coatings for flexible devices (Lead: Fraunhofer, Partner: CSEM, Smit TS, INURU)

• Water vapour transmission rate (WVTR) analysis of gas barrier films down to 10-6 g/m²*d for substrates and top encapsulations. Additionally, the WVTR measurements will be correlated by extrinsic degradation evaluations applying large area optical calcium tests. The Ca will be deposited under vacuum condition in Roll-to-Roll (300 mm width and several meters long). The coated and encapsulated Ca- samples will be stored under different climate condition up to 85°C/85% r.h. The degradation, like side ingress and dark spot nucleation will be quantified by image processing of recorded Ca sample images by high resolution. The pin hole density will be determined over large area of 10 x 10 cm² and will be correlated with the WVTR measurement under steady state mode.

• Validation of the pin hole density determination on 10 x 10 cm² OLED devices on barrier films to quantify the dark spot nucleation and side ingress behaviour.

• Determination of leak times, if desiccants with water trap functionality are implemented inside the devices.

• Defect image analysis recorded by hyperspectral imaging (HSI). HSI will be applied on barrier films, calcium test samples and flexible organic devices. The defect maps are correlated directly with pin hole densities, layer de-laminations and electro-optical behaviour. Afterwards, the devices will be stored under climate conditions and in shelves for operational long-time tests. All types of investigations will be correlated for rapid test inspections using HSI.

• Analysis of aging effects and device performance using hyperspectral imaging (HSI) to develop derive specific algorithms for an inline inspection system to identify just in time yield relevant defects.

• the measurement and quantification of the WVTR of ultra-barriers using the optical Ca test in S2S environment by CSEM, mapping of the water ingress.

T3.5 Large area printing quality measurement and comparison to design layout (Lead: VTT, Partner: ASPF, Coatema) A camera for high-resolution / micro scale mapping of the large printing area (one repeat unit) will be developed to follow printing defects. The image will be automatically compared with the printing design layout, defects will be marked, and errors highlighted in pictures will be shown to the printing machine operator. The error data with positions will be stored in cloud for the consequent process phases. The information is also possible to combine with R2R electrical characterization data. Printing quality measurements will be carried out by scalable 100%-micron scale inspection of the printed layers to R2R-produced printed electronics with high resolution (>10k pixel) line camera, high power led

illumination and ink-jet marking. This measurement system stores faulty printed position to file / cloud for the usage at the later process phases. Second part of the system is inkjet-printer which marks every printed repeat length of the printed roll so that faulty printed devices can be passed by combining repeat length info with high resolution line camera inspection result. Inspection is done by machine vision algorithms by comparing camera image to the design layout and differences will be highlighted for the printing machine operator so that the operator can optimize printing machine print units. The inspection data is also stored for consequent process phases to avoid further processing with non-working printed devices in expensive process phases. This scalable inspection system gives possibility to monitor 100% of the high volume printed electronics R2R-production. This will improve yield and prevent large amount of scrap in production, which is very important especially in case of the high-cost post-processing phases (add-on components and system level packaging). This in-line camera system development also expands machine vision in printing machines to micron scale detection at the fast-moving web by new kind of pulsed high power led illumination light source.

T3.6 Machine vision based micro meter level register accuracy measurement for visible and highly transparent layers (Lead: VTT, Partner: ASPF, Coatema) In this task, VTT will develop a camera system, which can detect the printed patterns with µm-scale resolution, even with transparent inks. VTT already has previous experience about similar cameras / machine vision-based measurements and about transparent material registration. This experience is utilized in the implementation of the task. Different printing layers of printed electronics are very different from each other. Some are opaque, and some are transparent. Some of them are so transparent that they can not necessarily even be seen by the naked eye in a stationary specimen. This will be even more challenging on the moving line. In practice, these different layers of materials require a different illumination geometry in the used camera system (up to 3 different illuminations). In this case, VTT's cheap LED technology is a natural solution taking into consideration that the minimum requirement is that one camera per printing unit is used in the production and one printing machine consists of several printing units. Coatema will give outlook for registration automation concept and process control. This is a study about future integration and upscaling. This is paper work based on experimental results from VTT, Coatema.

T3.7 Implementation of functional testing processes including aging testing and optical performance (Lead: LEITAT, Partners: SAL, CSEM, VTT, TNO, Fraunhofer, IRIS) In this task functional testing, WVTR and ageing testing processes will be developed and implemented. A library of test into the consortium will be developed taking into account, test for solar cells, for OLEDs and for whole integrated system. The test will include: accelerating ageing, natural ageing, in different conditions (including mechanical tests) depend on final uses. Correlation of test will be done between involved partners, as round robin test, in order to asses' production manufacture process and materials selection. LEITAT, CSEM, VTT will test solar cells following thin film standards (IEC61646 and ISOS), monitoring the performance of solar cells and OLEDs using the standard accelerated ageing tests. LEITAT will implemented different mechanical test in order to determine changes due to stress and bending (in line with the future IEC62899 standard), stability tests to determine aging effect and shelf life and validation study with synthetic analytes (following ISO 23640:2011) (LEITAT). Performance related to generation (solar cells) and luminescence (OLED) will be tested, before, during, and after each test (LEITAT).

LEITAT, VTT, ASPF will test solar cells, OLEDs and whole system in natural conditions for outdoor applications and indoor applications. All the results of test will feed to WP2 in order to minimize defects and improve whole performance. Further;

• Inline Roll-to-Roll and Sheet-to-Sheet inspection of barrier films using hyperspectral imaging (HSI) to validate the barrier performance determined in WP3.4

• Roll-to-Roll and Sheet-to-Sheet inspection on OLED and (O)PV devices to detect defects responsible for aging and device reliability issues, like sudden burn-ins (electrical short cuts).

VTT will develop a R2R setup to measure IV characteristics of OLED and solar cell rolls. The setup will contain also a method to mark unfunctional parts in the rolls, which allows removing the defective parts before further processing. CSEM: will perform other tests e.g. mechanical stress testing and Characterization of the optical performance by a) direct, b) indirect, or c) angular dependent reflection/transmission measurements.

T3.8 Development of machine learning based inspection data processing and defects detection software (Lead: BUL, Partners: IRIS) This task will deal with image processing, data analysis and GUI based software development for inspection systems based on the acquired NDT data (i.e. optical images).

• Image processing (De-noising (i.e. filtering), Feature enhancement through thresholding techniques, Morphological operation, Automated defect recognition and image fusion if required)

• Data analysis and processing/post-processing algorithms: Machine learning (deep learning, classification, improving probability of defect detection,decreasing false alarms,developing sufficient training data)

Further, the integrated software development is intended for driving the instrument, processing the data by means of predictive models and delivering the results through the user interface and towards the production line operators. Hardware drivers and the raw data acquisition layer will be set up for the core devices used in the previous tasks. A beta version including the required mathematical treatment of the raw data will be delivered for the lab test rig for each of the pattern types. The final version, adapted for inline configuration, will build upon this initial version while

incorporating all specific detailed analysis of the thickness and yield to be analysed by a plant operator for being used as a user friendly inline monitoring tool to control the correct part quality. As such, a Graphical User Interface will be set up. The data from the monitoring software will feed in the models regarding the process influence on thickness and yield.Upon integration, long term robustness trials, and further control system development, the communication capabilities for sending information to external systems (C/C++, SCADAs, PLCs, etc.), for example, the production line rejection system, or process optimization routines could be integrated in close loop.
T3.9 System level Testing and validation on production pilot lines (Lead: VTT, Partners: TNO, CSEM, ASPF, Advent) : The monitoring systems developed in the previous tasks will be tested in this task. The testing will be performed by the partner who has developed the monitoring system. The testing places will be decided during task 1.4 by all partners. The performance of the monitoring systems is validated in the roller system before installing them into the R2R pilot lines and R2R tester lines. Data measured at the pilot lines will be compared to the laboratory measurement data when needed.

Participation per Partner

Partner number and short name	WP3 effort
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	
6 - TWI	
7 - CSEM	
8 - Solibro	
10 - BUL	
11 - LEITAT	
12 - Coatema	
13 - DTF	
14 - SAL	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Optical inspection system	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	20

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.2	Inline NDT techniques and processes	6 - TWI	Report	Confidential, only for members of the consortium (including the Commission Services)	19
D3.3	Report on Functional Testing	2 - Fraunhofer	Report	Confidential, only for members of the consortium (including the Commission Services)	21
D3.4	Data processing, machine learning and defect detection software	10 - BUL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D3.5	Raman spectroscopy and photo-luminescence techniques	14 - SAL	Report	Confidential, only for members of the consortium (including the Commission Services)	19
D3.6	Large area printing quality measurement and registration accuracy	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	23
D3.7	Report on aging and performance testing	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	31
D3.8	Pilot line trials of inline inspection	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D3.1 Optical inspection system
D3.2 Inline NDT techniques and processes
D3.3 Report on Functional Testing
D3.4 Data processing and defect detection software
D3.5 Raman spectroscopy and photo-luminescence techniques
D3.6 Large area printing quality measurement and registration accuracy
D3.7 Report on aging and performance testing
D3.8 Pilot line trials of inline inspection

D3.1 : Optical inspection system [20]

Optical inspection system

D3.2 : Inline NDT techniques and processes [19]
Inline NDT techniques and processes

D3.3 : Report on Functional Testing [21]
Report on Functional Testing

D3.4 : Data processing, machine learning and defect detection software [34]
Data processing, machine learning and defect detection software

D3.5 : Raman spectroscopy and photo-luminescence techniques [19]
Raman spectroscopy and photo-luminescence techniques

D3.6 : Large area printing quality measurement and registration accuracy [23]
Large area printing quality measurement and registration accuracy

D3.7 : Report on aging and performance testing [31]
Report on aging and performance testing

D3.8 : Pilot line trials of inline inspection [34]
Pilot line trials of inline inspection

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS3	Validated inspection systems and software	6 - TWI	22	Validated inspection systems and software

Work package number [9]	WP4	Lead beneficiary [10]	4 - IRIS
Work package title	Development of AI driven production process automation and quality control system		
Start month	8	End month	32

Objectives

To develop sensor-based automation kit for production of opto-electrical parts.

Description of work and role of partners

WP4 - Development of AI driven production process automation and quality control system [Months: 8-32]
IRIS, VTT, ASPF, TWI, Solibro, BUL, Coatema, DTF, SMIT
T4.1 Identification of production automation requirements and process control system design (Lead: ASPF, Partner: Solibro, BUL, Smit TS, TWI,DTF, IRIS): In this task, identification of production automation requirements will be done for S2S and roll to roll manufacturing processes. These will inform the design of the process control system that will be composed of (i) the optical and NDT inspection systems developed in WP3, complemented by (ii) a set of sensors and (iii) a process control platform which will integrate the inputs from all process relevant information and produce recommendations or even closed loop control for improving the yield and critical process parameters.
T4.2 Sensor system development for automation and monitoring of production processes (Lead: IRIS, Partner: VTT, DTF, Coatema, Solibro, ASPF, TWI)
Subtask 4.2.1 Selection of suite of sensors and adaptation to process requirements
A suite of commercial sensors will be selected to follow the production process and automation parameters defined in T4.1 (e.g. temperature, pressure, material use rate, color, viscosity, power/energy consumption). It is envisaged that these sensors will detect and relay information about key strategic production-specific attributes in real-time and online or inline (depending on the process stage). The parameters measured will be carefully selected so as to correlate with each other and with the optical inspection systems developed under WP3 and enable an online and real-time quality control of production. In as much as possible the equipment used will integrate the following features to facilitate automation and the possibility of closed loop control: MicroElectroMechanical systems (MEMS)-based components; “smart” sensors that have intelligent readout electronics and the microsensor in a single chip; zero power ‘flow tracker’ sensors. Sensors used in this development will be acquired from commercial sources and will be strategically conditioned and tailored for providing affordable monitoring capability and enhanced product monitoring additional to the optical and NDT systems developed in the previous WPs.
SubTask 4.2.2 System integration and performance evaluation: A key performance marker of these sensors is their ability to be delivered as a single integrated system, seamlessly embedded in each of the pilot lines. This work will include the integration of the sensors on a common, single, multifunctional board that meets the requirement that each single sensor will have a standardized output in the desired outputs in the proper format for the use in the intelligent decision management and control software developed in T4.3. In addition, this board will provide the capability of both wired and wireless data transfer for maximizing flexibility and ensuring optimal integration with the process. In the context of this sub-task, IRIS will deliver one such board for each pilot and/or production line as necessary.
T4.3 Development of intelligent decision management and control software for automation and monitoring (Lead: IRIS, Partner: BUL, ASPF): In this task intelligent decision management software will be designed integrating inputs from manufacturing processes, inline inspection systems and production line sensors. This quality and control platform should be a common system for all pilot lines where sensors will be attached, and data will be acquired in order to apply, if needed, model treatment for statistical process control. In case a common system is not possible, minor tweaks will be applied to the platform for specific pilot lines to allow the sensors in that process to be integrated within the platform The platform will be composed of several layers such as the data acquisition layer, the data treatment and storage layer, data mining layer (designed to facilitate the machine learning algorithms developed under T3.8), and the User Interface. The output data will be used for process optimization and control with a process control layer added in case closed loop control is employed.
Work by ASPF relates to i) Monitoring of environment and processes, ii) Scheduling of tasks, machines, and employees, iii) Report generation for runs and machines, iv) Product data connected to machine, v) Machine parameter central collection, and vi) Inventory management and assistants, in order to establish a Smart Factory.
T4.4 Testing and validation on production pilot lines (Lead: IRIS, Partner: ASPF, DTF, Coatema, BUL, VTT): The developed automation kit including sensors and control and monitoring decision management software will be tested and validated in pilot and/or production line setting as necessary, within multi-process roll to roll manufacturing as well as in S2S setting. Pilot testing will be performed to validate the process control systems under controlled conditions.

This stage will be done before integration of all processes (WP6) and will be used to optimise the system before final deployment. Within this task, the control platform will be tested and optimised and real simulations of the final set-ups will be performed where robustness and stability of the platform will be applied. For this reason, during this stage the interaction with the pilot line owners will be essential to ensure successful and realistic tests prior to real environment deployment.

Participation per Partner

Partner number and short name	WP4 effort
1 - VTT	
4 - IRIS	
5 - ASPF	
6 - TWI	
8 - Solibro	
10 - BUL	
12 - Coatema	
13 - DTF	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Process control system requirements	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	10
D4.2	Sensor System Design	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D4.3	Intelligent decision management and control software design	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	13
D4.4	Technical report on the sensor system and process control platform	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	23

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.5	Process control system user and installation manual	4 - IRIS	Report	Confidential, only for members of the consortium (including the Commission Services)	32

Description of deliverables

D4.1 Process control system requirements – Including hardware and software
D4.2 Sensor System Design – Including sensor selection, board design, and specs
D4.3 Intelligent decision management and control software design – Including architecture, features (backlog), and hardware specs
D4.4 Technical report on the sensor system and process control platform – Final specs and features of the developed system
D4.5 Process control system user and installation manual – System ready for final deployment

D4.1 : Process control system requirements [10]

Process control system requirements

D4.2 : Sensor System Design [12]

Sensor System Design

D4.3 : Intelligent decision management and control software design [13]

Intelligent decision management and control software design

D4.4 : Technical report on the sensor system and process control platform [23]

Technical report on the sensor system and process control platform

D4.5 : Process control system user and installation manual [32]

Process control system user and installation manual

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS4	AI driven automation system,software and sensor kit	4 - IRIS	25	AI driven automation system,software and sensor kit

Work package number [9]	WP5	Lead beneficiary [10]	11 - LEITAT
Work package title	Development of processes for materials recycling, repair and re-use approaches for OLEDs and solar cells		
Start month	13	End month	36

Objectives

From a life cycle perspective, sustainability of OLEDs and solar cell products as well as their respective manufacturing process must be ensured from different approaches such as:
(1) the optimal selection and use of raw materials for manufacturing,
(2) avoiding and/or minimising waste generation along the production line, including the small production of discarded materials by the quality control system, and
(3) guaranteeing in the recyclability of these products and by-products during manufacturing from:
(3a) an easy dismantling/ separation of target materials for reuse into the own production line, and
(3b) to ensure their compatibility with potential existing pre-treatment technologies and recycling solutions for complete valorisation when possible (or valorisation of components defined and agreed with end-users) and reuse into the production line.
The WP5 has as a double objective (1) to ensure the recyclability of the end-of-life products and/or by-products generated by OLEDSOLAR manufacturing processes, and (2) the feasibility evaluation of existing and/or innovative recycling routes for the reuse of discarded materials as new raw materials into the own manufacturing process. Proposed recycling technologies will be tested and validated at laboratory scale and finally selected for scaled-up at a representative pilot scale according to the previous lab results.

Description of work and role of partners

WP5 - Development of processes for materials recycling, repair and re-use approaches for OLEDs and solar cells [Months: 13-36]
LEITAT, ASPF, CSEM, Solibro, DTF, Advent, INURU, SMIT
Description of work
This goal will be achieved by the development of the following tasks:
T5.1 Evaluation of potential existing pre-treatment technologies and recycling solutions for OLEDs and solar cells (M13-M26). (Lead: LEITAT, Partners: CSEM, ADVENT): Selected end-of-life products from OLEDSOLAR end-users and materials suppliers, and/or those discarded after inspection and quality control system (WP3) will be evaluated in terms of potential recyclability. For that, a first task focused on the identification of wastes along the manufacturing process will be performed. Identification of their generation and their recycling/reuse potential will be performed by using Key Performance Indicators (KPI). From selected wastes/materials, physic-chemical characteristics will be evaluated as well as their cost/economic potential. From results, procedures for correct handling and separation of wastes by categories defining potential reuse/recycling will be designed. Then, in order to guarantee the recyclability of separated wastes, the following recycling approaches will be proposed and evaluated:
- Manual sorting and selection of added value materials (pre-concentration) before external recycling (existing technologies),
- Grinding/homogenisation of wastes for external recycling (existing technologies)
Regarding the first two options, analysis of existing pre-treatment technologies and conventional recycling solutions will be performed. The analysis will include cost-competitive and mature solutions (TRL9) as well as novel technologies (TRL6-8). Both for mature and novel technologies, feasibility tests at laboratory and pilot scale (according to previous lab results) will be performed in Task 5.3. External treatment and recycling cost will be compared with the cost of the novel raw material used in OLEDSOLAR production processes.
T5.2 Definition of existing and/or innovative recycling routes into the own manufacturing process (M18-M36) (Lead: LEITAT, Partners: CSEM, Smit TS): According to the physic-chemical characteristics of those wastes already identified and quantified in Task 5.1, potentially generated from the scalable production line of OLEDSOLAR products (OLEDs and solar thin films), existing and/or innovative recycling routes into the own manufacturing process will be analysed. For that, manufacturing requirements will be checked, and potential reuse of wastes will be evaluated. Quality criteria such as potential loose of material properties (i.e. reuse of green solvents, semiconductors, reuse of anodes discarded by the quality control system, etc.) will be considered.
T5.3 Testing and validating the feasibility of recycling technologies at laboratory/pilot scale (M20-M36) (Lead: LEITAT, Partners: ASPF, CSEM, Smit TS, INURU): Mature and novel pre-treatment and recycling technologies proposed in

Task 5.1 will be firstly tested and evaluated at laboratory scale. These solutions may include mechanical (i.e. cutting, grinding), thermal (i.e. heating) and/or chemical (i.e. leaching) technologies. Feasibility of proposed solutions will be evaluated from techno-economic and environmental points of view in WP6 (T6.5) and quantified by using the KPIs defined in Task 5.1. Testing of recycled materials will be performed by LEITAT and CSEM, in order to evaluate the quality of the recycled material. Recovered materials may be semiconductors (e.g. ITO), electrodes (e.g. silver), substrates (e.g. PET), among others. Performance of selected recycled materials will be evaluated against reference material. Scale-up tests at a representative pilot scale will be performed according to the previous lab results.

Participation per Partner

Partner number and short name	WP5 effort
5 - ASPF	
7 - CSEM	
8 - Solibro	
11 - LEITAT	
13 - DTF	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Report on potential existing pre-treatment technologies and recycling	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D5.2	Report on potential recycling routes and waste reuse along the manufacturing process	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D5.1. Report on potential existing pre-treatment technologies and recycling [LEITAT]. In includes results from Task 5.1 and 5.3.
D5.2. Report on potential recycling routes and waste reuse along the manufacturing process. [LEITAT]. It includes results from task 5.2.

D5.1 : Report on potential existing pre-treatment technologies and recycling [36]

Report on potential existing pre-treatment technologies and recycling

D5.2 : Report on potential recycling routes and waste reuse along the manufacturing process [36]

Report on potential recycling routes and waste reuse along the manufacturing process

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS5	Recycling and re-use technologies tested	11 - LEITAT	36	Recycling and re-use technologies tested

Work package number [9]	WP6	Lead beneficiary [10]	17 - SMIT
Work package title	Integration, Testing, demonstration and performance validation		
Start month	26	End month	42

Objectives

WP6 will include integration, testing, demonstration and performance validation of developed processes and inspection techniques.

Description of work and role of partners

WP6 - Integration, Testing, demonstration and performance validation [Months: 26-42]
SMIT , VTT, Fraunhofer, TNO, IRIS, ASPF, TWI, CSEM, Solibro, BUL, LEITAT, Coatema, DTF, SAL, Advent, INURU
Description of work
T6.1 Integration of finalised manufacturing, inspection and automation processes on pilot & production lines (Lead: Lead: Smit TS, Partners: VTT, CSEM, TNO, Fraunhofer, IRIS, Inuru, ASPF, DTF): In this WP processes and solutions developed in WP2,3,4 will be integrated including deposition equipment, inline inspection systems, sensors, automation toolkit and developed software will be integrated within pilot and production lines for testing and demonstration. Integration into S2S inkjet, screen, or slot-die printing platforms at CSEM for testing and validation purposes.
Work by ASPF includes i) Redevelopment of the lamination line by a new automated system, automatic feeding system and automatic glue dispensing system, and ii) Upgrading of CO2 laser by installation of additional cameras, and by hardware and software updates.
T6.2 Testing and trials of integrated finalised OLEDSOLAR solutions on pilot lines (Lead: VTT, Partners: TNO, Fraunhofer, SAL): In this task testing and trials of finalised integrated processes developed and validated in WP2, WP3 and WP4 OLEDSOLAR will be done on pilot lines of OLED and solar cells, the process and inspection effectiveness will be evaluated and compared with state of the art and any optimisation or tuning in any of the developed technique and system will be updated to reach specific performance targets. Testing and trial will be done within pilot lines of VTT, Fraunhofer, TNO and CSEM. Solutions will be tested fully integrated in one site or distributed, as for the quality control of CIGS PV plates. I.e. manufacturing of CIGS PV plates will be at TNO and quality control measurements at SAL and FhG.
T6.3 Testing and trial of finalised OLEDSOLAR processes on production lines (Lead: ASPF, Partners: INURU, Smit TS, DTF, SAL, BUL, FhG, TNO, Coatema): In this task testing and trial of OLEDSOLAR will be done in industrial environment within production line of OLEDs and OPV solar cells within end users production facilities. Two production line demonstrations will be done including for OLEDs at INURU, and for OPVs at production facility of ASPF. Productivity yield, process adaptability and recycling metrics will be monitored and compared with current production rates.The demonstration for the in-line quality control of CIGS will not be on one site, but will be distributed. The (batch type) manufacturing of semi-finished and finished CIGS PV plates will be at TNO, while measurements on the semi-finished CIGS PV plates for forward quality monitoring (using models from Brunel and FhG-IWS) will be carried out at SAL (Raman, PL) and FhG-IWS (HSI). However, speed of measurements (without handling) and data processing will resemble that of monitoring in a CIGS production line.
T6.4 Scalability and performance validation of OLEDSOLAR processes and solution (Lead: Coatema, Partners: All): In this tasks after performance validation during industrial demonstration, scalability of developed techniques, systems, software will be checked and scale up plans will be devised to enable implementation of OLEDSOLAR in full production environment during commercialisation of developed technology. Coatema will perform following activities in collaboration with partners and end users S2S and R2R Upscaling & Process Transfer, Engineering Analysis for extended pilot lines ,Design concept & engineering solution for extended pilot line,Modification of existing pilot lines,Engineering analysis for innovative solutions,Trials and validation of scaled up processes at Coatema.The analysing of Impact on yield, cost and investment will be carried out in this tasks as well. VTT will evaluate the up-scaling processability of the Advent WF3 polymer. VTT to perform studies on the processing deposition procedures of WF3 : o-DTBR by ecofriendly solvents so that an active layer ink suitable for large area processing can be obtained at the end of OLEDSOLAR project.
T6.5 Quantification of the environmental and economic performance of the OLEDSOLAR solution (Lead: LEITAT, Partners: All): In this task quantification of environmental and economic performance will be done including Life Cycle Assessment and a Life Cycle Costing. A combination of the LCA and LCC is important to analyse the relation between

environmental and cost performance in order to prove the sustainability of the manufacturing process and the final products (OLED and organic solar cell).A comprehensive LCA based on international standards ISO 14040 and updated publications of the European Platform on Life Cycle Assessment and the European Commission (ILCD Handbook, PEF Guidelines) will be performed to quantify and analyse the potential environmental impacts of the products.The LCC, according to SETAC Guidelines and other existing initiatives, will evaluate the total costs of the system throughout its entire life cycle, considering all costs arising from raw materials to production, use phase and end-of-life. The LCA and LCC will allow characterising the sustainability performance of the OLEDSOLAR products generated in OLEDSOLAR project. The involvement of the rest of the partners is a key issue since they will provide the data obtained in previous tasks and WP.

T6.6 Standardisation, regulations, certification and safety codes (Lead: TWI, Partner: BUL): A standardization methodology (protocols) of opto-electronic component and packaged devices will be developed. To achieve a high-volume production, reliability testing and appropriate standards for qualification of optical devices e.g. OLEDs, CIGS and OPVs should be established. Those standards are mandatory as it serves the industry as a baseline for implementation of pilot-line production. However, in many cases reliability is not even studied or confirmed specially in R2R scale up production. As a result, the missing prove of long-term reliability of devices reduces their level of acceptance considerably. The standard protocols will be based on MIL standards already applied in microelectronics domain. The standard will cover the functionality of the components and will develop and apply accelerated aging tests to guarantee functionality of components for at least 10 years. Statistical analysis will be considered for representative testing during mass production. Ageing at accelerated conditions like high temperature, thermal cycling, cooling and/or heating in controlled cooling rate will be applied. Pass/fail criteria will be based on evaluation of the devices performance with respect to defined tolerances.

Participation per Partner

Partner number and short name	WP6 effort
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	
6 - TWI	
7 - CSEM	
8 - Solibro	
10 - BUL	
11 - LEITAT	
12 - Coatema	
13 - DTF	
14 - SAL	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	1

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Demonstration of integrated OLEDSOLAR system	17 - SMIT	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	39
D6.2	Finalisation of OLEDSOLAR and testing performance result report	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.3	Environmental and economic performance of OLEDSOLAR system	11 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	42
D6.4	Report on standardisation	6 - TWI	Report	Confidential, only for members of the consortium (including the Commission Services)	35
D6.5	Report on scalability and performance validation of OLEDSOLAR processes and solution	12 - Coatema	Report	Confidential, only for members of the consortium (including the Commission Services)	42

Description of deliverables

D6.1 Demonstration of integrated OLEDSOLAR system
D6.2 Finalisation of OLEDSOLAR and testing performance result report
D6.3 Environmental and economic performance of OLEDSOLAR system
D6.4 Report on standardisation

D6.1 : Demonstration of integrated OLEDSOLAR system [39]

Demonstration of integrated OLEDSOLAR system

D6.2 : Finalisation of OLEDSOLAR and testing performance result report [42]

Finalisation of OLEDSOLAR and testing performance result report

D6.3 : Environmental and economic performance of OLEDSOLAR system [42]

Environmental and economic performance of OLEDSOLAR system

D6.4 : Report on standardisation [35]

Report on standardisation

D6.5 : Report on scalability and performance validation of OLEDSOLAR processes and solution [42]

Report about results from T6.4.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS6	OLEDSOLAR technologies integrated,validated in production line	17 - SMIT	36	OLEDSOLAR technologies integrated,validated in production line

Work package number [9]	WP7	Lead beneficiary [10]	1 - VTT
Work package title	Dissemination, communication and Exploitation		
Start month	3	End month	42

Objectives

Objectives:Objectives of WP7 are for successful exploitation of project results to the scientific world, to industrial stakeholders and to the wider community of potential end users.
Specifically:
Promoting the transfer of technologies developed in the project to additional industrial end users with a emphasis on emerging European SMEs.
To develop and implement a structured exploitation plan.
To prevent possible conflicts arising with the consortium relating to knowledge management and intellectual property rights.
To track any innovations developed within the course of the project in order to support the protection of any IP generated by the consortium and individual partners.
To publish the project results in relevant forums and to arrange to present at the project on various relevant industrial and academic forums throughout Europe.
To potentially utilise existing methods that the Industrial End Users utilise to disseminate innovations across their business units.

Description of work and role of partners

WP7 - Dissemination, communication and Exploitation [Months: 3-42]
VTT, Fraunhofer, TNO, IRIS, ASPF, TWI, CSEM, Solibro, BUL, LEITAT, Coatema, DTF, SAL, Advent, INURU, SMIT
Description of work
T7.1 Plan for Exploitation and Dissemination of results (PEDR) (Lead: VTT, Partners: All)
Partner's individual dissemination and exploitation strategies will be merged to a coherent exploitation strategy on consortium level. Related information will be collected by VTT through innovation-related questionnaires. Particular, attention will be paid to the identification of risks and opportunities (including external factors) related to the potential use of results in further research and/or commercial activities, through regular patent search or early-stage Freedom-to-Operate (FTO) analysis by the industrial partners and the monitoring of potentially competing R&I projects by RTO partners. The results of this risk analysis will be periodically reassessed with any periodic report starting from Mo 18. The potential economic impact of project results will be defined on different scales from short-term to beyond the term of project. An exploitation workshop (D7.5) among relevant project partners and – if appropriate- with external stakeholders will be organized to identify and define exploitation opportunities and strategies. PEDR will include a table detailing each identifiable project result with potential economic value, the owner(s) of such result, sectors of potential application and use, an overall plan for the protection and exploitation of the result, and the timetable for action followed by a comprehensive market survey and business implementation strategy. IP Management activities also include guidance and support in case of conflict of IP-related interests between academic and industrial partners.
T7.2 Communication, advertising and conferences publications of technical outputs (Lead: VTT, Partners: All) TWI will elaborate a basic communication concept (M3) in close cooperation with all project partners, thereby to identify major target groups (scientific community, broad public and specific stakeholder/user groups) and appropriate communication and advertising channels and will conceive key communication messages and design elements (project logo, colour scheme, fonts) – thus defining the project's "brand identity". Templates to be used for presentations, reports, posters as well as for meetings and internal documents will be provided to all partners as a Dissemination Toolkit (D7.1). Additionally, VTT will design and produce a set of standard communication materials such as project leaflets and posters. The dissemination toolkit and all communication materials will be made available to the whole consortium through the internal communication and management platform allowing project partners to easily access and use the material for all future project-related communication and dissemination. Furthermore, VTT will be in charge of the set-up and maintenance of the external project website (D7.2), which will serve as an additional showcase for presenting the project's activities and achievements. To foster networking within the scientific community, particular attention will also be paid to the concentration with other related initiatives (other ongoing projects or networks on national and international level; particularly EU-funded initiatives on related domains. Both planned and implemented

communication and networking activities as well as publications that are under preparation or have already been submitted will be summarized in the periodically updated Plan for the Dissemination and Exploitation of Results (PEDR, D7.3). BUL as academic partners along with coordinator will encourage, assist and monitor the timely publication of results in peer-reviewed journals.VTT will provide an overview of the locally available repositories to each partner, as well as the information on the open access policies of the five most frequently used journals for each partner. An open access questionnaire to evaluate each partner's experiences with the process and their general opinion will be developed by VTT collecting partner feedback to be merged into an open access summary report (D7.4) at the end of the project. The periodically updated PEDR will include a current overview of the project's open access publication status.
T7.3 Valorisation of results (Lead: VTT, Partners: All): Assessment and valorisation activities within the project implementation will cover all relevant initiatives to identify and increase the economic value of foreground generated in the project, including the protection and exploitation of results into products, processes, services or any other form of commercial use. This can be done i.e. via licensing or assignment of IPR as well as the creation of spin-offs and start-ups.IP will need to be identified and captured at critical stages of the project. The consortium partners will log and capture IP as it emerges from the research work. Any foreground/deliverable generated will be labelled with regard to its confidentiality degree as well as in view to potential exploitability such as possible conflicts with third party IP rights. Furthermore, the consortium partners will regularly monitor standards/norms in the related field and, if necessary, start new initiatives for specific standards on innovative products and services.
Identified potential strategic partners for the uptake and commercialization of research outcomes will be contacted and invited to exploitation workshops and/or bilateral meetings at early stage. If appropriate, additional networking channels on EU level, such as the Enterprise Europe Network or business fairs and brokerage events, will be used to get in contact with potential partners for further technology and business cooperation.
T7.4 Post Project Business Plan and Commercialisation roadmap (Lead: Smit TS, Partners: All) OLEDSOLAR project Business Plan will incorporate a marketing strategy, route to market plan and operations plans to detail the sales plan, customer lists, costing, and route to market. Based on our analysis of the market and supply chains, plus the results of various trials and customer feedback, the information will be used to improve the Business Plan, i.e. Create contingencies, plans and alternative strategies & sensitivities. The consortium will further, scale up the system for industrially commercial ready system in cooperation with end users. Commercialisation roadmap plan will be developed incorporating supply chain of individual company's systems and all IPs involved.

Participation per Partner

Partner number and short name	WP7 effort
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	
6 - TWI	
7 - CSEM	
8 - Solibro	
10 - BUL	
11 - LEITAT	
12 - Coatema	
13 - DTF	
14 - SAL	
15 - Advent	
16 - INURU	
17 - SMIT	

Partner number and short name	WP7 effort
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Dissemination toolkit	1 - VTT	Report	Public	7
D7.2	Project website	1 - VTT	Demonstrator	Public	4
D7.3	Plan for the Dissemination and Exploitation of Results	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	41
D7.4	Open access summary report	1 - VTT	Report	Public	42
D7.5	Exploitation workshop	1 - VTT	Demonstrator	Public	24
D7.6	Business Plan	17 - SMIT	Report	Confidential, only for members of the consortium (including the Commission Services)	42

Description of deliverables

Deliverables (brief description)
D7.1 Dissemination toolkit
D7.2 Project website
D7.3 Plan for the Dissemination and Exploitation of Results
D7.4 Open access summary report
D7.5 Exploitation workshop
D7.6 Business Plan

D7.1 : Dissemination toolkit [7]

Dissemination toolkit

D7.2 : Project website [4]

Project website

D7.3 : Plan for the Dissemination and Exploitation of Results [41]

Plan for the Dissemination and Exploitation of Results

D7.4 : Open access summary report [42]

Open access summary report

D7.5 : Exploitation workshop [24]

Exploitation workshop

D7.6 : Business Plan [42]

Business Plan

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP8	Lead beneficiary [10]	1 - VTT
Work package title	Project Management and Coordination		
Start month	1	End month	42

Objectives

Objectives: Work package 8 will provide a clear organizational framework and all necessary support mechanisms to enable a smooth project workflow in OLEDSOLAR and to ensure that all contractual commitments can be met in time.
Provide optimal orientation to all partners through a quick set-up of effective management & communication structures, guidance and support
Transparency for consortium partners and the EC through proper project documentation
Maximize effectiveness of project activities: ensure the timely and qualitative achievement of project results through scientific and administrative coordination.
Ensure efficiency: use resources wisely, avoid duplication of efforts, reduce waste of time and energy to a minimum.

Description of work and role of partners

WP8 - Project Management and Coordination [Months: 1-42]
VTT, Fraunhofer, TNO, IRIS, ASPF, TWI, CSEM, Solibro, BUL, LEITAT, Coatema, DTF, SAL, Advent, INURU, SMIT
Description of work
T8.1 Financial and administrative management (Lead: VTT, Partners: All): The project coordinator, VTT, will be responsible for the achievement of technical and commercial progress in the project and its contractual and financial aspects. VTT will co-ordinate and arrange meetings and also manage any changes proposed to the workplan. VTT and their contracts team will also be responsible for the maintenance of the Consortium Agreement (CA). VTT will ensure that the CA is maintained throughout the project and changes, such as exploitation rights, are handled efficiently. In addition, the project coordinator will maintain regular contact with the appointed EC Project Officer.
T8.2 Risks, Deliverables, Milestones and IP management (Lead: VTT, Partners: All): A risk management plan for both commercial and engineering development will be carried out throughout the project. The consortium will apply PFMEA (Process Failure Mode and Effects Analysis) and assign quality risk levels to each step in a process (manufacturing or transactional). A table will be produced to indicate 3 levels of criticality (Green – Low, Amber – Medium, Red – High) in order to indicate the probability and consequences of each risk. Deliverables and Milestones will be monitored and IP management aspects including ownership of results and exploitation issues will be managed by project coordinator.
T8.3 Quality assurance, Innovation Management and Project Performance monitoring (Lead: VTT, Partners: All): The coordinator will make sure about the quality of work being undertaken and will continuously monitor performance of project tasks and partners. Coordinator will make sure all milestones and deliverables of all WPs are up to standard and fulfil quality criteria established in the project.
Innovation management of project including product development, finalisation and commercialisation will be monitored by BUL and Smit TS to take advantage of business opportunities and implementing creative ideas to achieve an enhanced product.

Participation per Partner

Partner number and short name	WP8 effort
1 - VTT	
2 - Fraunhofer	
3 - TNO	
4 - IRIS	
5 - ASPF	
6 - TWI	

Partner number and short name	WP8 effort
7 - CSEM	
8 - Solibro	
10 - BUL	
11 - LEITAT	
12 - Coatema	
13 - DTF	
14 - SAL	
15 - Advent	
16 - INURU	
17 - SMIT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Project Manual	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	3
D8.2	Project Innovation and Risk management report	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	41
D8.3	Report on cumulative expenditure	1 - VTT	Report	Confidential, only for members of the consortium (including the Commission Services)	39

Description of deliverables

D8.1 Project Manual
D8.2 Project Innovation and Risk management report

D8.1 : Project Manual [3]
Project Manual

D8.2 : Project Innovation and Risk management report [41]
Project Innovation and Risk management report

D8.3 : Report on cumulative expenditure [39]
This deliverable reports the cumulative expenditure incurred by each of the beneficiaries.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Establishing Requirements and specifications of OLEDSOLAR	WP1	5 - ASPF	4	Establishing Requirements and specifications of OLEDSOLAR
MS2	Validated manufacturing process and tools	WP2	3 - TNO	22	Validated manufacturing process and tools
MS3	Validated inspection systems and software	WP3	6 - TWI	22	Validated inspection systems and software
MS4	AI driven automation system,software and sensor kit	WP4	4 - IRIS	25	AI driven automation system,software and sensor kit
MS5	Recycling and re-use technologies tested	WP5	11 - LEITAT	36	Recycling and re-use technologies tested
MS6	OLEDSOLAR technologies integrated,validated in production line	WP6	17 - SMIT	36	OLEDSOLAR technologies integrated,validated in production line

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Changes in the Consortium	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Tasks assigned to the leaving participant will be reallocated among the Consortium or new participant will be invited and integrated to the Consortium.
2	Unclear roles & responsibilities between participants	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Review of CA and GA; effective communication, amendments of necessary
3	Risks related to participants' operational environment (bankruptcy, accidents, force majeure etc.)	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Effective communication; participants obligation to inform Coordinator and consortium; amendments to the Description of Action
4	Risks related to unidentified ethical risks	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Immediate actions are taken in on-going ethical review process to integrate the risks to the ethical procedures and codes of conduct
5	Insufficient financial reporting	WP8	Ensuring compliance with accounting rules, keeping records and timesheets; timely submission of financial statements (and possible certificate of the financial statements)
6	UK research financing not taking over after UK departure (replacing EU financing)	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	This leads to reallocation of remaining resources/ budget. (Current wisdom is that UK does not want to abandon EU research activities)
7	Risks related to ownership and user rights	WP1, WP2, WP3, WP4, WP5, WP6	Clearly defined background IPR in CA; transparent and efficient Exploitation Plan
8	Issues related existing patents and freedom to operate	WP1, WP2, WP3, WP4, WP5, WP6	The well maintained Plan for Exploitation and Dissemination of results (PEDR) will ensure that this issue is kept under control. Attention is paid to the regular updates as new patents are coming all the time.
9	Communication regarding the project and its results is not properly targeted (message is not clear)	WP1, WP2, WP3, WP4, WP5, WP6, WP7, WP8	Communication to be modified according to the audience and initial feedback
10	OLEDSOLAR processes scale up issues	WP1, WP2, WP3, WP4, WP5, WP6	3 pilot line testing and 3 production lines testing including system level testing in WP2, 3 and integrated full systems testing in WP6 has been planned to check scalability of developed processes making sure all processes proposed are scalable.
11	Pristine surfaces (e.g. OCS grade PET of DTF) cannot be exploited due to contamination due to exposure to environmental conditions prior to deposition	WP2, WP6	Special measures to reduce contamination sources; reduce attraction, and in-line surface inspection to ensuring a defect free surface will be coated.

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
12	Delays in preparing, modifying equipment for using or transport of specific foil or sheet material, for adding additional process steps and/or inspection techniques	WP2	Partners will carefully plan their resources, interactions and engagements with suppliers. Use a conventional production process as fall back
13	Developed techniques (output, input) don't match or interface with end user production lines and materials	WP1, WP2, WP3, WP4, WP5, WP6	Partners will define requirements in WP1 and will make advanced planning for demonstrations and validation (in WP6) and will monitor for changes during project duration.
14	Material Deposition: Lack of compatibility between mature sALD process and device	WP2	Via optimization and material handling system a Customer Requirement Specification (CRS) will ensure which will detail the intended layer(s), the parameter ranges to enable PE sALD.
15	Lack of deployment of metrology solutions in industrial environments	WP3, WP4, WP5	Tailored application of existing metrology lab equipment (Wyko WLI, AFM, XPS tools) for characterising surfaces. Deploying DuPont Teijin Films' patented large area metrology technology for assessment of A4's at nano scale for defects.
16	Lack of in-line and real time measurement system for quality control.	WP3, WP4, WP5	Different optical systems deployed, e.g. spectral reflectance and spectroscopic imaging ellipsometry. In addition, implementation and optimisation of NDT testing for zero-defect testing, in-line Raman spectroscopy and luminescence testing processes combined with the development of machine learning based inspection data processing will ensure the QC.
17	Weakness in data processing and integration	WP3	Specific technical objective dedicated to develop and implement machine learning based inspection data processing and automated defect detection software inline manufacturing process of OLEDs and solar cells.
18	Commercial availability of special purpose organic material (OM)	WP2	The partner in lead (VTT) with the help of the consortium network will preliminarily contact suppliers of candidate organic materials.
19	Lack of industrial equipment or any tool and related knowhow meant to demonstrate industrial suitability	WP2, WP6	The pilot tools and equipment already available and useful for the milestones of the project will be chartered.
20	Integration of existing equipment and existing characterization might create unexpected additional Non-Recurring-Engineering	WP6	Documentation (ie CRS) will take this compatibility as a first prio.
21	To define the system under research and design potential recycling routes, it is always	WP5	Potential difficulties arising from a lack of information, will be faced with a deeper literature research, conducted for that specific case and

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
	mandatory to analyse and/ or count on the samples chemical composition (e.g. defected and/or end-of-life OLEDS and/or OPV and/ or CIGS), as well as how materials are assembled. Since this is sensitive and confidential information for some end-users (i.e. for example, organic compounds nature will not be revealed, in the case of OLEDS and OPV), there´s a potential risk related to complete achievement of recycling/ reuse strategies, depending on each sample´s nature. Besides, some end-users have defined their interest in specific elements/materials to be recovered, instead of the whole solar cell; this reduces the recycling opportunities that could be explored for a complete module.		the development will be completed making use of theoretical background. Possibility of using non-confidential reference samples from the OLEDSOLAR research partners (e.g. VTT) will be examined.

Poster, Leaflets, Presentations, Dissemination Support Material	Research, Scientific & Academic Community, Commercial & Industry, Investors, General Public, Policy Makers	Month 3

3 Implementation

3.1 Work plan — Work packages, deliverables

Brief presentation of the overall structure of the work plan:

The brief overview of workplan and interconnection between different WPs is shown in PERT figure below. The project starts with WP1 establishing requirements of high yield manufacturing parameters, inspection needs and material challenges to achieve reliability of components and longer lifetimes promoting sustainability thus less waste and material to be recycled thus contributing to circular economy. Further, the sensors and automation of R2R and S2S lines will also be analysed in WP1. In WP2, innovative deposition, packaging and patterning techniques and process will be optimised for industrial scaled up implementation.WP3 will deal with inspection system, techniques and software development.WP4 will incorporate automation, sensor system, hardware and intelligent decision management software. WP5 will develop recycling and re-use strategies and technologies to promote sustainability. WP6 will include testing and trial at pilot scale as well as integration and validation activities in production lines of end user partners. WP7 will include dissemination and exploitation activities and WP8 will outline project management, coordination guidelines and implementation.

Graphical presentation of the components showing how they inter-relate (Pert chart):

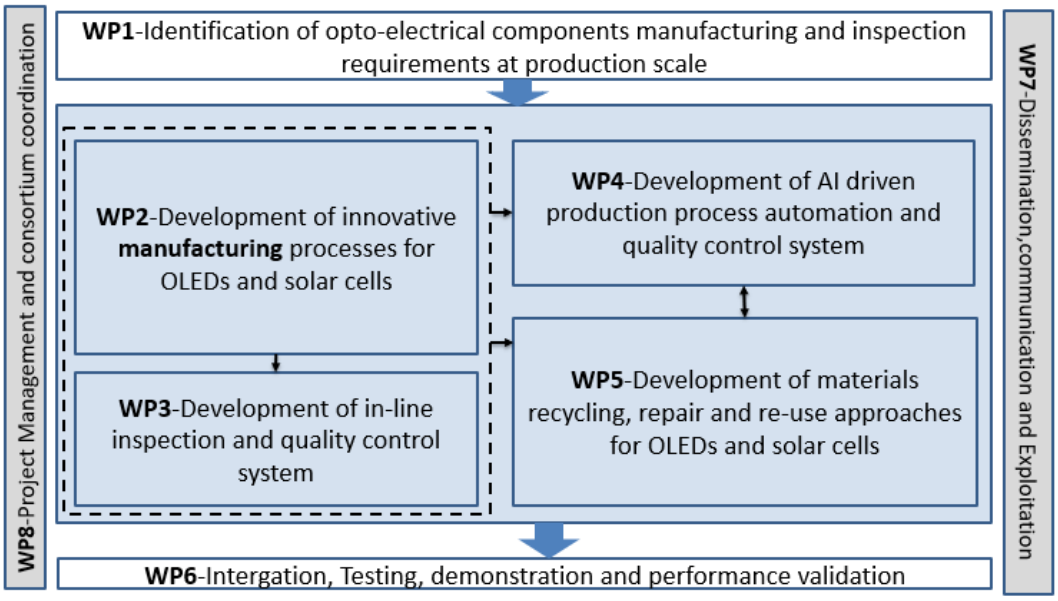


Figure 55 Project PERT diagram

Timing of the different work packages and their components (Gantt chart):

WPs and Tasks	Leaders	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
Identification of opto-electrical components manufacturing and inspection requirements at production scale	*ASPF*																																										
Identification of OLEDs manufacturing requirements	*INURU*																																										
Identification of Solar cells manufacturing requirements	*ASPF*																																										
Assessment of current production processes and process tools adaptability as per production requirements	*SMIT*																																										
Establishing requirements for inspection,quality and process control	*Solibro*					M1																																					
Analysis of material challenges for manufacturing process reconfiguration and recyclability	*Advent*																																										
Development of innovative manufacturing processes for OLEDs and solar cells	*TNO*																																										
Developing adaptive material handling system including intra equipment automation	*Coatema*																																										
Optimisation of scalable manufacturing processes for solar cells	*VTT*																																										
Optimisation of R2R and S2S atomic layer deposition techniques and processes in R2R and S2S production process	*TNO*																																										
Development of inkjet-based deposition technique for pre-encapsulation of solar cells and flexible OLEDs	*VTT*																																										
Development of reconfigurable and scalable lamination, joining and packaging process for OLEDs and solar cells	*VTT*																																										
Testing and validation on production pilot line for roll-to-roll fabricated barrier films	*FhG*																																										
Systems level Testing and validation on production pilot lines	*VTT*																													M2													
Development of in-line inspection and quality control system	*TWI*																																										
Development of in-line optical inspection system for component level testing at production scale	*IRIS*																																										
Implementation and optimisation of NDT testing for zero-defect testing at device level for OLEDs and solar cells	*TWI*																																										
Development of in-line Raman spectroscopy and luminescence testing processes and in-line prototype devices	*SAL*																																										
Qualification of thin film barrier coatings for flexible devices	*FhG*																																										
Large area printing quality measurement and comparison to design layout	*VTT*																																										
Machine vision based micro meter level register accuracy measurement for visible and highly transparent layers	*VTT*																																										
Implementation of functional testing processes including aging testing and optical performance	*LEITAT*																																										
Development of machine learning based inspection data processing and defects detection software	*BUL*																																										
System level Testing and validation on production pilot lines	*VTT*																													M3													
Development of AI driven production process automation and quality control system	*IRIS*																																										
Identification of production automation requirements and process control system design	*ASPF*																																										
Sensor system development for automation and monitoring of production processes	*IRIS*																																										
Development of intelligent decision management and control software for automation and monitoring	*IRIS*																																										
Testing and validation on production pilot lines	*IRIS*																													M4													
Development of materials recycling and re-use approaches for OLEDs and solar cells	*LEITAT*																																										
Evaluation of potential existing pre-treatment technologies and recycling solutions for OLEDs and solar cells	*LEITAT*																																										
Definition of existing and/or innovative recycling routes into the own manufacturing process	*LEITAT*																																										
Testing and validating the feasibility of recycling technologies at pilot scale	*LEITAT*																																				M5						
Intergation, Testing, demonstration and performance validation	*SMIT*																																										
Integration of finalised manufacturing, inspection and automation processes on pilot line	*SMIT*																																										
Testing and trials of integrated finalised OLEDSOLAR solution on pilot lines of OLED and solar cells	*VTT*																																										
Testing and trial of OLEDSOLAR on production lines	*ASPF*																																										
Scalability and performance validation of OLEDSOLAR processes and solution	*Coatema*																																										M6
Quantification of the environmental and economic performance of the OLEDSOLAR solution	*LEITAT*																																										
Standardisation, regulations, certification and safety codes	*TWI*																																										
Dissemination and Exploitation	*VTT*																																										
Plan for Exploitation and Dissemination of results (PEDR)	*VTT*																																										
Communication, advertising and conferences publications of technical outputs	*VTT*																																										
Valorisation of results	*VTT*																																										
Post Project Business Plan and Commercialisation roadmap	*SMIT*																																										
Project Management and Coordination	*VTT*																																										
Project Management and Coordination	*VTT*																																										
Risks, Deliverables, Milestones and IP management	*VTT*																																										
Project reporting to EC and periodic progress meetings	*VTT*																																										
Quality assurance, Innovation Management and Project Performance monitoring	*VTT*																																										

4.1 Participants (applicants)

	Participant 1 **Organisation Name: Teknologian tutkimuskeskus VTT Oy (VTT)** **Website: http://www.vttresearch.com/**

Organisation Description:
VTT Technical Research Centre of Finland Ltd is a state owned and controlled non-profit limited liability company established by law and operating under the ownership steering of the Finnish Ministry of Employment and the Economy.
VTT is an RTO whose activities are focused on three areas: Knowledge intensive products and services, Smart industry and energy systems, and Solutions for natural resources and environment. VTT is impact-driven and takes advantage from its wide multi-technological knowledge base to strengthen Finnish and European industrial competitiveness. VTT can combine different technologies, produce information, upgrade technology knowledge, and create business intelligence and value added for its stakeholders.
VTT has a staff of 2128, net turnover in 2016 was 162,6M€ and other operational incomes were 86,4M€. Over the years, VTT has gained vast experience from participation and coordination of numerous European projects including R&D Framework Programme projects and other thematic frameworks and programmes. VTT is ranked among the leading European RTOs.

Relevant Publications:

- Välimäki, M., Apilo, P., Po, R., Jansson, E., Bernardi, A., Ylikunnari, M., Vilkman, M., Corso, G., Puustinen, J., Tuominen, J., Hast, J., R2R-printed inverted OPV modules – towards arbitrary patterned designs, Nanoscale, 7 (2015) 9570-9580, DOI: 10.1039/c5nr00204d
- Vilkman, M., Hassinen, T., Keränen, M., Pretot, R., van der Schaaf, P., Ruotsalainen, T., Sandberg, H.G.O, "Fully roll-to-roll processed organic top gate transistors using a printable etchant for bottom electrode patterning", Organic Electronics, 20 (2015) 8-14.
- Kopola, Pälvi; Tuomikoski, Markus; Suhonen, Riikka; Maaninen, Arto, "Gravure printed organic light emitting diodes for lighting applications", Thin Solid Films. Vol. 517 (2009) No: 19, 5757 – 5762
- Vilkman, M., Apilo, P., Välimäki, M., Ylikunnari, M., Bernardi, A., Po, R., Corso, G., Hast. J., "Gravure-Printed ZnO in Fully Roll-to-Roll Printed Inverted Organic Solar Cells: Optimization of Adhesion and Performance", Energy Technology (Printed Energy Technologies special issue), 3 (2015) 407-413
- Apilo, P., Välimäki, M., Po, R., Väisänen, K.-L., Richter, H., Ylikunnari, M., Vilkman, M., Bernardi, A., Corso, G., Hoppe, H., Roesch, R., Meitzner, R., Schubert, U.S., Hast, J., "Fully Roll-to-Roll Printed P3HT/Indene-C60-Bisadduct Modules with High Open-Circuit Voltage and Efficiency", Solar RRL, 2018, DOI: 10.1002/solr.201700160.

Relevant Projects:

- EU_THIME: Thinfilm measurements on organic photovoltaic layers (2012-2014)
- EU_ArtESun: Efficient, large-area arbitrary shape solar energy (2013-2016)
- EU_Monoscribe: Roll-to-Roll Monolithic Interconnection of Customizable Thin-film Solar Modules (2015-2018)

- EU_IN-LIGHT: Development of printed transparent OLEDs integrated into window applications (2012-2015)
- EU_Polaric: Printable, organic and large-area realisation of integrated circuits (2010-2014)

Infrastructure/Equipment:

- MAXI R2R Printing line
  - 4 interchangeable printing unit slots
  - Available deposition units: forward and reverse gravure, rotary silk screen, flexography and slot die coating
  - Plasma substrate treatment unit
  - Lamination unit
  - R2R hot embossing unit
  - Die cutting unit
  - Drying units (air, UV)
  - Automatic registration system
  - Max. web width 300mm
  - Max. web velocity 30m/min

Figure 1: MAXI R2R printing line

- R2R Functional tester
  - 192 measurement channels
  - Minimum test pad size 2mm x 2mm
  - Maximum test fixture size 408mm x 290mm
  - Automated test fixture alignment
  - In-house test fixture design and assembly
  - Different measurement possibilities between test points
- DELTA converting line
  - 6 interchangeable converting unit slots
  - kiss cut (top and bottom)
  - lamination
  - slitting
  - CO2 laser
  - Automatic registration system for lamination
  - Max. web width 300mm
  - Max. web velocity 90m/min

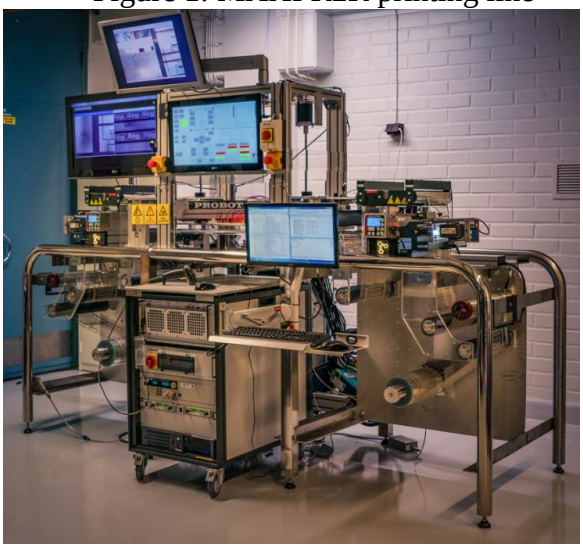


Figure 2: R2R Functional tester

- S2S gravure (Labratester), flexo (RK Flexo), inkjet (PiXDRO LP50 & Fujifilm Dimatix DMP-2800) & screen (EKRA) printers, gravure & flexo with 10 µm alignment accuracy (Superproofer 220)
- PiXDRO inkjet has Fujifilm Dimatix S-Class and Konica Minolta KM512 industrial scale printheads
- Encapsulation/lamination inside N2 glove box for sheets and short (1-2 m) rolls
- Aging measurements (ISOS-L-3) for solar cells
- Electrical characterization of solar cells/modules and OLEDs

Figure 3: DELTA converting line

Key personnel

Name: Marja Vilkman **Gender: Female**

Background:

Marja Vilkman has worked with printed electronics since 2001 when she started as a research trainee at VTT. Her research has focused on functional materials and processing methods for flexible electronics - both in S2S and R2R scale. She has been steering several international R&D projects related to organic solar cells, transistors and scale-up of printing/patterning processes.

Education:

2017: Adjunct professor at Aalto University (Field: Printed Electronics)
2010: D.Sc. in Physics (Field: Polymer physics)
2002: M.Sc. in Chemistry (Field: Polymer chemistry)

Work Experience:

Senior Scientist at VTT 2012: Responsible for project planning & management, scientific coordination & supervision related to S2S and R2R printed solar cells and transistors, etching processes etc.

Research Scientist at VTT 2003-2012: Responsible for ink formulation, printing processes and electrical and chemical characterisation of thin films.

Product Development Experience:

Working with industrial clients to increase the TRL of materials and processes.

Intellectual Property Management:

Two patents:

1. Vilkman, M., Mäkelä, T., Lehtinen, K., "Time and humidity sensor and the use thereof" EP 2009432 B1, filing date Jun 3, 2008.
2. Calabrese, A., Carbonera, C., Vilkman, M. "Inverted polymer solar cells and process for producing the same" WO 2015068102, filing date Nov 2014 (priority: Nov 2013 IT 2013-MI1831).

Innovation Management:

WP leader for Dissemination and Exploitation in two EU FP7 projects (EU_Polaric & EU_Rotrot).

Name: Kim Eiroma **Gender: Male**

Background:

Kim Eiroma has worked as an inkjet printing and materials deposition expert at VTT since 2006. He has contributed in publicly funded collaborative and privately commissioned research projects

in several application areas, ranging from graphic applications to diagnostic and sensor devices to electronic component fabrication and integration.

Education:

2005: M.Sc. (Tech.) Graphic Arts Technology (Printing)

Work Experience:

Research Scientist at VTT 2006 - present: Research and development in applications, materials and processes for printed functional devices and graphic applications, specializing in the inkjet deposition process.

Product Development Experience:

Working with industrial clients to increase the TRL of materials and processes.

Name: Tuomas Happonen **Gender: Male**

Background:

Tuomas Happonen has worked with printed and flexible electronics since 2010 at VTT and University of Oulu. He has solid expertise on functional and reliability testing on several application areas in printed and hybrid electronics. He has strongly contributed this field by developing new test equipment and test methods for project use.

Education:

2016: D.Sc. (Tech), University of Oulu, Electrical engineering

2008: Lic.Sc. (Tech), University of Oulu, Applied electronics

2005: M.Sc. (Tech), University of Oulu, Electronics

Work Experience:

Senior Scientist at VTT 2016 - present: Automated roll-to-roll functional testing of printed electronics, environmental reliability testing of printed and hybrid electronics, bending and stretching tests of printed and hybrid electronic structures

Several positions in research and teaching staff at Optoelectronics and Measurement Techniques Unit, University of Oulu 2004 – 2016: Reliability testing of printed electronics under environmental stresses (especially cyclic bending tests), development of printed and flexible sensors, remote and embedded testing of electronics, planning and realization of study modules in the degree program of electrical engineering.

Product Development Experience:

Working with industrial clients to increase the TRL of system integration in printed and hybrid electronics.

Fraunhofer	**Participant 2** **Organisation Name: Fraunhofer Gesellschaft (FhG)** **Website: www.fraunhofer.de**

Organisation Description:

The Fraunhofer Institute for Organic Electronics, Electron Beam and Plasma Technology (FEP) is one of 60 research institutes within the Fraunhofer Society, the largest society for applied research in Europe. In July 2014 Fraunhofer FEP was merged with the former Fraunhofer COMEDD. The core competences of the FEP are novel device technologies in the field of organic electronics based on vacuum coating technologies, like thermal evaporation, pulse-magnetron-sputtering, plasma activated evaporation and plasma enhanced chemical vapour deposition. FEP concentrates

on applied, industry-oriented research both in public-funded joint projects as well as in direct research contracts with industrial companies. One major field of FEP is the coating of flexible products like polymer webs, ultra-thin glass and metal foils. Both, high throughput plasma-assisted evaporation as well as high quality magnetron-sputtering are used to coat polymer surfaces with transparent conductive oxides, permeation barrier layers and optical layer stacks as well as metal surfaces with scratch resistant, electrical insulation or chemical protection layers. Based on the existing infrastructure FEP can offer product developments starting with the system concept till pilot fabrication level.

Together with CSEM, TNO and MBNT, Fraunhofer (FEP and IWS) will providing feedback for barrier film performance analysis an large areas for flexible organic devices with high barrier requirements down to 10-6 g/m²*d. The final goal is to implement an in-situ inspection system (hyperspectral imaging) for rapid in-line tests to reveal yield relevant issues under pilot production scale. The barrier performance will be validated and qualified in the existing roll-to-roll OLED line.

FEP and IWS will contribute to WP2, WP3, and WP6

Relevant Publications:

- Stefan Mogck et. al., “Present status of Roll-to-Roll OLED fabrication and encapsulation”, J. Jpn. Soc. Colour Mater., 86 (12), 461–465 (2013)
- Christian May, “Flexible OLED – new possibilities for automotive lighting”, Intelligent Automotive Lighting, Düsseldorf, Germany, 27 – 29 January 2015
- Christian May, “Next generation flexible lighting by OLED technology “, LpS 2013 – 3rd LED Professional Symposium + Expo, Bregenz, Austria, 24 – 26 September 2013
- Hoffmann, M.; Kristukat, C; Diekmann, K.: „OLED lifetime determination procedure and proposal for standardisation”, in: A Buckley (ed.): Organic light-emitting diodes (OLEDs): materials, devices and applications, Woodhead Publishing, August 2013, ISBN: 978-0-85709-425-4, p. 601-634
- M. Fahland, T. Vogt, W. Schoenberger, J. Fahlteich, and N. Schiller, Transparent Conductive Electrodes for Flexible Solar Cells, 51st Annual Technical Conference Proceedings (2008), 785-788

Relevant Projects:

- EU FP7 project TREASORES OLED device development on novel transparent electrodes for large area illumination (10x 10 cm², > 25 lm/W, lifetime > 10000 hrs.)
- EU FP7 project SMARTONICS focusing on pilot lines for organic solar cells and OLED devices. In this project Frauhofer FEP (together with the Fraunhofer-POLO® alliance) develops roll-to-roll manufacturing processes for ultra-high barriers
- German funded project FLEX25 about pilot-scale manufacturing of outdoor stable encapsulation films for photovoltaic applications
- German funded project FlexSOL focusing on transparent electrodes for flexible silicon solar cells.
- German funded project R2flex for the development of roll-to-roll OLED process technology of small molecule OLED devices on flexible substrates.
- German funded project R2D2 (Roll to device) is the focus on reliability and long term stability for roll-to-roll OLED fabrication.
- H2020 project PI-SCALE Bringing flexible organic electronics to pilot innovation scale (> 1m OLED lighting stripes with efficacy of 30 lm/W, lifetime > 15 years)

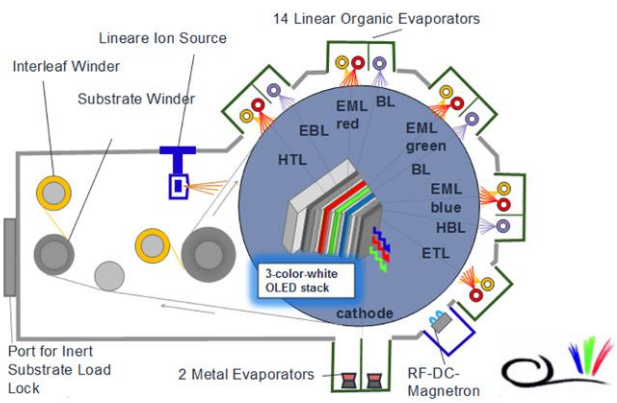
Infrastructure/Equipment:

The Fraunhofer FEP has the following main items of technical equipment for OLED lighting:

- R&D and pilot line for rigid and flexible substrates in ISO class 4 clean room environment (200x200 mm² substrates, some processes up to 370x470 mm²) for lighting and organic solar cells applications
- Coflex600 and novoflex600 for pilot production of TCO electrodes and ultra-high barriers on transparent polymer films with a maximum web width of 600 mm.
- A roll-to-roll OLED line fabrication for flexible substrates (foil width of 300 mm for metal, ultra-thin glass and polymer films) including small molecule vacuum deposition, inspection system and lamination under protective nitrogen atmosphere for the barrier film encapsulation.

Figure 1: Process flow of the R2R prototype line from substrate inspection up to the final electro-optical characterization of large area OLED devices.

Figure 2: On the left hand side a schematic drawing of the R2R vacuum coater and the indication of the white p-i-n OLED stack processing which requires 14 organic linear evaporators is depicted. On the right hand side shows a photograph of the substrate winding unit and the deposition drum of the R2R vacuum coater with winders for substrate and protective liner foil to minimize substrate damage after the OLED deposition process.



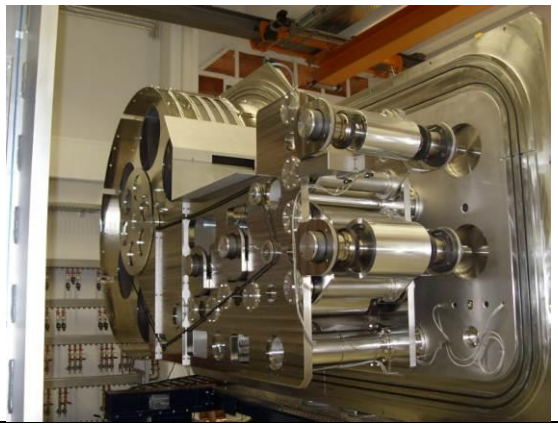


Figure 3: Photograph of R2R lamination system (left) which is encased in an inert box at Fraunhofer FEP (right)

Figure 4: Ro rbine ISO6.

Key Personnel

Name: Dr. Stefan Mogck

Gender: Male

Background:

Dr Stefan Mogck studied physics at the University of Tübingen, Germany in 2000. In 2004 he received his PhD at the Institute of Applied Physics (Materials science and engineering) of the University of Groningen (NL). Afterwards, he joined to Advanced Micro Devices in Dresden as a defect engineer. In 2007 he was engaged at MicroEmissive Displays (MED) in Dresden as process engineer for the sputtering and the PLED spin coat process. Since 2008, he changed to Fraunhofer Institute for Photonic Microsystems. He is now leading the department "Roll-to-Roll Organic Technology".

Name: Dr. Wulf Grählert **Gender: Male**

Dr Wulf Grählert obtained a degree in chemistry from the Institute of Technology (Leuna Merseburg) and doctorate in the field of spectroscopy of solids and layered systems in 2002. He has been staff member at the Fraunhofer Society since 1995 and leads the group "Optical Inspection Technology" since 2004 at FhG-IWS, Dresden. His current research interests include gas analysis, particle sensing, multi-spectral sensing for industrial applications. Dr. Grählert has been involved in several European projects and has coordinated the Large-scale integrating project "nanoMOF" from 2009 – 2013.

TNO innovation for life	**Participant 3** **Organisation Name: Nederlandse Organisatie voor Toegepast Narrturwetenschappelijk Onderzoek (TNO)** **Website: www.tno.nl**

Organisation Description:

TNO (Nederlandse Organisatie voor toegepast-natuurwetenschappelijk Onderzoek TNO) is one of the major contract research organisations in Europe with a staff of circa 3500 and an annual turnover of 580 million Euros. TNO carries out research in order to achieve impact in themes: Healthy Living, Industrial Innovation, Transport and Mobility, Energy, Built Environment, Information Society, and Defence, Safety and Security. TNO acts as an intermediary between basic research and industry. By translating scientific knowledge into practical applications, TNO contributes to the innovation capacity of businesses and government.

TNO has extensive experience in EC research and project management through involvement in coordinating and participation in hundreds of EU projects. In the theme of Industrial Innovation the focus is on innovation in the areas of high-tech systems and materials, manufacturing, the sustainable chemical industry and space applications. The Thin Film Technology research group will contribute to this project. It has as main activities development, modelling and control of deposition, patterning and interconnect processes for economical and durable thin film PV cells. The group is on the forefront of process and equipment development of spatial (including R2R and S2S) ALD of barrier, buffer and electrode layers, and front electrode grid applications, and of advanced tests development for cell and module stability testing. TNO has expertise and facilities concerning the growth and the physical/electrical characterisation of CIGS-based solar cells.

TNO's main involvement will be in WP2 (which will be coordinated by TNO) and WP6, but TNO will also participate in WP1, WP3 and WP7. TNO's technical role is to contribute to specifications and validation of the metrology systems for quality control of (optionally Plasma Enhanced or PE) spatial ALD barriers and optimization of the barrier coating process itself tools.

Relevant Publications:

- A. Illiberi, P. Poodt, P. Bolt, F. Roozeboom, Recent Advances in Atmospheric Vapor-Phase Deposition of Transparent and Conductive Zinc Oxide, Chemical Vapor Deposition 20 (2014)
- A. Illiberi, R. Scherpenborg, Y. Wu, F. Roozeboom, P. Poodt, "Spatial Atmospheric Atomic Layer Deposition of AlxZn1–xO", ACS Appl. Mater. Interfaces, 2013, 5 (24), pp 13124–13128
- S. Sneck, R. Knaapen, H. Heezen, A. Smith, "Rolling out the sunshine: advanced solar cell production via continuous coating systems", Surface Coating International, 96(3):137-139, June 2013
- C. Frijters, P.J. Bolt, P. Poodt, R. Knaapen, J. van den Brink, M. Ruth, D. Bremaud, A. Illiberi, "Atmospheric spatial atomic-layer-deposition of Zn(O, S) buffer layer for flexible Cu(In, Ga)Se2 solar cells: From lab-scale to large area roll to roll processing", Conference Record of the IEEE Photovoltaic Specialists Conference, Volume 2016, 18 November 2016, Article number 7749856, pp 1449-1451
- V. Zardetto, B.L. Williams, A. Perrotta, F. Di Giacomo, M.A. Verheijen, R. Andriessen, W.M.M. Kessels, M.Creatore, "Atomic layer deposition for perovskite solar cells: research status, opportunities and challenges", Sustainable Energy and Fuel, 2017, 1, 30

Relevant Projects:

- **Clean4yield** - Contamination and defect control for increased yield for large scale R2R production of OPV and OLED; FP7 program, grant agreement 281027 (ended 30-04-2016)
- **INREP** – Towards Indium free TCOs; H2020 program, grant agreement 283974 (ended 31-01-2018)
- **R2R-CIGS** - Roll-to-roll manufacturing of high efficiency and low cost flexible CIGS solar modules; FP7 program, grant agreement 283974 (ended 30-09-2015).

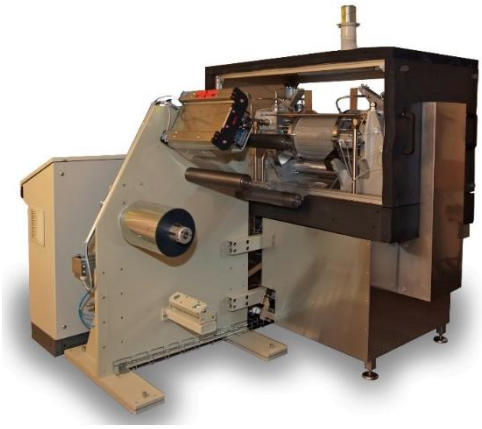
- **Mujulima** – Innovative materials for high performing OPV; FP7 program, grant agreement 604148 (ended 31-12-2016)
- **PI-SCALE** - Bringing flexible organic electronics to pilot innovation scale; H2020 program, grant agreement 688093 (ends 31-12-2018)

Infrastructure/Equipment:

- Atmospheric spatial atomic-layer-deposition (S-ALD) tools: two rotary (lab scale, with/without remote or direct plasma), sheet-to-sheet (32.5cm x 40cm, with/without remote plasma) and roll-to-roll (30 cm wide) tools.

Rotary spatial ALD tool

Roll-to-roll spatial ALD tool



Sheet-to-sheet spatial ALD tool

- Test line for manufacturing of (10 cm x 10cm) CIGS modules on rigid (glass) substrate using co-evaporation and (30 cm x 30 cm) using 2 step process route (with sputter and CBD tools for precursor coating) to support in-house the development and validation of passivation, front window and barrier deposition processes and techniques for CIGS modules.

- R2R multi-layer slot-die coating facility for 30 cm wide foil.
- S2S slot-die (1-10 m/min, 15x15 cm2 and 30x30 cm2) tool.
- Techniques to characterize the optical, electrical and morphological properties of thin films; spectroscopic ellipsometry, spectrophotometer, photoluminescence, X-ray diffraction, Hall measurements, four points probe, high resolution scanning electron microscope (SEM), high resolution transmission electron microscope (TEM), energy dispersive X-ray spectroscopy (EDX), X-ray fluorescence (XRF).
- Climate chambers, partly with in-situ illumination and real time monitoring of cell and module performance.

Key Personnel

Name: Dr Pieter Bolt **Gender: Male**

Background:

MSc. Applied Physics, Groningen University and PhD Technical Sciences, Eindhoven University of Technology. He worked for 2 years in Japan at AIST and at Hitachi Ltd. Since 1991 employed by TNO in field of materials processing and application research and knowledge transfer, in projects with automotive, steel & aluminium, building, electronics, medical, lighting and PV industries. Since 2008 his research is focused on thin-film photo-voltaics. He has been and is involved in a large number of national and European collaboration projects, as partner and as coordinator (of the FP7 R2R-CIGS and FP7 SSL-erate / Lighting for People projects).

Name: Dr. Fieke van den Bruele **Gender: Female**

Joined TNO after receiving in 2012 a PhD in Chenistry from the Radboud University Nijmegen in the Netherlands. Research scientist responsible for spatial ALD high performance barriers for solar and battery applications.

Name: Dr. Valerio Zardetto **Gender: Male**

MsC and PhD degrees in Electronic Engineering at Università degli Studi di Roma. After a post-doc position at Eindhoven University of Technology since 2016 working as research scientist at TNO specialized in spatial ALD for photovoltaic solar cell technology.

Name: Dr. Paul Poodt **Gender: Male**

M.Sc. degree in chemistry in 2003 and Ph.D. degree in physics in 2008 from Radboud University Nijmegen. In 2008 he joined TNO, working on thin film deposition for solar cell applications, and since 2009 as lead process scientist spatial ALD and since 2013 as manager of the spatial ALD research program for large area electronics (solar, battery, lighting) applications.

Participant 4

Organisation Name: INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL (IRIS)

Website: www.iris.cat

Organisation Description:

IRIS (www.iris-eng.com) is a technology SME that was established in Barcelona in 2007 and specialises in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analysers for at-line and in-line process monitoring.

On the one hand our solutions are based on the latest developments in photonics-in the fields of MEMs Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we

leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analysers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloud-based for mobility, remote monitoring and multi-site applications.

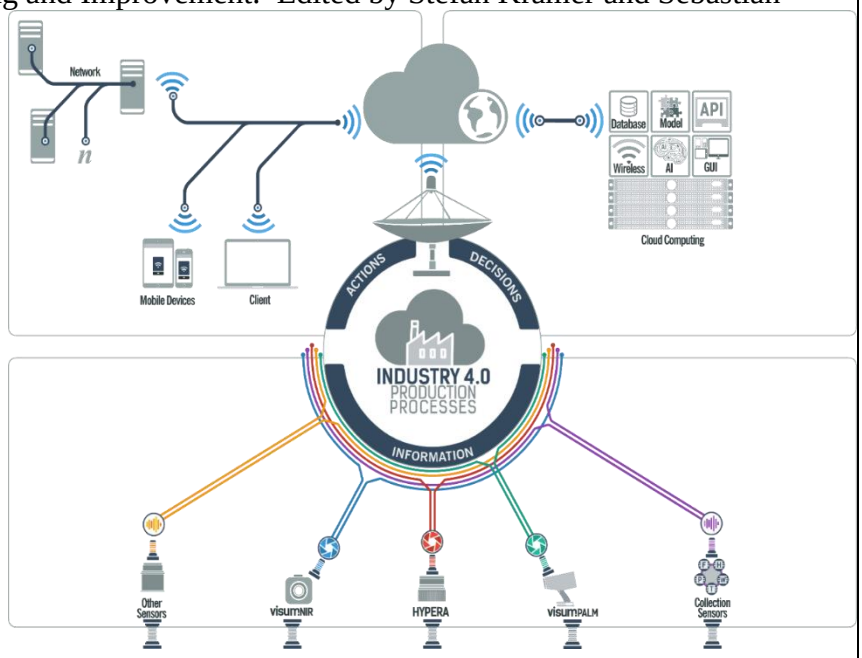
Our Innovation Division has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimisation
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorisation

Our Innovation Dvision is experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners) and principally in the SPIRE, FoF, and BBI programmes. The company was established in 2007 and currently employs 56 staff with a turnover of €3.8 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics & telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification).

Relevant Publications:

- A. Rosales, O. Mc Nerney. Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3 in press
- Commercial Data Mining, Processing, Analysis and Modeling for Predictive Analytics Projects, David Nettleton, Elsiever book, ISBN9780124166028, 304p
- Conference ICIEMPM 2016: Towards Automatic Calibration of In-Line Machine Processes, 18th International Conference on Industrial Engineering and Manufacturing Process Modeling, D.F.Nettleton, E.Bugnicourt, C. Wasiak, A. Rosales, London, UK, July 28-29, 2016.



- Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings by Optical Technologies, E. Bugnicourt, T. Kehoe, M. Latorre, C. Serrano, S. Philippe and M. Schmid, Nanomaterials 2016, Review, 6(8), 150; doi:10.3390/nano6080150
- Injection moulding process optimization by online material monitoring. E. Bugnicourt et al., October 2012. Technical article in Packaging connections.

Relevant Projects:

- OptiNanoPro - Processing and control of novel nanomaterials in packaging, automotive and solar panel processing lines (H2020- NMP-PILOTS, Grant Agreement no: 686116, 2015-2018).
- ProPAT - Robust and affordable process control technologies for improving standards and optimising industrial operations (H2020- SPIRE, Grant Agreement no: 637232, 2014-2018).
- Visum® Inline - cloud-enabled IR spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries.
- Hypera® -Hyperspectral Imaging (HSI) System for on-line foreign body detection and chemical composition analysis (H2020- SMEInst, Grant Agreement no: 726572, 2016-2018).
- Thime - Thinfilm measurements on organic photovoltaic (OPV) layers (FP7, Grant Agreement no: 315665).

Infrastructure/Equipment:

- Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).
- Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.
- IRIS' optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Key Personnel

Name: JOAN PUIG LAGO **Gender: Male**

Background:

Engineering Degree in Electronics from Universitat Politecnica de Mataró and in Industrial Organization from Escola Universitat Politécnica d'Enginyers Industrials de Terrassa. Executive Education in Program for Management Development from ESADE. Joan comes from an operations background and he is an expert in the design of efficient process and the implantation of continuous improvement programmes. Since joining IRIS in 2012, he has implemented new operational procedures based on lean and agile methodologies and he has successfully led the company in securing ISO 9001:2008 certification, in addition to UNE-EN-ISO 166002: Management Systems R&D&i certification. Prior to IRIS, he has 12 years industry experience in automation of industrial processes for the main automotive & aeronautics companies: Mercedes, VW, SEAT, EADS, BOEING, etc. Joan is a master in the definition of strategies and targets, and in aligning systems and workflows according to operations schedules and within agreed costs and high quality levels. He has extensive experience in managing Technical Departments and Operations such as organization and monitoring of production and processes, development of operational plans, and in providing guidance to design and implementation teams.

Name: EDUARD BOSCH COCH **Gender: Male**

BSc in Industrial Engineering (2010) from University of Girona (UdG), M.Sc. in Astrophysics, particle physics and cosmology from the University of Barcelona (UB) and M.Sc. in Mechanics of Materials and Structures. Prior to joining the Production Processes group at IRIS as Mechanical Team Leader, he gained wide experience in structural design, as a project engineer (stress analyst) developing projects including design, testing, manufacturing, calculations and finite element

simulations for the aeronautical sector. He also was the simulation manager of the department, which provides him with the appropriate skills to leader a team. He has also held a position as façade engineer and as an assistant teacher in the University of Girona (UdG) teaching material science. He is expert in design for manufacturing, he is proficient in SolidWorks (Computer Aided Design program) and has a wide experience with Abaqus (Finite Element Method program) among many others. He also have experience as a researcher on advanced materials science. He is currently working as the Team Leader of the Mechanical Engineering department in IRIS, where he has also developed some private R&D projects for the food industry that have successfully become products.

Name: DIANA GALILEA Gender: Female

Holds a Bachelor's degree in Industrial Engineering with specialty in Industrial electronics from the Universidad of La Rioja. Her professional background comprises technical experience in the areas of embedded microcontrollers programming for wireless communications working with different range of sensors. Her software expertise includes Matlab, Visual Studio, Autocad, Protel DXP. Her programming skills include C++, C#, .NET among others. She is working currently as software engineer in IRIS, developing software for chemo-metric analysis, translating chemo-metric algorithms from researchers source code as could be R or Matlab to c++ or C# in order to integrate it with a user interface that carries out the analysis, as well as controls the system. She also has been working in monitoring systems where information about sensors and actuators are displayed on the user interface.

Name: DAVID NETTLETON Gender: Male

Data mining analyst at IRIS. Has a Ph.D in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years experience of applying data mining algorithms (including neural networks, rule induction) and statistical techniques (including ANOVA, regression, principal components) in a diversity of data analysis and modeling projects, such as medical diagnosis/prognosis, fraud detection, customer loyalty and segmentation, online user web search behaviour and data privacy. As well as commercial projects, he has conducted research and lectured at two universities. He has published a diversity of articles in specialized data analysis journals/congresses and three books. He has participated in several competitive multi-partner research projects and has a European patent application for a graph matching algorithm. His IT Industrial experience includes project management, consultancy, programming and analysis, database development and business startups. He has developed software with a diversity of programming languages, (including Java, C++ and COBOL/CICS/DB2), database systems (such as Oracle, Ingres, MySQL) and data mining systems (IBM Intelligent Miner, Clementine, Weka). In IRIS he is currently involved in two major European projects (OpenMind and Sharebox), as a data mining specialist. The work in OpenMind involves modeling a complex industrial production process for producing high precision medical instruments. The work in Sharebox involves building a symbiosis system for industrial collaborations between companies for the recycling of waste products using case based reasoning and agent modeling.

Name: MANEL SANCHEZ VELANDO Gender: Male

Software developer at IRIS. Degree in Informatics Engineering and Diploma in Computer Systems from "Universitat Politècnica de Catalunya". English conversation level (B2). Having worked over 20 years in the field of information technology in different companies he has reported both organizational and management experience. He has great experience in analysis of customer needs and project management. Unix and Windows system administrator for 10 years. As a software engineer, he has done many projects using several languages: C, C++, C#, ASP, ASP.NET, PHP, Java, Javascript, SQL. Starting MEAN stack. He uses design patterns like: MVC, subject-observer, status, representative, factory, facade, etc. GUI design experience in both desktop and web(responsive) and in conformance to web content accessibility levels. UML design. Great experience in MS SQL Server and Mysql databases. He has experience in CAD/CAM and artificial vision control quality systems too. He enjoys analyzing projects with both customers and

team members in order to get the best solution. He likes to listen and learn from everyone around him and share his knowledge with them. Team results are the most important for him.

Name: Dr. MARIANO PERALVAREZ **Gender: Male**

PhD in Physics from the University of Barcelona (2009). Currently Project Technical Responsible and Optical Engineer at IRIS, Mariano is specialised in the fields of LED technology, Lighting, Optoelectronics and Photonics. He has been responsible of Photometric, Photonics and Optoelectronics labs, focusing, among many other activities (Spectroscopy, Materials Science…), on the design (3D-CAD, ray-tracing and CFD), development, prototyping and performance assessment of luminaires and light-engines (photometric, thermal and electric), collaborating closely with manufacturers in the Lighting field. He also has a deep knowledge of Concentration Photovoltaics especially in terms of raytracing simulations of optical concentrators and indoor/outdoor tests of optoelectronic performance. Mariano also participated in 18 R&D projects, funded through both public competitive calls (at National and EU level) and private institutions such as the Project LED4ART, devoted to the refurbishment of the Sistine Chapel Lighting system.

Participant 5

Organisation Name: OPVIUS GmbH (OPVIUS)

Website: www.opvius.com

Organisation Description:

OPVIUS GmbH, founded in 2012 and located in Nuremberg and Kitzingen (INNOPARK Kitzingen), is among the world-leaders in the field of organic photovoltaics. OPVIUS develops and manufactures organic solar cells with a focus on client-specific OPV solutions. In addition, the company is also active in the field of research and development, in order to continue to provide its clients with creative and innovative solutions. With a team of scientists, engineers and manufacturing experts, OPVIUS creates customized OPV solutions. In this regard, OPVIUS uses a unique production process, combining printing, lamination, and laser structuring techniques. This advanced technology is easily scalable and allows the production of the utmost individual, customer-specific designs. OPVIUS also helps clients with system solutions in order to integrate OPVs into already existing or new products.

Relevant Publications:

1. Press release, **12.12.2017**, OPVIUS develops new production process for freedom OPV modules
2. S. Berny *et al*., Adv. Sci. **2016**, 3, 1500342, Solar Trees: First Large-Scale Demonstration of Fully Solution Coated, Semitransparent Flexible Organic Photovoltaic Modules

Relevant Projects:

1. OptiNanoPro: Processing and control of novel nanomaterials in packaging, automotive and solar panel processing lines, Grant Agreement no: 686116, H2020- NMP-PILOTS-2015
2. Infinity: Indium-Free Transparent Conductive Oxides for Glass and Plastic Substrates, Grant Agreement no: 641927; H2020-SC5-2014
3. ETFE-MFM: finished FP7 project on the development and demonstration of a flexible multifunctional ETFE module for architectural façade lighting, Grant agreement no: 322459, FP7-ENERGY-2012-2

4. Sunflower: finished FP7 project on the development of highly efficient, long-lasting, cheap and environmentally friendly printed organic photovoltaics, Grant agreement no: 287594, FP7-ICT-2011-7
5. BMBF (German Federal Ministry for Education and Research) project "POPUP" (finished German national project on building integration of organic photovoltaics)

Infrastructure/Equipment:

- R2R slot-die coating line for OPV manufacturing.
- Screen-printers (flat bed and rotary).
- Lasers.
- Laminator.
- Characterization equipment.

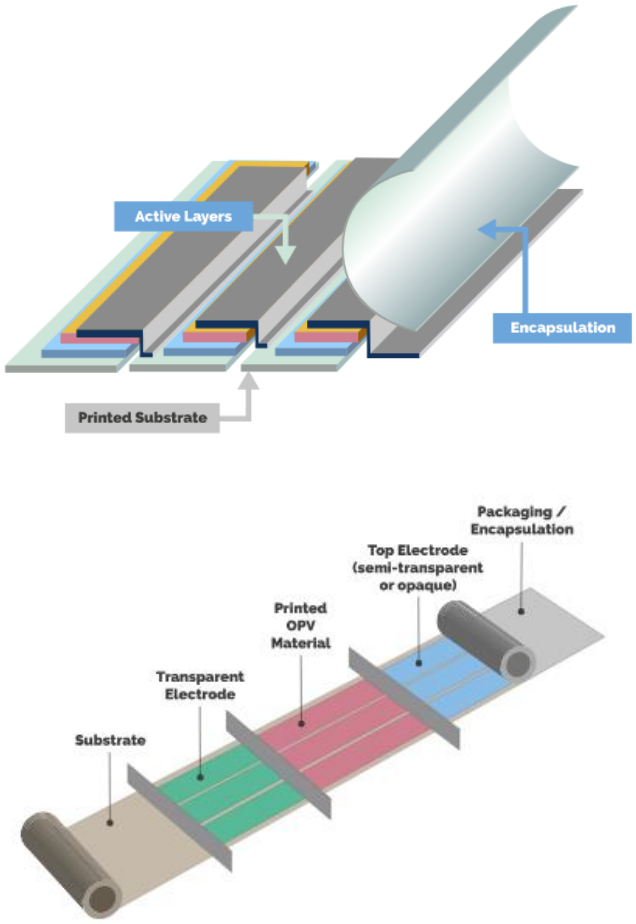


Figure 1: OPVIUS R2R OPV manufacturing process

Key Personnel involved

Name: Dr. Sebastian Meier **Gender: Male**

Background:

Dr. Sebastian Meier is Director of R&D at OPVIUS GmbH. He holds a Diploma degree in Materials Science and a Ph.D. in Engineering from the Friedrich-Alexander University of Erlangen-Nuremberg, Germany. Sebastian has 8 years of industrial experience in the field of organic electronics and is author and co-author of 10 scientific publications in recognized international journals and 5 patents in the field of organic light-emitting devices and solar cells. Currently he is involved in several European and German projects in the field of organic and printable photovoltaics as well as building integrated photovoltaics.

Education:

- 2009: Diploma degree in materials science (Friedrich-Alexander University of Erlangen-Nuremberg, Germany)
- 2014: PhD in Engineering (Friedrich-Alexander University of Erlangen-Nuremberg, Germany)

Work Experience:

- Oct 2009 – Dec 2012: Industrial PhD at Siemens Corporate Technology, Erlangen, Germany
- Jan 2013 – June 2013: Postdoctoral Researcher, University of Valencia, Spain
- July 2013 – April 2014: Research Scientist, BELECTRIC OPV GmbH, Nuremberg, Germany
- May 2014 – May 2016: Senior Research Scientist and Project manager R&D, BELECTRIC OPV GmbH, Nuremberg, Germany
- June 2016 - present: Director R&D, OPVIUS GmbH (formerly BELECTRIC OPV GmbH), Nuremberg, Germany

Product Development Experience:

Dr. Sebastian Meier was involved in the successful realization of several lighthouse projects within OPVIUS GmbH demonstrating the market readiness of organic photovoltaics (OPV). Examples include among others, the German Pavilion at the EXPO Milan (2015), the Security and Peace Building of the African Union in Addis Ababa, Ethiopia (2015), External Lift Shaft at a pharmacy building in Marburg, Germany (2016), Climate Pavilion of the State Horticultural Show, Apolda, Germany (2017) (see press releases at www.opvius.com).

Intellectual Property Management:

- WO 2012/172025 A1 (also published as DE 10 2011 104 169 A1)
- WO 2014/013030 A1 (also published as EP 2 688 119 A1 and US20150208482)
- WO 2014/076278 A2 (also published as WO 2014/076278 A3, EP 2 733 188 A1, EP 2 733 188 B1, US20150303394 A1 and DE 10 2013 202 250 A1)
- WO 2014/124804 A1 (also published as DE 10 2013 202 250 A1)
- WO 2016/009019 A2 (also published as WO 2016/009019 A2, EP 3 170 210 A2 and DE 10 2014 213 978 A1).

Innovation Management:

- Regular participation as an invited speaker at recognized international conferences
- Author and co-author of 10 scientific publications in recognized international journals

Name: Dr. Pavel Schilinsky **Gender: Male**

Background:

Dr. Pavel Schilinsky is Senior Project Manager at OPVIUS GmbH. He holds a PhD degree in physics from the University of Oldenburg. In August 2004, he joined Konarka Technologies. Since October 2012 he leads the activities at OPVIUS (formerly BELECTRIC OPV) related to all aspects of module architecture and system layout. He is the author and co-author of more than 30 scientific publications and of more than 10 patents and patent applications.

Education:

- 2001: Diploma in Physics from the University of Oldenburg
- 2004: PhD in Natural Sciences from the University of Oldenburg

Work Experience:

- 2001 - 2004: Industrial PhD at Siemens Corporate Technology, Erlangen, Germany
- 2004 - 2012: Scientist at Konarka Technologies, Nuremberg, Germany
- 2012 - present: various positions at OPVIUS GmbH (formerly BELECTRIC OPV GmbH)

Product Development Experience:

Dr. Pavel Schilinsky was involved in the successful realization of several lighthouse projects within OPVIUS GmbH demonstrating the market readiness of organic photovoltaics (OPV). Examples include among others, the German Pavilion at the EXPO Milan (2015), The Security and Peace Building of the African Union in Addis Ababa, Ethiopia (2015), External Lift Shaft at a pharmacy building in Marburg, Germany (2016), Climate Pavilion of the State Horticultural Show, Apolda, Germany (2017) (see press releases at www.opvius.com). His main responsibility has been the interface between the OPV-modules and the outer world, e.g. inverters, batteries, etc.

Intellectual Property Management:

- WO2009137141A3 (US20090211633)
- WO2010102116A2 (EP2404333A2 or US20100224252)
- WO2007104039A2
- WO2004025746A3 (US7306968)
- US7413997
- US7781254
- US7304361
- US7868405
- EP1997219A4
- DE102004036734A1
- US20070193621

Innovation Management:

Dr. Pavel Schilinsky is the author and co-author of more than 30 scientific publications and of more than 10 patents and patent applications.

Name: Tobias Sauermann **Gender: Male**

Background:

Tobias Sauermann is Director of Production at OPVIUS GmbH. In his position he leads the production of customized OPV modules, development of new module layouts and is also involved in project management. He obtained his Diploma degree in Materials Science from the Friedrich-Alexander University of Erlangen-Nuremberg, Germany in the end of 2010. He is author and co-author of 10 scientific publications and has industrial experience in OPV for 6 years.

Education:

- 2010: Diploma degree in Materials Science (Friedrich-Alexander University of Erlangen-Nuremberg, Germany)

Work Experience:

- 2011 - 2012: Industrial PhD at Konarka Technologies GmbH, Nuremberg
- 2012 - 2014: Continuing Industrial PhD at Belectric OPV GmbH, Nuremberg
- 2014 - 2016: Project management at Belectric OPV GmbH, Nuremberg
- 2017 - present: Director of Production at OPVIUS GmbH, Kitzingen

Product Development Experience:

Tobias Sauermann was involved as a project manager in the successful realization of several lighthouse projects within OPVIUS GmbH demonstrating the market readiness of organic photovoltaics (OPV). Examples include among others, the German Pavilion at the EXPO Milan (2015), the Security and Peace Building of the African Union in Addis Ababa, Ethiopia (2015), External Lift Shaft at a pharmacy building in Marburg, Germany (2016), Climate Pavilion of the State Horticultural Show, Apolda, Germany (2017) (see press releases at www.opvius.com).

Innovation Management:

- Regular participation as an invited speaker at recognized international conferences
- Author and co-author of 10 scientific publications in recognized international journals

Participant 6

Organisation Name: TWI Limited (TWI)

Website: www.twi-global.com

Organisation Description:

Established in Cambridge, UK in 1946 and with several facilities across the globe, TWI Ltd (www.twi-global.com) is one of the world-leading not-for-profit research and technology organisations with a turnover in 2016 of £60m (€69million). From bases in the UK, South East Asia, India, the Middle East, Central Asia and the USA, nearly 800 staff provide expertise in joining and fabrication, material science and structural integrity. Services include generic research; contract R&D; technical information; engineering services and advice; standards development; and training and qualification services. With over sixty years of service to industry, TWI has worked in almost every conceivable sector, adding value to such disparate areas as offshore wind turbines and oil-platforms; motorsports; bridges; microchips; nuclear power plants; advanced aircraft, both civilian and military; medical implants; solar-panels and many more. TWI's business is built around the development and exploitation of advanced technologies. It is a company on the cutting edge of technological advancement and is recognised as one of the world's leading research and technology organisations. TWI's technologies fall into the following areas:

- Structural Integrity and Corrosion Management cover a wide range of area, focused around reliability assurance and failure resistance
- Non-destructive testing and condition monitoring cover all methods of testing and inspection which are performed without damaging the test subject.
- Welding and joining technologies range from the familiar, such as laser and arc welding, to the more specialised, like ultrasonic and diffusion bonding.
- Materials processing includes cutting, surfacing and surface engineering, and use of specialist materials.
- Manufacturing support provides an integrated approach to consultancy on all aspects of manufacturing, advising on both business and technical aspect.

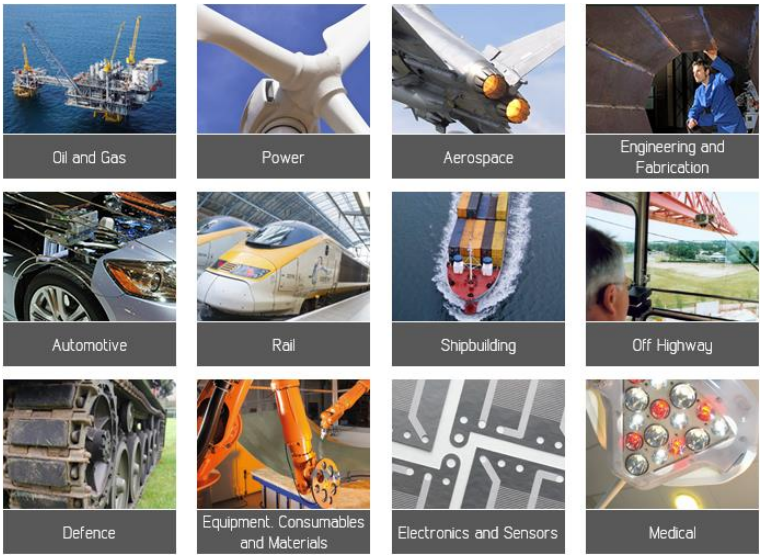
TWI is internationally renowned for its ability to employ multidisciplinary, impartial teams to implement both established and advanced joining technologies, or to solve problems arising at any stage of the product life cycle – from initial design, materials selection, production and quality assurance, through to in-service performance, repair and life extension. TWI also has representation on over 100 standards committees.

TWI therefore has the ideal credentials to deliver a successful project, on time and within budget with a higher than normal chance of product exploitation at the end of the project and particularly where the project developments are highly relevant to their industrial membership that currently consists of 1800 organisations in 70 countries.

Summary of R&D Investment:
A large proportion of TWI's activities (65%) are contract research, funded by a combination of TWI Industrial Members or UK and EU public funding bodies, and built around the development of advanced technologies. TWI is an experienced participant in collaborative research projects in Europe and the UK. TWI has successful coordinated a large number of these government funded projects (~50%) and has gained extensive expertise in being project or technical project leader.

TWI's Industrial Members:
TWI is a membership-based organisation with 1,800 Industrial Member companies from 70 countries, representing every sector of the manufacturing industry. Many of the largest engineering companies in the world are amongst its diverse Industrial Membership base, including:

- Rolls-Royce, Boeing, Airbus, BAE SYSTEMS, GKN, Lockheed Martin and Honeywell from the aerospace sector
- BP, Exxon Mobil, Shell, Chevron, Petrobras, Lloyd's Register, Kawasaki Shipbuilding and Samsung Heavy Industries from the oil and gas and marine sector
- Areva, EDF, E·ON, Alstom, Scottish Power, SunPower and ITER from the power sector
- Honda, Jaguar Land Rover, Yamaha, London Underground, Network Rail, Tata Steel and Hitachi from the automotive and rail sectors
- JCB, Caterpillar, Laing O'Rourke, Tata Steel, ESAB, Linde Group and BOC from the construction and fabrication sectors
- DePuy, Smith & Nephew, Siemens, Danfoss and Emerson from the Medical, electronics and sensors sectors,



Added to this, TWI's dedicated team of industry specialists add value to almost every conceivable industry sector – from offshore oil and gas, civil engineering and nuclear power, to electronics, advanced aircraft (both civilian and military), medical equipment and solar power. Because it serves such a broad client base TWI understand and appreciate that different industries have different needs. Therefore, TWI's research programme is market driven.

Innovation:
TWI has a rich and well-publicised history of innovation and invention dating back to the time of the British Welding Research Association in the 1940s. Memorable highlights from the past include the CTOD method for measuring fracture toughness, the Wells wide plate test, types of high powered CO_2 laser, and many innovations in the use of laser, friction, arc, adhesive bonding and other technologies related to welding, joining, coating and testing. In particular, TWI's NDT activities focus on bespoke and technically challenging inspection solutions.
As well as undertaking research and project work on behalf of its Members, TWI also produces a number of products and software packages:

- PolyTest system for plastic pipe inspection
- Railect phased array inspection system for rail welds
- RiskWISE 5 – RBI code-compliant plant inspection and maintenance software
- CrackWISE – fatigue and fracture assessment software in accordance with BS 7910
- IntegriWISE – fitness for service software compliant with industry standards

TWI is actively exploiting 5 distinct inventions and has over 180 granted patents in its portfolio. TWI has granted over 250 licences world-wide. Income from licensing exceeded £2.3m in 2014. Inspired by a need for highly skilled next generation engineers, TWI is supporting PhD students with national and European Universities, who will deliver research that has academic merit and industrial and societal impact. Our students will be working with our experienced Project Leaders and their PhD Supervisors to broaden their research and technical knowledge and will have the opportunity to get exposure to transnational collaboration with other academics, SMEs and other organisations.

Relevant Publications:

1. Development of a comprehensive in-line quality control system for printed circuit board assemblies, P. Ian Nicholson et al. Circuit World (2007), Volume 33, Issue 4, pp. 31-42
2. Theories and industrial applications of optical interferometric NDT techniques: A Review. HX Shang and JX Gao, Insight, 51(5): 240-251, (2009)
3. Deformation analysis of MEMS structures using modified digital moiré methods. ZW Liu, XH Lou, JX Gao, Optics and Lasers in Engineering, 48(11), p1067-1075, (2010)
4. NDT capability of digital shearography for different materials. ZW Liu, JX Gao, et al, Optics and Lasers in Engineering. 49(12): 1462-1469, (2011)
5. Inspection of thin steel gauge welds for the shipping industry using laser guided inspection robot' G. Asfis, C. Carpentier, D. Panggabean, C. Nageswaran, V. Papadimitriou, I. Roditis, P. Chatzakos, A. Chong, J. Ferrando Chacon, Z. Qu, V. Kappatos ', ECNDT 2014, October 6-10, 2014, Prague, Czech Republic.

Relevant projects:

1. CITCOM – (H2020/2017, grant agreement no 768883) - A Complimentary Inspection Technique based on Computer Tomography and Plenoptic Camera for MEMS Components
2. ChipCheck– (FP7/2007-2013, grant agreement no 262212) – Development of novel X-ray system for fast automated detection of counterfeit plastic board components. TWI was responsible for the digital radiography development and hardware and software integration. www.chipcheck.eu
3. Plastronicsspec - (FP7/2007-2013, grant agreement no 286531) - Development of an automated digital radiography system for the inspection of plastic electronics. TWI was responsible for the digital radiography work package and systems integration.www.plastronicsspec.eu
4. AutoInspect– (FP7/2007-2013, grant agreement no 283288) - Automated inspection for sintered parts by non-destructive techniques for improved quality in production. TWI was responsible for the digital radiography work package and systems integration.
5. Microscan (FP6 Coop-ct-2003-508613) – Development of Comprehensive In-line Quality Control System for PCB Assemblies. TWI was responsible for the digital radiography work package and systems integration and automated defect detection software

Infrastructure/Equipment:

In 2015, TWI opened three new laboratories at its Cambridge headquarters, including state-of-the art facilities for the National Structural Integrity Research Centre, a dedicated postgraduate centre for industry-led engineering research (see images below) TWI also houses a professional institution, The Welding Institute, with a separate membership of 6000 individuals. TWI has offices in Europe, North and South America, India and Far East, and thus offers global connections to its customers.

TWI currently operates from 54,000 square metres (581,000 square feet) of manufacturing, testing and training space; five UK and 13 overseas facilities serve both its Industrial Membership and its training and examination needs. A successful international Training and Examinations programme sees 25,000 students trained each year in welding and inspection technologies.

TWI possess NDT equipment and knowledge across a wide range of techniques for Digital Radiography development, testing and validation and packaging. This includes, but is not limited to:

- Digital radiography laboratory,
- Computed tomography (CT),
- Radiography simulation software and work station,
- General purpose walk in radiography bay for development of prototype X-ray systems
- Automated/robotic inspection
- A wide range of digital X-ray detectors (linear and flat panel) and different X-ray sources
- Electronic assembly workshop
- 3D PLA printer
- A wide range of digital X-ray detectors (linear and flat panel) and different X-ray sources (material characterization, failure analysis, process optimization)
- Precise non-contact laser-ultrasonic system for testing, measuring, flaw detection and material characterisation (process monitoring, post-process evaluation, in-service inspection, internal flaws and surface defects, detection of foreign particles)
- Seven-axis ultrasonic immersion tank (inspection of critical components, in-situ inspection of large components)
- 3D laser scanning vibrometer (material fatigue and defects, acoustic optimization, dynamic testing and vibration analyses)
- Radiography simulation software and work station (computed radiography digital imaging)
- 3D non-contact profilometer (3D roughness parameters, physical surface characterization)
- Permeation testing facility (electronic encapsulation)
- X-Ray diffractometer (analyse and characterization of crystalline and semi-crystalline organic, inorganic and metallic materials)
- Non-destructive characterization methods: Dark and illuminated lock-in thermography (LIT) and respective data analysis and processing/post-processing software;
- Non-destructive testing and optical quality control technique: Laser shearography (flaws detection - voids, disbands, delaminations; quantitative measurement of strains, stresses, vibrations);

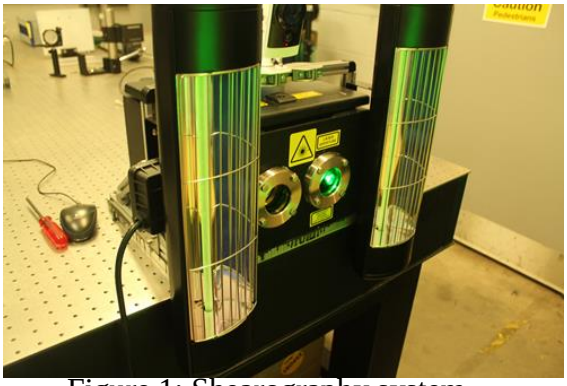


Figure 1: Shearography system

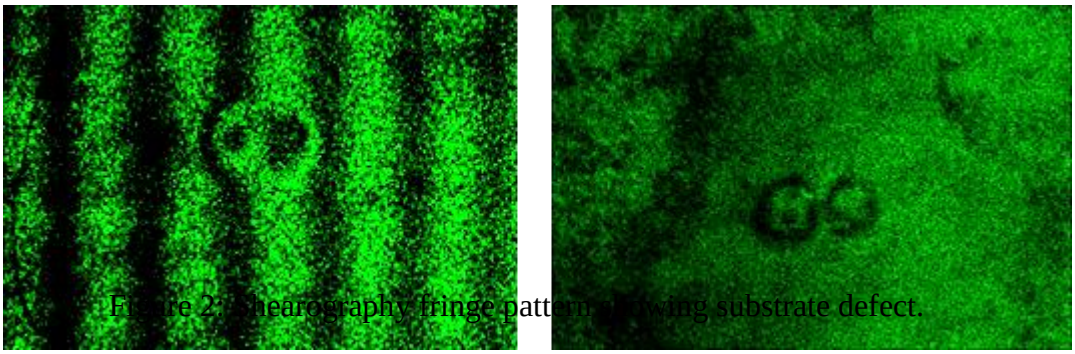


Figure 2: Shearography fringe pattern showing substrate defect.

Key personnel

Name: Giorgos Asfis Gender: Male

Background:

Giorgos Asfis is a principal project leader in TWI's Non-destructive-testing section. In this role, Giorgos has developed extensive experience integrating complex electronics with multiple data streams and control equipment, including for the development of robotics and drone projects focusing on industrial inspection activities. He is currently working on automating NDT methods, manufacturing of ultrasonic instrumentation, optical equipment design and wave propagation simulation. Giorgos has a Diploma in Mechanical Engineer from National Technical University Athens focusing on structural and machine dynamics, vibration analysis, predictive maintenance and modal analysis. He is currently carrying out research in the field of Non Destructive testing and condition monitoring. His work in public and private R&D institutes has provided significant experience in ultrasonics, guided waves, radiography, finite element modelling and high performance computing. He is also certified as a Level III inspector in Ultrasonic Testing, Phased Array Ultrasonic Testing and Radiography.

Education:

Jun. 2007 **Diploma in Mechanical Engineering** (MSc & BSc) School of Mechanical Engineering, National Technical University Athens (NTUA), Grade: 7,04/10 (Upper second class)

Dissertation Title: "Study and Construction of a PC - Accelerometer Interconnection for use in Vibration Analysis Based Maintenance"

Work Experience:

April 2014 – Present **TWI Ltd** (**The Welding Institute)**, Cambridge, NDT Principal Project Leader

Sep 2012 – April 2014 **TWI Ltd** (**The Welding Institute)**, Cambridge, NDT Senior Project Leader

Jul 2011 – Aug 2012 **Brunel University (BIC)**, Cambridge, Research Assistant

Nov 2009–Jun 2011 **Center for Research and Technology of Thessaly,** Research Assistant

Nov 2009–Jun 2011 **Innora Ltd,** Athens, Structural Dynamics Engineer - Project Leader

Nov 2008–Oct 2009 **MVS & Associates,** Rendsburg, Germany, Assistant Project Manager

Product Development Experience:

- Mechanical design, Electrical & electronics design and programming skills
- Designed new optical module prototypes for medical dermatoscope
- Developed a fully functional 8ch prototype phased array ultrasonics prototype
- Developed probe holder for various applications
-

Name: Dr Jianxin Gao Gender: Male

Background:

Dr Gao obtained his PhD at Tsinghua University, China in 1989 on Experimental Mechanics and have since led numerous research projects. He has supervised or co-supervised over 20 MSc and PhD students in universities in China and the UK. Prior to joining in TWI, Jianxin worked on biomechanics and biomedical engineering at Imperial College London (2001-2004) and the University of Surrey (2004-2005). Since joining TWI in 2005, he have been involved in a number of FP6, FP7 and Horizon2020 collaborative projects, acted as the coordinator of two FP7 projects which involved developing optical and image processing techniques. Jianxin is also the Scientist in Charge of four Marie Curie International Incoming Fellowship projects. Currently, he is the coordinator of a Horizon2020 project SocketMaster (2015-2018), and the scientific manager of another two H2020 project WInspector (2016-2018) and SheaRIOS (2018-2020). So far he has published over 70 technical papers in journals and at conferences.

Education:

PhD in Engineering Mechanics, Tsinghua University, China.
Msc in Engineering Mechanics, Tsinghua University, China.
BSc in Solid Mechanics, Zhejiang University, China.

Work Experience:

2005 – Present: Principal Project Leader, TWI Ltd, UK.
2004 – 2005: Research Fellow in Center for Biomedical Engineering, Surrey University, UK.
2004 – 2004: Professor, in Department of Biomedical Engineering, School of Life Science and Technology, Tongji University, Shanghai, China.
2001 - 2004: Research associate in Computing Department, Imperial College London.
1999 - 2000: Associate professor in Institute of Biomedical Engineering, Tongji University, Shanghai, China.

Name: Lu Zhao Gender: Female

Background:

Lu joined TWI in November 2009 as a Project Leader in NDT Validation Centre in Wales where she has been managing and technically delivering projects involving inspection via laser generated ultrasound as well as assisting colleagues in projects on digital radiography. After relocating to the NDT Technology Group in Cambridge in August 2011, she has been working on modelling, condition monitoring, and phased array inspection projects and is experienced in data and electronics integration. She obtained her certificate as Associate LabVIEW Developer in 2012. She performed the structural health monitoring field trial of a crane and designed the software for machine condition monitoring of wind turbine. She has also been working on phased array inspections projects in railways and wind turbines. They including site inspection, design inspection procedures, developing inspection system, and modelling. She has substantial experience with simulations software PZFlex and CIVA. She has been working on research project for thick austenitic weld inspection using phased array since 2013 and using AU techniques on WTB since 2014. Since 2011, she has been working with Dr Charles Schneider on various statistical analysis projects and has been to the external training course for statistics and experiment design.

Education:
BA(Hon)\MEng in Chemical Engineering from University of Cambridge
Charted Engineer and Member of The Welding Institute
Associate LabVIEW developer (2012-2014)
Acoustic Emission Level 2
CSWIP Level 2 in Manual ultrasonic inspection of plates
CSWIP Level 2 in phased array ultrasonic inspection of welds
CSWIP Level 2 in Radiographic Interpretation
Cambridge Overseas Trust Scholarship 2005 - 2009
Singapore Ministry of Education Scholar 2001 - 2004
Work Experience:
2009-now TWI Ltd Cambridge, NDT Project Leader

Name: Dorothee Panggabean **Gender: Female**
Background:
Dorothee is an engineer working within the NDT department of TWI where she has had experience of working in a variety of different industries including railway, wind energy and oil and gas. She is involved in projects where several NDT techniques are applied. However the work is not mainly technical; time/resource management is required and working in collaboration with many project partners is key for the success of the project. Dorothee has worked on various collaborative and single client projects, with some industrial experience from performing site inspection works.
Education:
MSc in Mechanical Engineering obtained in the University of Bordeaux. The degree has a specialization in NDT.
CSWIP Visual Weld Inspector
CSWIP Eddy Current (welds) Level 2
CSWIP ACFM Level 2
CSWIP Fitness for Service Level 2
Work Experience:
Previous to working in TWI, I was completing my degree in universities. In order to obtain the degree, several internships were carried out: an internship in the IMS laboratories focussed on the optimization of coatings in solar cells. Also an internship in the I2M in Bordeaux where the topics of thermography and ultrasonic were use jointly in order to characterise material properties.

Participant 7

Organisation Name: Centre Suisse D'Electronique Et De Microtechnique SA (CSEM)

Website: www.csem.ch

Organisation Description:

The CSEM is a privately held, R&D center established in 1984 in Neuchâtel, Switzerland. From the beginning, CSEM has been committed to both applied and order-related R&D. This bridge function between economics and science is still a core aspect of the company, and CSEM has continuously adapted its research areas to the needs of industry. In addition to its initial proximity to the watchmaking industry, CSEM has expanded its fields of activity and now offers a wide range of technologies for various markets.

The CSEM has currently more than 400 employees and is mainly active in the fields of micro/nanotechnology, microelectronics, systems engineering, and information and communication technologies carrying out applied research, development, prototyping and small series production. The CSEM has the competencies required to support its clients at each stage of development and provide the following services according to their needs:

- Feasibility studies
- Concept design
- Prototype or full product developments
- IP licensing and any combination of the above

The department "Center Muttenz" involved in this project is located in Muttenz (Basel-Landschaft canton) and provides expertise in the fields of photovoltaic, thin film optics, micro and nanostructures, printable electronics. During the past years, CSEM has developed their printing and device fabrication capability from lab-scale single cells made by spin-coating in an inert gas on small 2.5x2.5cm glass substrates, through doctor blade coating in the ambient environment on increasingly larger substrates, to complete modules printed on flexible substrates. Fully printed, semi-transparent, flexible, encapsulated modules were demonstrated at LOPE-C. Organic tandem solar cells with 10% efficiency were developed in project Sunflower, as well as a thin, robust, nanostructured films for enhanced light incoupling in thin film solar cells. Tandem module demonstrators were realized in collaboration with the end-user in the project. In other collaborative projects, CSEM has validated novel materials and substrates and integrated these in prototype technology demonstrators. Current research activities include the integration of organic and hybrid devices on flexible film, and the design and process development of free-form organic devices and printed optics.

Relevant Publications:

- G. Basset, T. Offermans, M. Chrapa, G. Nisato, Combinatorial Studies of Organic Solar Cells Based on Robotic High Throughput Fabrication and Characterization, October 2010 ISOS 3
- T. Offermans, J. Schleuniger, G. Nisato, Inkjet Printing of Polymer Solar Cells, June 2010 LOPE-C
- T. Offermans, M. Chrapa, J. Schleuniger, M. Zinggeler, W. H. Hughes, R. Ferrini, G. Nisato, Printing polymer tandem solar
- modules, June 2013 LOPEC 2013
- M. Pfannmöller, H. Heidari, L. Nanson, O. R. Lozman, M. Chrapa, T. Offermans, G. Nisato, S. Bals, Quantitative tomography of organic photovoltaic blends at the nanoscale, September 2015 Nano letters
- P. Chabrecek, M. Aubert, U. Sonderegger, O. Fernandez, M. Chrapa, G. Nisato and R. Ferrini, New Fabric-Based Electrodes for Low-Cost, Large-Area Flexible Organic Light-Emitting Diodes (OLED), 2013 LOPEC2013

Relevant Projects:

- FP7 European collaborative project SUNFLOWER (FP7-ICT-2011-7 number 287594), www.sunflower-fp7.eu
- INTERREG project Rh(e)in-Solar, www.rhinsolar.eu
- DURSOL, http://www.ccem.ch/dursol
- CTI project with SEFAR, https://www.csem.ch/Page.aspx?pid=43814
- EU project INGRESS, http://www.ingress-project.eu/

Infrastructure/Equipment:

- Clean room environment with gloveboxes
- Coating and printing platforms (S2S): slot-die, inkjet, gravure, screen, spin coating
- Simulation, origination, replication, and upscaling (step & repeat replication) of micro and nano structures
- S2S hot embossing, S2S nano-imprinting, S2S UV casting,S2S and R2R hot embossing

- Optical characterization (angular and wavelength resolved reflection/transmission, photoluminescence, phosphorescence lifetime, UVVIS, optical coherence tomography, infrared imaging)
- AFM, SEM, profilometer
- Photonic sintering (S2S, R2R compatible)
- Accelerated lifetime testing (damp/heat, light soaking, mechanical stress tests)

Fig 1: Mask Aligner MA6 –Karl Suss

Fig 2: Essemtec Exhibit SP900 In-Line Screen

Fig 3: MICRODROP single nozzle inkjet

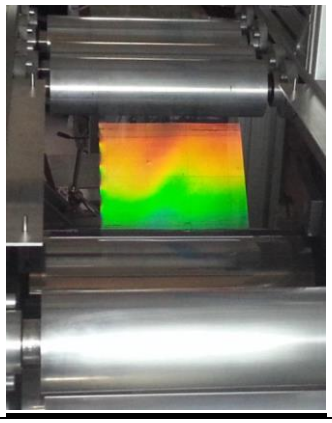


Fig 4: Hot embossing R2R pilot line

Fig 5: Hot embossing S2S pilot line

Fig 6: Gloveboxes dedicated to OPV, OLED, and OTFT

Key Personnel

Name: Ton Offermans **Gender: Male**

Background:

Ton Offermans is a senior R&D engineer in the Polymer Optoelectronics Section of CSEM. He studied applied physics at the University of Technology Eindhoven and received his PhD degree in the field of organic photovoltaics. Before joining CSEM he has worked as researcher on light emitting tile at the Materia
Nova research center. At CSEM he is leading the research on OPV. Current research activities include light management solutions for incoupling, and the upscaling of the fabrication of nano and micro structures on large area films.

Education:

PhD, physics. University of Technology Eindhoven

Work Experience:

R&D Center, Materia Nova, Belgium, 2005-2007
R&D Center, CSEM, Switzerland, 2007-2018.

Product Development Experience:

Development of solar cells/modules, starting at TRL3, arriving at TRL6 within the Sunflower project. Near yearly fabrication of technology demonstrators (LOPEC)

Intellectual Property Management:

WO/2015/022424

Innovation Management:

- Various dissemination events at conferences such as LOPEC, Swiss e-print
- FSRM course tutor Organic Photovoltaics

Name: Oscar Fernandez **Gender: Male**

Background:

Oscar Fernandez received his MSc and PhD degrees in Physics from the University of Valladolid (Spain) in 2000, and the University of North Wales, Bangor (UK) in 2004, respectively. After finishing his PhD he started his postdoctoral research in the Molecular Electronics Group in Bangor where he investigated the interfacial origins of threshold voltage instabilities in organic thin-film transistors (OTFT). In 2006 he joined
Cambridge Display Technology (CDT) where he led the device physics activities in the lighting program. In 2011 he joined Polymer Vision BV in Eindhoven (Netherlands) where he worked on the development of OLED processing and characterization for oxide TFT-based rollable OLED displays. Since May 2012 Oscar is part of the MicroNano Integrated Optics and Photonics group at CSEM in Muttenz (Switzerland).

Education:

PhD, Physics. University of Valladolid (Spain)

Work Experience:

CDT, UK
Polymer Vision, Eindhoven, Netherlands
R&D Center, CSEM, Switzerland

Product Development Experience:

Prototyping of printed OLEDs, light management for (O)LEDs in industrial and publicly funded projects LASSIE, DELIGHT

Innovation Management:

- Various dissemination events at conferences
- FSRM course tutor Organic Light-Emitting Diodes

Name: Sören Fricke **Gender: Male**

Background:

Sören Fricke received his Diploma degree in Physics from the Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland in 2004. He joined Sonys Fusion Domain Laboratory in Tokyo, Japan, where he prepared his Diploma Thesis. In 2005, he continued his research on micro-sensors at the Corporate Research Centre of EADS in Ottobrunn, Germany and received his PhD degree from Saarland University, Germany in 2010. In 2008 he joined Robert Bosch GmbH and worked in the field of semiconductor and MEMS packaging in the business unit Automotive Electronics. From 2012 to 2013, he was team manager of an international engineering team responsible for the volume production of automotive pressure sensors. Since November 2013, he is working at CSEM Center Muttenz, where he is heading the section on Large Area & Flexible Systems since 2015.

Education:

PhD, Physics. Swiss Federal Institute of Technology (ETHZ)

Work Experience:

2013 – Present: CSEM Center Muttenz
2008-2013: Robert Bosch GmbH, Germany
2005-2008: EADS Innovation Works, Germany

Product Development Experience:

Harsh environment pressure sensors starting from TRL2 to TRL4. Automotive ASIC and Sensor Packages TRL8 to TRL9. Immersed gratings for space-borne spectrometer TRL2 to TRL3.

Intellectual Property Management:

WO 2008/003374
WO 2007/101426
DE 102005047535

Innovation Management:

Various dissemination events at conferences, Member of program committee and session chair of the SPIE conference on Smart Sensors, Actuators and MEMS.

Participant 8

Organisation Name: Solibro Hi-Tech GmbH (Solibro)

Website: http://solibro-hitech.com/en/home/

Organisation Description:

The Solibro Hi-Tech GmbH located in Solar Valley is a pioneer in the field of CIGS thin-film technology. Its focus is on the continuous improvement of all products and production systems, as well as the expansion of the existing production capacities in the context of international turnkey projects. In cooperation with the Solibro Research AB in Sweden and the Solibro GmbH at the same location in Germany, joint work is carried out aimed at the advancement of our products and processes, in order to always be able to offer our customers state-of-the-art modules and production facilities.

The improvement of inline measurement and control capabilities elaborated in the project will contribute to more efficient production systems (higher production yield) and better products (increased conversion efficiency).

Relevant Publications:

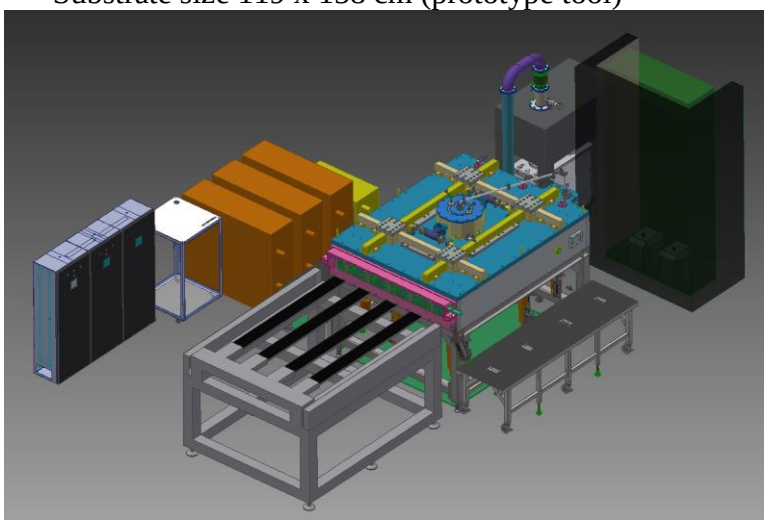
- Solibro Hi-Tech press release, February 2018: "Solibro Hi-Tech sets new world record of 18.72 % for CIGS PV modules" (http://solibro-hitech.com/en/news/news/)
- P. Kratzert: "17% Total Area Efficiency at Commercial Size CIGS Module", Plenary Talk at the 33rd EU PV Solar Energy Conference, Amsterdam, September 25th 2017
- D. Herrmann, P. Kratzert, S. Weeke, M. Zimmer, J. Djordjevic-Reiss, R. Hunger, P. Lindberg, E. Wallin, O. Lundberg, L. Stolt, "CIGS Module Manufacturing with High Deposition Rates and Efficiencies", Proc. 40th IEEE PV Specialists' Conference (2014).
- P. Kratzert, S. Jander, P. Leyendecker, H. Schenk, V. Naumann, K. Kaufmann, C. Hagendorf, J. Djordjevic-Reiss, S. Weeke, E. Wallin, O. Lundberg, D. Herrmann, L. Stolt, Improved Performance by Mo-CIGS Interface Control, 29th EU PV Solar Energy Conference (2014)
- R. Hunger, U. Malm, J. Djordjevic-Reiss, T. Jarmar, E. Wallin, O. Lundberg, S. Jander, M. Zimmer, S. Weeke, P. Kratzert, D. Herrmann, L. Stolt, "Homogeneity of Ga Grading in CIGS Solar Modules", 27th EU PV Solar Energy Conference (2012).

Relevant Projects:

- EFFCIS: German BMWi-funded collaborative project, focussing on efficiency improvements
- Pliant: Process Line Implementation for Applied Surface Nanotechnologies. Project ID: 309530 funded under: FP7-NMP
- LL-CIGS : German collaborative project in the cluster "Sollarvalley Mitteldeutschland" funded by the country Saxony-Anhalt focussing on lifetime & efficiency improvement (2009-2012)

Infrastructure/Equipment:

- Existing inline (intermediate) product characterization tools: CIGS thickness + composition measurement
- PV module flasher, electrical I-V characterization + PV module light soak stabilisation chamber
- Low Pressure Chemical Vapor Deposition (LPCVD) Tool for Industrial Size PV module manufacturing,
  Substrate size 119 x 158 cm (prototype tool)



- Industrial CIGS coating line for PV module manufacturing

<u>Key Personnel</u>

Name: Dr. Philipp Kratzert **Gender: Male**

Background:

Philipp Kratzert received his doctoral degree in physics at the Humboldt University Berlin in 2002, focusing on the MBE of II-VI quantum dots. After a period of basic research he changed into semiconductor industry and worked several years as a manager in the development and production ramp of different NAND & DRAM memory technology nodes. In 2009 Mr. Kratzert changed into the photovoltaics and took over the responsibility for the efficiency optimization at the Solibro manufacturing site. Since then he is engaged as a driving motor for the efficiency roadmap and Technology development at the Solibro manufacturing site in Thalheim Germany. Currently he is supporting Solibro as a Vice President for Technology and Process Development.

Education:

Diploma in physics from Technische Universität Berlin
PhD in physics from Humboldt University of Berlin
Lean six sigma Black belt

Work Experience:

2014 – present: Vice President Solibro
2011 – 2014: Director Solibro
2009 – 2011: Team leader Solibro
2005 – 2009: Lead engineer semiconductor Quimonda.
2004 – 2005: Project leader semiconductor Infineon Technologies
2002 – 2004: Device Engineer semiconductor, Infineon Technologies
1998 –2002: Research Scientist, Humboldt University Berlin.

Product Development Experience:

- Lead R&D department (4 Teams, ca. 50 engineers)
- Plan and execute all R&D projects within the R&D department
- Organize and lead the cross functional supports on the efficiency, yield, product performance & reliability improvements

Innovation Management:

- Integration and successful transfer the R&D technology, process, equipment, CIGS sources to domestic overseas Production sites

- Definition the Turn Key Line efficiency roadmap, lead the development of the technology, process, and CIGS tool to meet the annual Turn Key Line sales business targets

Name: Dr. Ralf Hunger **Gender: Male**

Background:

Dr. Ralf Hunger had studied Physics at the Berlin University of Technology, where he had also acquired his PhD. For 12 years, he had conducted research on chalcopyrite thin films at the Helmholtz-Zentrum-Berlin (Germany), the Advanced Institutes of Industrial Science and Technology Tsukuba (Japan), and the Darmstadt University of Technology (Germany). Thin film growth and characterization, deposition and vacuum technology, semiconductor surfaces and interfaces are his fields of expertise. 10 years ago, Dr. Hunger had joined the Solibro GmbH and had transferred the Cu(InGa)Se2 deposition processes to industrial scale, in the beginning as senior process engineer CIGS, later as team leader CIGS development, now as team leader process integration.

Education:

Diploma in physics from Technische Universität Berlin
PhD in physics from Technische Universität Berlin

Work Experience:

2014 - present Team Leader Process Integration, Solibro Hitech GmbH
2007 - 2014 Senior Process Engineer, Solibro GmbH, Bitterfeld/Wolfen, Germany.
2001 - 2007 Darmstadt University of Technology, Department of Materials Science, Surface Science Division, Germany.
1999 - 2001 Advanced Institutes of Industrial Science and Technology, Tsukuba, Japan
1995-1999 Helmholtz-Zentrum Berlin (formerly: Hahn-Meitner-Insitute), Germany

Product Development Experience:

Project management of development projects, e.g. prototyping of advanced, alternative PV module encapsulation concepts.

Innovation Management:

Ralf Hunger et al., Method for producing sub-solar modules using electrically insulating insulation trenches in a thin-layer solar module, and method for producing a thin-layer solar module with such insulation trenches, WO2015027996A1, applied 2015-03-05

MEYER BURGER	**Participant 9** **Organisation Name: Meyer Burger (Netherlands) B.V. (MBNL)** **Website: https://www.meyerburger.com/nl/en/**

Organisation Description:

Meyer Burger (Netherlands) B.V. (MBNL) is a member of the Swiss Meyer Burger Group and is located in Eindhoven. The main activities of MBNL are the development, manufacture and sale of industrial production equipment for applications in various markets such as photovoltaics, organic electronics and batteries. The current equipment portfolio consists of industrial inkjet printers and thin film deposition equipment (plasma deposition and atmospheric atomic layer deposition). Meyer Burger's systems and production equipment provide sustainable added value to our customers in the photovoltaic (solar), semiconductor and optoelectronic industries as well as other selected high-end markets based on semiconductor materials.

The strategy of MBNL is to develop and commercialize state of the art industrial equipment that can serve a wide variety of applications. In particular, within this project, the aim of MBNL is to

develop and test plasma enhanced (PE) spatial atomic layer deposition (sALD) compatible with an R2R system.

Our plan is to use the know-how developed within this project to offer added value to existing systems and as a building stone for next generation systems (R2R). A successful development will help market penetration of high quality flexible devices (OLED, OPV, CIGS) and will generate sales of R2R sALD systems and business expansion for MBNL and help create additional employment opportunities in the region.

Additionally, there are many other applications and markets that can as well benefit from this technology development to improve product quality and increase throughput. This will also contribute to increased revenue and can create additional expansion opportunities.

Relevant Publications:

1. FLEx ALD R2R and S2S. https://www.meyerburger.com/nl/en/technologies/specialized-technologies/spatial-atomic-layer-deposition/
2. TFE CONx (in particular JETx). https://www.meyerburger.com/nl/en/technologies/specialized-technologies/product-detail/product/conx-tfe/
3. Presentation for ALD (example). http://www.semi.org/en/dana-borsa-flexible-functional-devices-mass-production-level-flex-r2r-sald-platform
4. Presentation for barriers (example - Industrial scale manufacturing of transparent and flexible moisture barriers). https://www.fep.fraunhofer.de/en/events/rueckblick_2017/sidme17.html#tabpanel-2

Relevant Projects:

1. OPZuid. https://www.stimulus.nl/op-zuid/project/industriele-productie-dunne-film-barriers/
2. Enso. http://enso-ecsel.eu/
3. Collaboration with Holst: Development of a roll-to-roll PECVD deposition tool to be used for flexible OLED lighting and organic solar cells. https://www.oled-info.com/roth-rau-joins-holst-centre-research-barriers-flexible-electronics

Infrastructure/Equipment:

- Spectral ellipsometry
- Profilometer
- Low resolution SEM (>1µm)
- Climate chamber / ovens

Figure 1: Spectral

Figure 2:

Figure 4: Climate chamber /

Key Personnel

Name: Luca Gautero Gender: Male

Background:

Dr. Luca Gautero is presently a senior process engineer at Meyer Burger (Netherlands) B.V. During his career he has focussed on micro and nano-technology in fields as semiconductors, photovoltaic, functional thin films and recently (moisture) encapsulation. His innovation maker approach, both in academics and professional domain, resulted in several patents, publications and conference presentations. His European academic path features university titles from Italian, French and Swiss institutions, a PhD degree in renewable energy in 2010 developed in a German research centre and later, in 2015, an Executive Master in Energy Management from a leading business school in UK.

Education:

MSc Nanotech – Institut polytechnique de Grenoble
PhD micro and nanotechnology – École polytechnique fédérale de Lausanne
MSc in Energy Management. - ESCP Europe London Campus

Work Experience:

2010 – present : Meyer Burger (Netherlands) B.V. – senior process development engineer
2007 – 2010: Fraunhofer ISE GmBH - Ph.D. student.

Product Development Experience:

- Coordination in the development of solutions for PV, OPVs and OLEDs
- Definition and translation of requirements from market level to equipment level.
- Engineering and development of functionality in integrated production equipment.
- Optimization of solar power management, also as consultancy to customers.
- Active contribution to sales team communication to customers.
- Product management enabler (accomplished 1st sale of new production equipment – 2014)
- Engaged in the processing of moisture barriers (WVTR below 10E-6 g/(m2*day))
- Manage customer expectation with regular reporting on present and planned activities. Intervention on supply chain to guarantee prompt and adequate performance of components. Definition, execution and reporting of acceptance tests in collaboration with the customer and a board of scientific experts (Holst Centre).

Name: Dana Borsa Gender: Female

Background:

Dr. Dana Borsa is Manager Research of the Thin Film Coating and Projects Business Unit at Meyer Burger (Netherlands) B.V. She is responsible for all process and application developments. Her PhD in physics from the Rijks Universiteit Groningen was focused on the growth and characterization of nitride-based thin films and multilayers. After some years of research on switchable mirrors and hydrogen storage materials at the Vrije Universiteit Amsterdam she moved

to ECN and worked on high efficiency solar cell concepts before joining OTB Solar (currently Meyer Burger (Netherlands) B.V.) in 2009.

Education:

PhD – Centro de Microanalisis de Materiales, Universidad Autonoma de Madrid, Spain & Nuclear Solid State Physics Group, University of Groningen, NL

Work Experience:

2012 – present: Meyer Burger (Netherlands) B.V., Eindhoven, NL – Manager Research
2009 – 2012: Meyer Burger (Netherlands) B.V., Eindhoven, NL – Project Leader
2007 – 2009: Energy Research Centre Nederland, Petten NL – Researcher Silicon PV
2004 – 2007:Vrije Universiteit Amsterdam, Amsterdam, NL – Researcher

Product Development Experience:

- Project manager customer project for the fabrication of a prototype system for battery applications; responsible for all phases
- Project leader for INREP H2020 project for the design, fabrication and commissioning of an injector head for sALD S2S tool
- Project leader 'SiOx for market implementation': budget and resource planning; technology scouting, process development and platform upgrade for 24/7 production
- Coordination CIP-SiNx program: definition and continuous topic update, resource planning (FTEs, tool time and consumables); progress monitoring and supervision of process development (>40 WP); field implementation and customer support
- Coordination customer sampling program: communication with customers and local sales team; define test program; realization and after care
- Sales support at HQ and on-site; regular customer visits and process and product marketing.

Participant 10

Organisation Name: Brunel University London (BUL)

Website: http://www.brunel.ac.uk/

Organisation Description:

Brunel University has been highly successful in creating an industrial innovation research resources, named Brunel Innovation Centre (BIC), which sits between the knowledge base and industry, supporting partners in industry to transfer academic research into industrial application. Brunel has a wide range of knowledge and expertise in the following research areas: (i) Power ultrasonics; detection, cleaning of fouling and fouling prevention techniques non-invasively software, anti- or de-icing application (ii) Smart Non Destructive Testing (NDT); autonomous devices, miniaturised multifunction systems for NDE in bespoke environments; marine, high temperature, aerospace, pattern recognition hardware / software, (iii) Active Structural Health Monitoring (SHM) and Condition Monitoring (iv) Design of associated hardware and software (v) Signal and image processing and numerical modelling - Noise reduction, signal condition and control (including active vibration control), data interpretation and analysis, (vi) Automation and robotics of inspection system, (vii) Improved or new and robust sensors (Piezoelectric, Electromagnetic) e.g. high temperature inspection, transduction (viii) Electronics - Low power consumption for multiple channels, wireless, high speed, wideband, fast bps, energy harvesting, (ix) Structure integrity assessment of offshore facilities, (x) Fatigue & fracture analysis. BIC have coordinated as well as delivered over 60 projects since 2011 across both the UK and EC platforms. In year 2015, BIC has received the Highly Commended Innovation Awards from The Institution of Engineering and Technology (IET) for two award categories: (i) Measurement in Action and (ii) Manufacturing Technology. The collaborative European project (FP7) was put forward to the competition and was chosen from over 300 entries from 28 countries (globally).

Relevant Publications:

- J.T. Sun, I. Szabo, M. Livadas, V. Kappatos, C. Selcuk and T-H. Gan, “Automatic defect detection of powder metallurgy (PM) parts using three dimensional X-ray computed tomography imaging”, Proc. Euro PM2015 Congress & Exhibition, 4–7 October, Reims, France
- I. Szabo, J. Sun, V. Kappatos, C. Selcuk, A New Automated in Line Quality Control System based on Non-Destructive Evaluation for Additive Manufacturing of Net-Shape Parts from Particulates, Additive Manufactured Metallic Materials Properties & Structures Conference, IOM3 Institute, Coventry, UK, 14th May 2015
- J. Sun, A.Y.B. Chong, O. Gilmour, G. Foulds, M. Thorton, V. Kappatos, C. Selcuk, T.H.Gan, “Automated defect characterization on composite materials using enhanced image processing technique for ultrasonic test”, BINDT Aerospace Event 2016, Bristol, United Kingdom
- I. Szabo, G. Logeswaran, V. Kappatos, C. Selcuk and T.H., Gan, X-ray enclosure design and prototype for non-destructive testing of net shape parts from particulate materials, Euro PM2015 Congress & Exhibition, 4 – 7 October 2015, Reims, France

Relevant Projects:

- CITCOM: A Complimentary Inspection Technique based on Computer Tomography and Plenoptic Camera for MEMS Components funded by H2020. Project ID: 768883.

- AutoDISC: Automated Ultrasonic Inspection of Aerospace Composites with Enhanced Defect Detection Probabilities Aided by Gantry Deployed, CAD Controlled Robotics, funded by Innovate UK. Project website: www.autodisc-project.com/
- Qualinet: Automated In- Line Inspection And Quality Control Of Net Shape Powder Metallurgy Components Using Microfocus Three Dimensional X Ray Computed Tomography Imaging, funded by Innovate UK. Project website: www.qualinet-project.com/
- AutoInspect: Automated inspection for sintered parts by non-destructive techniques for improved quality in production (AutoInspect) funded by FP7. Project website: www.autoinspectproject.eu/project
- AssureNet: Automated 100% Production Quality Assurance Of Net Shape Manufactured Components Using In Line Micron Resolution X-Ray Stereographic Imaging (Assurenet),Funded by Birmingham Council: Advance Manufacturing Supply Chain Initiative (AMSCI) www.assurenetproject.co.uk/

Infrastructure/Equipment:

Brunel's vast equipment and test areas include, but are not limited to:

- Several powerful work-stations including signal and image processing software (Matlab, Labview, C++, C#), finite element (Ansys, Abaqus, COMSOL), computing power for algorithm derivations and ability to set up guided user interfaces.
- Power amplifier, oscilloscope and impedance analyser
- Optical tracking device and high resolution cameras
- State of the art 3D Laser scanning vibrometer for: vibration measurement (wave propagation and transducer analysis); condition monitoring purposes as well as for sensor, materials, behaviour & characterisations
- NDT and Structural Health Monitoring hardware i.e. Acoustic Emission, Fibre Bragg Grating(temp, strain, vibrations), strain gauge equipment
- PCB design and manufacturing facilities.
- Experimental industry specific set-up rigs for detection and cleaning of fouling, rotating machinery; mock-ups; inspection, monitoring systems demonstrated in full scale.

Key personnel

Name: Jamil Kanfoud Gender: Male

Jamil Kanfoud (MEng, MPhil) is the Technical Manager at BIC. Jamil has worked on the development of novel passive noise cancelling solution for the special industry. He has had a 3-year experience in the academic world as a faculty member including 2 years heading the electromechanical department at an engineering college. As head of a department, he managed 40 permanent faculty staff and 850 students. Under his leadership, the electromechanical and civil engineering departments developed collaboration with the industry. He also worked on the implementation of innovative pedagogies (problem based learning and project based learning). Jamil joined TWI Ltd in 2014 where he worked mainly on collaborative projects delivering technical work (numerical modelling, analytical modelling, sensor selection, data processing, prototype design and assembly) and managing the collaborative projects. He continued working in the aerospace field working on the de-icing of aerodynamic surfaces using transient vibrations and electro-thermal heaters. He also worked on passive and active control solutions using spring-dampers, piezoelectric actuators and PID Control. Most of his efforts were dedicated to the development of condition and structural health monitoring technologies for pipes and wind turbines machinery and structures. During his twenty-month experience, Jamil developed experience in collaborative projects (H2020, FP7 and InnovateUK). He also led technically 12 proposals, among which he secured an H2020 thematic and 2 H2020 FTI.

Jamil joined Brunel Innovation Centre in 2016 as technical manager of the innovation centre. His responsibilities include looking after the successful delivery of the portfolio of projects BIC is involved in as well as the development of research strategy for the innovation centre and the identification of funding opportunities for business development.

Name: Wenbo Duan **Gender: Male**

Wenbo (BSc (Distinguished), MSc, PhD, CEng, MIMechE) has been working on the mathematical modelling of sound wave propagation in pipeline systems with the aim of detecting blockages or defects since 2006. He was responsible for developing analytic models to study coupling between acoustics and structures when he was a PhD student in the University of Manchester. After that, he moved to Brunel University, and developed new stand-alone finite element software to implement fast and accurate mathematical models. A hybrid finite element method was proposed which combines modal expansion solutions with the conventional finite element solutions. He is currently working on the development of advanced numerical models to investigate wave propagation and scattering in complicated conditions, such as coated, buried and/or immersed pipes etc. He also has experience in product development and innovation management. He has published 10 referred journal papers in top ranking journals, including the Journal of the Acoustical Society of America, Journal of Sound and Vibration etc. He received the 'Outstanding Reviewer' award from Journal of Sound and Vibration in 2016, and he is a reviewer for a number of acoustic related top journals.

Name: Istvan Szabo **Gender: Male**

Istvan (BSc, BEng, MSc, CEng, MWeldI) has over 10 years of mechanical and design engineering experience across the Defense, Renewables, Building Services and R&D Industry. Prior to his current position he led and been involved in thermal energy recycling, energy efficient heating system and industrial air conditioning system design projects.Through his career he obtained in depth knowledge of materials and manufacturing processes, Powder Metallurgy, 3D printing and rapid prototyping in particular, and also significant experience in product design from concept to final product including designing for manufacturing He is an expert user of multiple CAD packages and has in depth knowledge of Finite Element Analysis, Computed Fluid Dynamics and Thermal analysis methods. He is an expert user of multiple CAD packages including SolidWorks, Catia V5, Autodesk Inventor and Revit, Pro E. He is Research Assistant at BIC, leader of two X-ray image processing device enclosure design projects, actively involved in activities varying from research and development in the areas of novel technique development, sensor applications.

Participant 11

Organisation Name: Acondicionamiento Tarrasense Associacion (LEITAT)

Website: projects.leitat.org

Organisation Description:

LEITAT Technological Center, founded in 1906, is a private non-profit RTO with more than 110 years of deep knowledge and valuable experience in industrial innovation. We transform technological solutions and scientific results into economic and competitive values for our clients and collaborating entities. LEITAT is strongly committed to over 1500 customers that benefit from our creative and innovative solutions. LEITAT provides added value to customers by delivering technological innovations through applied research and development activities in areas including advanced materials, advanced manufacturing, renewable energies, storage system, circular economy, sustainability and biotechnology. LEITAT rely upon over 240 highly skilled team members who deliver flexible and innovative solutions in the face of any industrial challenge.

LEITAT has a strong presence and valuable experiences in carrying out research under different international and national funding schemes including FP7 and Horizon 2020. Participation in 43 FP7 projects and 55 H2020 projects are proof of LEITAT capabilities in coordinating, managing and executing of international projects. LEITAT is recognized by the Spanish Government as a CIT (Centre of Technological Innovation) and is one the 5 Technology Centers that is accredited by the Regional Government of Catalonia. Finally, it is member of FEDIT (Federation of Technological Centers in Spain) and the IT Network of the Catalan Regional Government.

The Circular Economy Business Unit offers innovative technologies and strategies for the efficient management of natural resources, a sustainable production, and the optimal treatment and valorization of residual streams. Main research areas are:

- Waste management and valorization: biomass, industrial & municipal wastes/byproducts.
- Water treatment & reuse: industrial & municipal waters.
- Air treatment & control: indoor & outdoor environments.
- Biotechnologies: Greener products & processes.
- Sustainability: Life Cycle Assessment, communication & responsibility, sustainable production.

Relevant Publications:

1. Torrentellé, M., Energy Efficiency, a Sustainable strategy for the improvement of competitivenesss. Revista de Química e Industria Textil 2012 (Article)
2. C. Hidalgo. Environmental and Economic assessment of innovative medical devices with a Life Cycle Approach. (Restoration project). SETAC LCA Case Study Symposium. Nova Sabi (Serbia) November 2014 (Oral Presentation).
3. G. Ferrer, C. Hidalgo, M. Escamilla. Life Cycle Assessment as a methodology to develop an easy tool, Bookdaper, to improve the environmental performance of printed products produced by SMEs. SETAC Europe 19th LCA Case Study Symposium.Rome (Italy) 11-13 November 2013 (Poster).
4. M. Della Pirriera. Stability of Dye Sensitized Solar Cells for indoor applications: A comparison with Amorphous Silicon under Low Irradiation, European PV Conference and Exhibition, 3 October 2013, Paris, France (Oral presentation)
5. M. Della Pirriera New PV technologies: What about their recyclability?, Workshop on Ecoinnovative challengue for PV sector, 29 September 2016, Terrassa, Spain (Oral presentation)

Relevant Projects:

1. CITCOM: A Complementary Inspection Technique based on Computer Tomography and Plenoptic Camera for MEMS Components H2020/768883, 2017-2020.
2. MatHero - Developing Environmentally Compatible Organic Solar Cells, FP7/604603, 2014-2016.
3. REMANENCE - Rare Earth Magnet Recovery for Environmental and Resource Protection FP7/310240, 2013-2016.
4. EPHOCELL: New materials for thin film solar cells, development, integration and testing, FP7/227127, 2009-2013.
5. ERG. New materials for hybrid thin film solar cells, development and testing. JTI-ENIAC/270722-2 , 2011-2014.
6. URWASTECH: Innovative Urban Waste Treatment: A Sustainable Approach Using Hydrodynamic and Biological Technologies, LIFE+/LIFE10ENV/ES/530, 2011-2014

Infrastructure/Equipment:

LEITAT holds licenses of the two most recognised LCA software:

- SIMAPRO (Pré Consultants), professional LCA software tool to collect, analyze and monitor the sustainability performance of products and services.
- GaBi Professional (PE International), combines modelling and reporting software with data content and expertise.

LEITAT holds license of the main LCA Databases such as Ecoinvent and Gabi Professional databases, and the rest of commercial Databases included in the softwares. LEITAT works with different impact methods according to European guidelines: ILCD method, CML, ReCiPe, among others. LEITAT has chemical and physical laboratories, testing and development equipment in the industrial workshop.

LEITAT has dedicated equipment for standard and innovative tests:

- Test following current standards. More critical test are available, using a portfolio of climatic chambers.
- Sun simulator AM1.5G (2x2 cm2) and UV and metal halide light sources, full monitoring during tests.
- UV-Vis-NIR spectrometer
- Pyranometer and calibrated c-Si solar cells
- Battery of different ageing chambers (metal halide lamp- climatic, UV-climatic, Xenotest, saline fog, etc.) up to 2.2 x 3.4 x 2.3 m (width x length x height) and a flat empty space of 3x2 m to host big prototypes
- AutoLAB and multichannel VMP3 potentiostat / galvanostat
- Set of optical components: lenses, beam splitters, filters, spectrometer of wide range, white laser from 400 to 1200 nm, thermo graphic camera, etc.
- Outdoor platforms, one with suntracker and others static, for natural aging with weather station sensors, full monitoring during tests.
- Physic-chemical characterization: IC, TGA, DSC, ATR-FTIR, BET, GC, ICP-MS, SEM...
- Mechanical tests equipment (bending, stretching, dynamometer, etc.)

Key personnel

Name: Marta Escamilla Monell Gender: Female

Marta Escamilla is a graduate of Environmental Science from University of Girona. Has a widespread experience in environmental management, especially on the implementation of Environmental Management Systems in SMEs and large companies, audits, life cycle assessment and ecodesign, Ecolabel, environmental awareness and training workshops among others. As the Sustainability Division Manager of LEITAT Technological Center, is in charge of coordinating projects and managing initiatives, at regional, national and European level, promoting sustainable development, making a special effort on energy efficient, rational natural resources consumption, waste management, to preserve and improve the environmental performance as well as contributing to substantial economic savings of the industrial sector.

Name: Sergio Martínez Lozano Gender: Male

Sergio Martínez hold a PhD in Environmental Science (University of Girona) and his career has been mainly focused on the evaluation of biomass as a renewable resource, covering aspects like biomass quantification, studying biomass conversion technologies, the economic feasibility of power generation using biomass, Life Cycle Analysis, physico-chemical and thermal analysis of biomass and the evaluation of atmospheric emissions. He has directly and indirectly participated in several national and international R+D projects. He has also participated in numerous conferences and he published articles in national and international scientific journals. His scientific experience during his PhD has been combined with teaching experience in the University of Girona, where he punctually participated in some lectures in Environmental Science Degree. After finishing his studies, he worked as plant manager in a biomass composting plant (treating organic fraction of urban solid wastes and sewage sludge). He works as a researcher in R+D+I projects for the development of new technologies for solid waste and wastewater treatment and valorisation using

advanced technologies such as anaerobic digestion, pyrolysis and pre-treatment technologies. He enjoyed a post-doctoral fellowship (Torres Quevedo, 2011-2013), and today, he works as a Senior Researcher in LEITAT in the preparation, negotiation, execution, dissemination and management of European, national and industrial projects. He is responsible of the Energy Valorisation and Waste recycling Unit in LEITAT.

Name: Monica Della Pirriera **Gender: Female**

Monica Della has a PhD in Electronic Engineering from the Polytechnic University of Catalonia and a degree in Physics from the University of Buenos Aires. Her research career has focused on the study of organic semiconductors for printed electronics mainly organic photovoltaics. She has participated in various conferences and has publications in international journals in the field of materials science for energy. Also noteworthy is her teaching experience as an assistant professor in the Department of Physics, Faculty of Natural Sciences at the University of Buenos Aires. She is developing their professional career at LEITAT Technological Center involved in R + D + I projects since 2010, specializing in the development of new technologies in the field of Photovoltaic. Currently she leads the team of Renewable Energies, and participates in European and Spanish platforms such as EU PV cluster, EMIRI, Photonics 21, Fotoplat, among others.

Name: Julia García-Montaño **Gender: Female**

Julia García-Montaño holds a PhD in Chemistry (Universitat Autònoma de Barcelona) and is the Group Leader of the Environmental & BIO Technologies Division in LEITAT. Her professional career has been mainly centred in the study of alternative water and waste treatment & valorisation technologies: Advanced Oxidation Processes (AOPs), advanced electrochemical treatments, biotechnologies, nanotechnologies, membrane processes, high added value products recovery, water/waste-Energy nexus, among others. She has developed two research stays at the Plataforma Solar de Almería, has developed and coordinated several innovative national and international projects, has participated in various congresses and has numerous scientific publications in the water/wastes decontamination area. It is also noteworthy her educational experience as she combined her scientific career with her career as an assistant teacher in several Universities.

Participant 12

Organisation Name: Coatema Coating Machinery GmbH (Coatema)

Website: www.coatema.de

Organisation Description:

COATEMA stands for high quality coating, laminating and printing plants and has been successful in the field for more than 40 years. It offers equipment solution for laboratory R&D, for pilot plants and custom made production equipment. The company is a leading supplier of variable multifunction coating plants with many economic advantages (e.g. the modular Click&Coat® system), do their own R&D in important application fields, hold patents for pioneering solutions, and have been awarded with innovation prizes.

Coatema has Europe's largest lab for R&D in coating technology and is able to offer Know-how, full service solutions and to control whole production processes. The unique combination of know-how and facilities makes them first choice to successfully work on the project tasks with respect to inline coating and pilot line production. Within the R&D Centre the demonstration and realisation of production processes are possible using existing pilot lines.

Relevant Projects:

- PI-SCALE - Bringing flexible organic electronics to pilot innovation scale, (Grant Agreement No. 688093).

- PhotonFlex: Photonische Prozesskette zur Fertigung flexibler organischer Solarzellen im Rolle zu Rolle Verfahren – PhotonFlex, (EFRE-0800069).
- HEA2D: Herstellung, Eigenschaften und Anwendungen von 2D-Nanomaterialien, (EFRE-0800148).
- InTres: Innovative Trägermaterialien zur Optimierung der Stromableiter von elektrischen Speichern, (EFRE-0800645).
- GreenSense: Sustainable, Wireless, Autonomous Nanocellulose-based Quantitative DoA Biosensing Platform, (Grant Agreement No. 761000).

Infrastructure/Equipment:

- Coating and equipment solutions
- Large R&D centre for the experimental verification and demonstration of pilot line processes – 9 pilot lines, 1 production line, 1 click & coat line
- Coating, printing and lamination technologies for upscaling and production
- Coatema pilot line, which is used for the pilot line production of electrochromic devices. This was realized in the EU FP7 Project EELICON! (www.eelicon.eu)

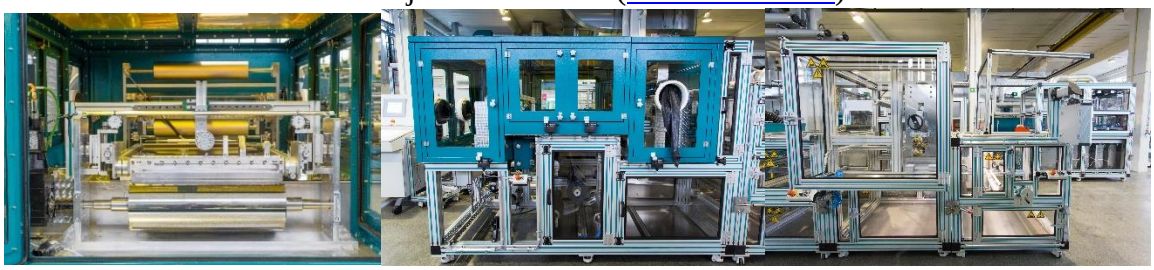


Figure 1: Coatema´s Engineering solution for inert slot-die coating (left), lamination, LED-curing (center) and its final cutting (right), which was realized in the EELICON-project for the electrochromic film R2R-Pilot line

Figure 2: Coatema pilot line for the production of electrochromic devices

Key Personnel

Name: Thomas Kolbusch **Gender: Male**

Background:

Thomas Kolbusch, Vice President of Coatema Coating Machinery, has a degree as business economist in 1997. Worked for 3M, Germany, alpi GmbH, and from 2006 till 2009, as Managing Director for Solarcoating Machinery GmbH, a company specialised on R2R equipment for gen3 PV. Since 2009, member of the board of the OE-A, leading the working group Upscaled production and is chair of the advisory board of the OPE journal. He has been speaker on over 100 conferences and is chairman of an international coating conference.

Name: Dr. Klaus Peter Crone **Gender: Male**

Background:

Dr. Klaus Peter Crone, R&D Manager worked as project manager and later as head of the division "Process Analysis 2" of Bayer AG in Leverkusen. He was the head of the division "Measurement and Analysis" of Agfa Photo GmbH. In 2009 he became head of the technical centre at Coatema. Since 2016 he is Consultant for Coating Technology at Coatema.

Name: Dr. Nico Meyer **Gender: Male**

Background:

Nico Meyer received his Ph.D. in chemistry from RWTH Aachen in 2000. He joined AIXTRON as Process Engineer for Organic Electronics, e.g. OLED and OPV. He became Project Manager for R&D projects in 2003 and was responsible for the coordination and execution of funded Projects. In 2013 he joined Coatema to manage and coordinate EU and national funded Research Projects. He is author and co-author of many publications in international journals and conferences.

Name: M. Sc. Thomas Exlager **Gender: Male**

Background:

Thomas Exlager, R&D Project Manager received his Master of Science in Nano-Opto-Electronics in 2017 from the University Duisburg-Essen. In 2017 he joined Coatema as a Project Manager. He is interested in upscaling of opto-electronic processes for production.

DuPont Teijin Films	**Participant 13** **Organisation Name: DuPont Teijin Films UK Ltd (DTF)** **Website: http://www.dupontteijinfilms.com/**

Organisation Description:

DuPont Teijin Films is a leading global producer of PET and PEN polyester films, specialising in film products and related services for the Healthcare, Photovoltaics, Durable Labels, Imaging Media, Flexible Electronics, Packaging, Electrical Insulation, Capacitors Industries and many more. We have dedicated marketing, technical and manufacturing operations worldwide.

One of the market areas we focus on is the Flexible Electronics market and have spent significant efforts in developing specialised substrates to meet the needs for dimensional stability at elevated temperatures, UV absorbtion, low haze high transmittance, light outcoupling, thermoformability and last but not least clean and smooth surfaces.

Relevant Publications:

1. See Sunflower website publicity: http://www.sunflower-fp7.eu
2. See Clean4Yield www.clean4yield.eu
3. See Flex-o-Fab website publicity: http://www.flexofab.eu/

Relevant Projects:

1. HiQSurf – Development of clean processes for film handling in flex electronic applications
2. Clean4Yield - Contamination and defect control for increased yield for large scale R2R production of OPV and OLED
3. Sunflower - Printed OPV with high efficiency
4. Flex-o-Fab - Flexible OLEDs from Lab to Fab
5. Pi Scale – development of a foundry based on a distributed pilot line for providing the market with OLED solutions

Infrastructure/Equipment:

- A strong R&D centre with head count of 80 engineers, scientists and labs.
- Metrology lab equipment (Wyko WLI, AFM, XPS tools) for characterising surfaces.
- Patented large area metrology technology for assessment of A4's at nano scale for defects.
- A Polyester film pilot manufacturing line for product and process development.
- A manufacturing line in Scotland equipped with an ISRA laser scanner for in line defect detection.

Key Personnel

Name: Valentijn von Morgen **Gender: Male**

Background:

Valentijn started as a Process Engineer in manufacturing of PET films with ICI in Holland. Since then he worked in the same business in a wide range of technical roles spanning Continuous Improvement, process, product, manufacturing and business development. Recently involved with some high profile EU projects developing a new product platform with substrates for the flexible electronic markets.

Education:

Master in Physics from Eindhoven Technical University

Work Experience:

He has experience with developing process technology for manufacturing of PET films and new products having worked at both manufacturing plants and in the central R&T department. Further he has had extensive exposure to Continous Improvement of product development, manufacturing and business processes. Managed many projects and most recently involved with EU project initiatives for the Flexible Electronics industry.

Product Development Experience:

He has a broad experience with product developments from a manufacturing perspective and developed scale and product transfer methodology for the business. Recently developed a new product platform with grades for the Flexible Electronics market from concept to commercial viability. Sofar sold 000's of m2 and expected sales are in the region of 50:000 m2 this year.

Intellectual Property Management:

In process of filing a patent related to grades developed for Flexible Electronic applications.

Innovation Management:

As a qualified 6 Sigma MBB developed methodologies and tools for product development, transfers (between manufacturing assets) and scale up from pilot production.

Name: Julian Robinson **Gender: Male**

Background:

Julian has been working for ICI / DuPont / DuPont Teijin Films in their Technology group based at their R+D centre in the UK. A physical chemist by training, he has developed and applied his knowledge of colloid and surface science, particularly towards coatings on polyester film. He is now a Business Research Associate and team leader, developing new products and the technology platforms for next generation films, including those for future flexible electronics markets.

Education:

PhD in Chemistry from St. Andrews University

Work Experience:

New product and technology platform development in the Technology group at DTF

Product Development Experience:

He has a broad experience of new product development, from concept, through laboratory and pilot scale prototyping, through to full manufacturing scale up and commercialisation. Product developed over the last 28 years will equate to many thousands of tonnes of film.

Intellectual Property Management:

Many patents filed over the years.

Innovation Management:

He is involved with idea generation and selection, as well as developing the product technology strategy for DTF

Name: Thane Gough **Gender: Male**

Background:

Thane started career in ICI Plastics & Petrochemicals division as a Research Technician working in the Polyester Films business. Roles included development of novel polymer systems, and surface treatments. Since then, worked in the same business in a range of roles including Technical Support and Market Management . Current role is Business Development Manager looking for opportunities in the flexible electronics market.

Work Experience:

After working in the Melinex PET business for two years, spent the following two years in the Propafilm PE business, developing new polymer blends. Moving back to the Melinex business, was then involved in the development of white polyester films, high metal adhesion films, and non-slip packaging films. In a Technical Service role in the Packaging business, provided product support, and identified new product opportunities, with both existing and new customers. More recently had Market Management responsibility for the Flexible Electronics sector, and now Business Development Manager looking at opportunities for novel Polyester films in this market.

Product Development Experience:

Extensive experience in New Product Development, from laboratory scale, through manufacturing scale-up, to commercialisation and customer support. Was part of the development team that invented and commercialised white polyester films, which over 30 years has grown from nothing to a global annual volume of 12,000 with a sales revenue in excess of 50 million euro.

Whilst working in the packaging market was instrumental in developing and launching both high metal adhesion films and non-slip films. These films were widely sold in Europe and US, at volumes in excess of 600tpa, with sales revenue of more than 5 million euro per annum.

In the current role, provides commercial guidance and market input to several Product Development Teams involved in innovation of new substrates for electronics applications

Intellectual Property Management:

Patent publication : CA2005598 A1 – Polymeric film

Patent publication : CA2206240 A1 - Polymeric film coated with a polyester composition

Participant 14

Organisation Name: CTR Carinthian Tech Research AG (CTR)

Website: www.ctr.at

Organisation Description:

CTR Carinthian Tech Research AG is an Austrian non-university research centre focusing primarily on targeted research into Advanced Sensor Systems and Smart System Integration. Located in Villach, and with a permanent staff of currently more than 50 researchers, CTR covers the entire R&D chain from proof of concept to prototype development. Conducting R&D projects both on the national and the international level since 1997, CTR has achieved a number of internationally acclaimed innovations and an extensive, dependable network of partners. The research interests of the involved departments Photonic Systems and Smart Systems include the development and system integration of advanced photonic metrology systems and emerging energy systems, from fundamental concepts to the realization and characterization of fully operational, application-specifically integrated functional prototypes. CTR holds comprehensive competence in the R&D of sensors and sensor equipment for in-line quality control in different processing steps, even under harsh environments, covering a broad range of sensor technologies

(magnetic, thermal, optical, electric) . With a focus on (spatially resolved) spectroscopic methods for industrial quality control, CTR has significant experience in bringing lab established technologies in an industrial environment, which addresses one of the main requirements of this H2020 call.

CTR owns the appropriate R&D infrastructure to carry out various fault detection strategies in the field of destructive and non-destructive testing, as well as the characterisation of final products within this project (e.g. spatially resolved industrial micro Raman spectrometer, FTIR/NIR spectrometers, IR microscopes, THz spectroscopy and imaging, state of the art PV metrology and characterisation equipment, including electro-luminescence analysis, climate-controlled test chambers etc.).

Within several projects involving partners from science and industry alike, CTR gained significant experience in the material and quality analysis of photovoltaic cells, modules and electrical subsystems. Besides expertise in design and implementation of metrology systems, CTR thus has significant experience in the qualification and quantification of PV quality defects, understanding the interactions and relationships between material components and environmental conditions by using intelligent measurement and detection approaches, including numerical simulation and statistical data analysis tools. For instance, the project "EVAnetz" the successfully designed, implemented validated a non-destructive, inline-capable analysis method for determining the degree of crosslinking of the elastomeric ethylene/vinyl acetate encapsulants in PV modules, thus increasing manufacturing efficiency and reducing costs in PV module production while warranting optimal product quality.

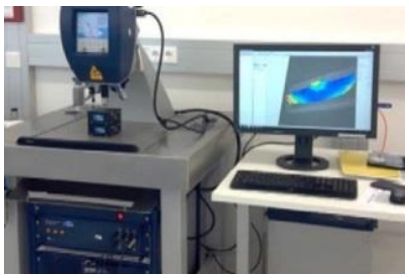
Relevant Publications:

- Ch. Hirschl, L. Neumaier, W. Mühleisen, M. Zauner, G. Oreski, G. C. Eder, S. Seufzer, Ch. Berge, E. Rüland and M. Kraft, "In-Line Determination of the Degree of Crosslinking of Ethylene Vinyl Acetate in PV Modules by Raman Spectroscopy", Solar Energy Materials and Solar Cells 152: 10-20 (2016), http://dx.doi.org/10.1016/j.solmat.2016.03.019
- W. Muehleisen, G. C. Eder, Y. Voronko, M. Spielberger, H. Sonnleitner, K. Knoebl, R. Ebner, G. Ujvari, C. Hirschl, "Outdoor detection and visualization of hailstorm damages of photovoltaic plants", Renewable Energy 118: 138-145 (2018) https://doi.org/10.1016/j.renene.2017.11.010
- M. De Biasio, M. Kraft, M. Schultz, B. Goller, D. Sternig, R. Esteve, M. Roesner, "Micro-Raman spectroscopy as a Tool for the Characterization of Silicon Carbide in Power Semiconductor Material Processing", Proc. SPIE 10210: 1021014 (2017); http://dx.doi.org/10.1117/12.2259930
- L. Neumaier, W. Mühleisen, C. Hirschl, G.C. Eder, M. Aichinger, L. Plessing, A. Martin, A. Zimmerman, "Monitoring of moisture ingress in PV module laminates during accelerated aging tests", Proc. EU PVSEC (2017)
- M. De Biasio, M. Kraft, E. Geier, B. Goller, Ch. Bergmann, R. Esteve, M. Cerezuela-Barreto, D. Lewke, M. Schellenberger, M. Roesner, "Raman micro-spectroscopy as a non-destructive key analysis tool in current power semiconductor manufacturing", Proc. SPIE 10210: 102100U (2017); http://dx.doi.org/10.1117/12.2259927

Relevant Projects:

- 2015 – 2022: COMET Competence Center **ASSIC** – Austrian Smart Systems Integration research Center
- 2015 – 2018: Energy Research Flagship Project **Infinity** - Climate Sensitive Long-Time Reliability of Photovoltaics

- 2015 – 2017: Production of the Future project **OPtima**– Optimized Production for Polymer Parts in Semiconductor Machines



- 2014 – 2017: Energy Research project **PV@Fassade** – Façade-elements with integrated PV-active layers
- 2014 – 2016: SOLAR-ERA.NET project **InnoModu** - Leadfree modules with low silver content and innovative busless cell grid
- 2013 - 2015: Production of the Future project **EVAnetz** – In-line analysis of the degree of crosslinking of ethylene/vinyl acetate encapsulants in PV modules

Infrastructure/Equipment:

- **Renishaw inVia Qontor confocal Raman Microscope**

The Raman microscope facilitates the Raman spectroscopic investigations of both macroscopic and microscopic samples, down to sub-µm resolutions. Applications range from the investigation of the molecular composition of samples to e.g. contact-free direct optical measurement of stress and strain in semiconductor materials. Equipped with 4 separate excitation lasers from the blue to the beginning near-IR, polarisation optics etc. the instrument enables 2D and 3D maps as well as bulk analysis and – using an external arm or fibre optics - even of samples too large to fit into the sample compartment.

- **Bruker LUMOS FTIR Microscope**

The LUMOS is a fully automated stand-alone mid-IR microscope facilitating the measurement of both macroscopic and microscopic samples in transmission, reflection or attenuated total reflection (ATR). Applications include e.g. the identification of particles, contaminations and coatings as well as the determination of the chemical composition of complex samples. The instrument enables point analysis as well as highly resolved 2D maps. In addition, the instrument features a MACRO unit with various accessories for bulk analysis.

- **Polytec Micro System Analyser MSA-500-TPM**

The Micro System Analyzer is a premier measurement tool for the analysis and visualization of structural in-plane and out-of-plane vibrations as well as surface topography in micro structures such as MEMS. Combining a microscope with scanning laser doppler vibrometry, stroboscopic video microscopy and scanning white light interferometry, the MSA is designed as an all-in-one combination of technologies that clarifies real microstructural response and topography.

- **UV/VIS Spectrometer Cary 60**

Laboratory UV/VIS spectrometer with wide spectral coverage from the atmospheric UV to the initial near-IR range at good spectral resolution and spectral dynamics. Applications include measuring colorimetric assays as well as the characterisation of optical components (filters, AR coatings, layers etc.).

Dedicated PV Instrumentation (PL, EL, …)

The defined high level goals of the project may furthermore benefit from CTR's

- State of the art photovoltaics in-lab and in-field testing infrastructure, including electroluminescence, thermal imaging, fluorescence imaging, twilight electroluminescence, outdoor power meter, cell flash tester etc.,
- Optical measurement systems (laser profiling, microscopy, white light interferometer), and
- Harsh environment characterisation equipment as well as a cryogenic chamber complete the ability of carrying out high performance industrial research in this project.

Furthermore CTR is also equipped for smart sensor integration in devices and owns a research PV power plant to carry out field tests under real environmental conditions.

Key Personnel

Name: Dr. Martin Kraft **Gender: Male**

Background:

Dr Martin Kraft is CTR's area managers for Photonic Systems, and the centre's IPR manager. Holding a PhD in Technical Chemistry, his scientific expertise ranges from chemical process analytics to comprehensive computer-based multi-parameter system simulation and optimisation. Current key research interests include applied vibrational (IR, Raman) spectroscopy, process analytical technologies (PAT) and applied process control, photonic (nano-) particle sensors and multi-method simulation-based system & process elucidation and optimisation. Further key qualifications include extensive experience in the management of complex multi-partner R&D projects and in IP rights.

Education:

Engineering degree ("Ing.") in Polymer Technology
M.Sc. in Technical Chemistry, curriculum Analytical Chemistry and Physico-Chemistry (TU Wien; Univ. Cambridge)
PhD in Technical Chemistry
various courses in project management, IPR, legal contracting in R&D, research management

Work Experience:

2004 – present: Carinthian University of Applied Sciences, Area Engineering & IT: Industry Lecturer (Chemistry, Physics and Chemistry of Semiconductor Manufacturing, IPR for Engineers)
2001 – present: CTR Carinthian Tech Research AG: Key Researcher Chemical Sensors & Applied Spectroscopy
Division Manager Photonic Systems & ASSIC Area Manager Smart Systems Integration (SYS) (2015 -)
1998 - 2001: Institute of Analytical Chemistry, TU Wien: Full-time research assistant.

Product Development Experience:

15+ years in applied research, with joint projects typically ranging from TRL 2-3 to TRL 6-8; latest example: "AVL CPC" condensation particle counter, https://www.avl.com/-/apcplus-avl-particle-counter

Intellectual Property Management:

- Centre IP manager, covering the field from patent searches to patent filing coordination; IP portfolio management for the PHOS area, advisory function for decision making for both CTR and company partners
- Inventor of several original inventions/patents, from MEMS-based FTIR spectrometry to high-temperature CPCs
- Editorial author of 20+ original patent applications

Innovation Management:

Responsible area manager in charge of customer interactions and national and international networking, scientific as well as commercial dissemination.

Name: Katja Presich **Gender: Female**

Background:

Katja Presich has a strong background in technical/medical physics with the focus on imaging technologies. She will support the CTR team in the field of signal processing and data analysis.

Education:

2014-2015: Paracelsus Medical University Salzburg – PhD programme Medical Sciences
2011-2014: Technical University Vienna –Biomedical Engineering / Medical Physics and Imaging (M.Sc.)
2007-2011: Technical University Vienna – Study of Technical Physics (B.Sc.)

Work Experience:

2017 - present CTR Carinthian Tech Research AG
2015 - 2017 KTM AG, NVH Engineer
2014 - 2015 radART Institute, PMU Salzburg, Medical Physicist - research assistant
2013 - 2014 Physicist at the federal hospital in Salzburg, employed by EBG MedAustron GmbH

Product Development Experience:

- Accompanying the whole product development process at KTM AG, conducting measurements of structure-borne noise in different phases of prototypes
- Input on the prototype design based on these measurements which required changes, in agreement with the project manager of specific model or component
- Physical characterisation and input on the design of a new in-room CT for ionizing radio therapy application

Innovation Management:

- In-room CT for ionizing radio therapy: prepare necessary documentation and quality assurance protocols to comply with official regulative
- Conduct acceptance tests at the customer side

Name: Lukas Neumaier **Gender: Male**

Background:

Lukas has an extended knowledge in electrical engineering and biomimetics/bionics (university education). He is also an experienced researcher in the field of renewable energy systems, especially photovoltaics focusing in the area of: (a) quality control and characterisation of PV-modules and components and (b) PV plant monitoring. He has carried out several research projects in the area of smart systems specifically (a) data acquisition, analysis and evaluation and (b) sensor integration. Lukas is also a certified project manager Level D (IPMA/pma) - Project management for national (FFG) and international (Interreg) funded projects as well as Project management of contract research projects.

Education:

2013-2015: Carinthian University of Applied Sciences, Villach - Study of Bionik/Biomimetics in Energy Systems
2007-2012: Technical University Graz - Study of Electrical Engineering and Power Engineering
1998-2006: Secondary School Feldkirchen – Focus: Natural Science

Work Experience:

2012-present: CTR Carinthian Tech Research AG – Researcher & Project Manager Smart Systems and Photovoltaics

Product Development Experience:

- Development of an Raman-based inspection tool for non-destructive, inline quality control to enable smart production in photovoltaic industry

- Development, analysis and verification of various sensor-based prototypes for industry based applications (environmental sensing, identification, energy harvesting)

Intellectual Property Management:

- Knowledge in professional Patent Literature Research (DEPATISnet)
- Invention disclosure concerning the avoidance of potential induced degradation for photovoltaic modules

Innovation Management:

Extensive publication activities on national and international conferences (more than 50 publications as author and co-author)

Participant 15

Organisation Name: ADVANCED ENERGY TECHNOLOGIES S.A. (Advent)

Website: http://www.advent-energy.com/

Organisation Description:

ADVENT TECHNOLOGIES S.A. (AT) develops new materials and systems for renewable energy technologies, such as plastic photovoltaics and high temperature PEM fuel cells. The Company, founded by researchers from the Foundation for Research & Technology-Hellas (FORTH-ICEHT) and the University of Patras, is a spin-off operation from these two academic institutions and is funded by financial institutional investors (Connecticut Innovations, Piraeus Capital Management, Dolphin Capital PLC), industrial partners (Systems Sunlight S.A., Velti PLC, ILPRA S.A) as well as angel investors. The activity in plastic photovoltaics and related materials goes back to 2001 where the cofounders of Advent were engaged in an R&D contract activity from University of Patras and FORTH/ICEHT with the world industry leader Konarka Technologies of Lowell, MA. Regarding the R&D activity in relationship to government grants and contracts, the Company is the recipient of 11 competitive R&D grants during the last few years, a testament to the excellent level of the cofounders and the scientific personnel of the Company. The competitive R&D grants are in the area of plastic photovoltaics, fuel cells and catalysts.

Relevant Publications:

- Chochos, C. L.; Singh, R. et al. “Enhancement of the power conversion efficiency in organic photovoltaics by unveiling the appropriate polymer backbone enlargement approach” Adv. Funct. Mater. 2016, 26, 1840
- Chochos, C. L.; Ameri, T. et al. “An Alternative Strategy to Adjust the Recombination Mechanism of Organic Photovoltaics by Implementing Ternary Compounds“ Adv. Energy Mater. 2015, 5, 1501527.
- Chochos, C. L.; Scherf, U. et al. “Ultra Low Band Gap α,β-unsubstituted BODIPY-based Copolymer Synthesized by Palladium Catalyzed Cross-Coupling Polymerization for Near Infrared Organic photovoltaics” J. Mater. Chem. A 2015, 3, 16279.
- Chochos, C. L.; Brabec, C. J. et al. ““Photophysics of molecular weight induced losses in indacenodithienothiophene-based solar cells” Adv. Funct. Mater. 2015, 25, 4898.
- Chochos, C. L.; Heeney, M. et al. “Influence of the Electron Deficient Co-monomer on the Optoelectronic Properties and Photovoltaic Performance of Dithienogermole-based Co-polymers” Adv. Funct. Mater. 2014, 24, 678.

Relevant Projects:

- “Development of smart machines, tools and processes for the precision synthesis of nanomaterials with tailored properties for Organic Electronics (Smartonics)” EU FP7-NMP large at ADVENT Technologies SA (2013-2016 Budget 125.000 Euro); Grant agreement no.: 310229.
- “New materials for highly efficient and reliable organic solar cells (MatHero)” FP7-PEOPLE-2013-NMP small at ADVENT Technologies SA (2013-2016 Budget 220.000 Euro); Grant agreement no.: 604603.
- “Development of New Low Band Gap Conjugated Polymers for Organic Photovoltaic Applications (ADVEPOL)” “Support for businesses employing highly scientific training" through the "Human Resources Development" of General Secretariat for Research and Technology in Greece (GSRT) at ADVENT Technologies SA (2012 – 2014 Budget 45.000 Euro)
- “Organic Semiconductors for NIR Optoelectronics (OSNIRO)” Marie Curie Initial Training Networks (ITN) Call: FP7-PEOPLE-2013-ITN at ADVENT Technologies SA (2013-2017 Budget 330.000 Euro); Grant agreement no.: 607585.
  ITN-ETN OSNIRO is the development of optoelectronic materials and devices (organic solar cells, photodetectors and image sensors as well as organic light-emitting diodes)

Infrastructure/Equipment:

The infrastructure capabilities at Advent Technologies facilities, includes pilot production space at the Patras Science Park. The Patras Science Park location is an ideal place for Advent Technologies due to the proximity to both FORTH/ICEHT and University of Patras where major instrumentation available to Advent Technologies exists (NMR, MS, XPS, Raman etc.) As far as the instrumentation at Advent Technologies itself is concerned, the Company is very well equipped to perform the work described in the project. Specifically, a series of chemical synthesis stations exist where synthesis in small quantities is possible in addition to the presence of 50 L chemical reactors (2 Buchi reactors connected to 2 Huber units; models Huber Unistat 815W and 385W) that is capable of Kg level synthesis. Regarding characterization, size exclusion chromatography (GPC), UV-Vis spectrometry and Photoluminescence spectroscopy (PL) are some of the techniques available in house.

Figure 1: 50L Chemical

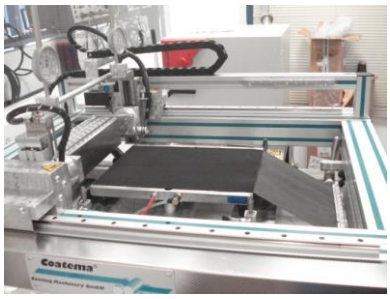


Figure 2: Coating machinery

Figure 3: Analytical facility

Key Personnel

Name: Dr Christos Chochos **Gender: Male**

Background:

Christos L. Chochos holds a degree in Chemistry (2001), a M.Sc (2004) and a PhD (2006) in Polymer Science and Technology from the Chemistry Department of University of Patras. His research interests include the design, synthesis and commercialization activities of organic semiconducting materials for organic electronic applications. Recently, he received a Marie Curie (2013-2015) Intra European Fellowship (FP7-PEOPLE-2012) with an excellent mark of 98,4/100 for the development of new high performance conjugated polymers with environmentally friendly methodologies for application in organic photovoltaics. At the same time, as manager of organic electronic materials in Advent Technologies SA, he is the scientific supervisor in two European projects on the development of organic materials for applications in the near infrared and organic

photovoltaics. He is the author of 32 scientific publications [h-index=15; Web of Science (March 2015)] and 3 US patents. All related to the development of conjugated polymers for various optoelectronic applications. In the past, Dr Chochos was postdoctoral researcher at the Centre National de la Recherche Scientifique (CNRS) in Strasbourg, Department of Mechanical Engineering and Materials Science and Engineering of the Cyprus University of Technology and the Institute of Chemical Engineering Sciences (ICE-HT) in Patras.

Education:
PhD: 2006, Interdepartmental (Chemistry, Physics, Chemical Engineering, Materials Science and Engineering), University of Patras, Greece, Prof. Joannis Kallitsis, Synthesis and Characterization of New Conjugated Polymers for Optoelectronic Applications. Co-supervisor: Dr. Vasilis Gregoriou
M.Sc.: 2004, Interdepartmental (Chemistry, Physics, Chemical Engineering, Materials Science and Engineering), University of Patras, Greece, Prof. Joannis Kallitsis, Synthesis and Characterization of Rod-Coil Diblock Copolymers with Oligofluorenes as Initiators. Co-supervisor: Dr. Vasilis Gregoriou
B.Sc.: 2001, Department of Chemistry, University of Patras, Greece, Prof. Dionyssios Papaioannou, Total synthesis of prototype cyclic polyamine derivatives with biological interest
Work Experience:
2015 – Present: ADVENT Technologies SA, Patras, Greece, Manager of Organic Electronic Materials
Previous Positions:
2013 – 2015: Department of Materials Science and Engineering, University of Ioannina, Greece, Marie Curie Intra European Research Fellow at the Polymeric Materials' Laboratory
2011 – 2012: Institute of Chemical Engineering Sciences (ICE-HT), Foundation for Research and Technology Hellas (FORTH) Patras, Greece, PostDoc Fellow.
2008 – 2011: Department of Mechanical Engineering and Materials Science and Engineering of the Cyprus University and Technology, Limassol, Cyprus, PostDoc Fellow.
2008 Institute of Physical Chemistry of the National Research Centre Demokritos Athens, Greece, PostDoc Research Fellow.
2007 – 2008: Laboratoire d'Ingénierie des Polymères pour les Hautes Technologies (LIPHT), Université Louis Pasteur, Ecole Européenne de Chimie, Polymères et Matériaux (E-ECPM), and Centre National de la Recherche Scientifique (CNRS) Strasbourg, France, PostDoc Fellow.
2007 Research and development department of ADVENT Technologies SA, Patras, Greece, Research Scientist.
2007 Department of Chemistry, University of Patras, PostDoc Fellow.
Product Development Experience:
Scale up in multigram quantities organic polymeric semiconductors as active layer components for various optoelectronic applications by traditional organic synthesis in chemical reactors.
Intellectual Property Management:

- "Acid-doped polyelectrolyte modified carbon nanotubes and their use in high temperature PEM fuel cell electrodes" US Patent (US 2009/0081487 A1) 26 March 2009 and WO Patent (WO 2009/087484 A2)
  Christos Chochos, Nora Gourdoupi, Nikos Triantafyllopoulos, Joannis Kallitsis

- "Novel polymer networks based on benzimidazole moiety for high performance polymer electrolyte membrane fuel cells" US Provisional Patent (US 61/812,368) 16 April 2013
  Christos Chochos, Vasilis Gregoriou

- "Novel zwitterionic polyelectrolytes as efficient interface materials for application in optoelectronic devices" US Provisional Patent (US 61/815,788) 25 April 2013
  Christos Chochos, Vasilis Gregoriou

- “Conjugated polymers containing fused electron rich and electron poor units, preparation method and uses thereof” US Provisional Patent (US 61/817,386) 30 April 2013
  Christos Chochos, Vasilis Gregoriou

Innovation Management:

Key exploitable product: Efficient donor polymers will be synthesized and characterized, which can be applied as component for the active layer as well as in the electrodes of OPV devices and as semi-conducting material for other organic electronic applications.

- Some joint IPR on the synthesis of semi-conducting polymers and interlayer materials
- Public website to inform about the objectives, consortium, events, open position, and publishable achievements. The public project reports and events will be available on this website and updated on a regular basis.
- Publications in Scientific Journals.
- Communication and interaction with other EC Projects (e.g. MatHero, SunFlower, RoleMak, ROTROT, FlexNet (NoE), COLAE (CA), e-LIFT, FLEXIBILITY, TDK4PE, HIFLEX, POLARIC, COMBOLED, Chip2Foil, PLACE-it, Smartonics) and exchange non-confidential information with them to assure coherence of the EC research and avoid useless replication of work.
- Educational programmes: To ensure a long term impact, several training materials based on the newest and most advanced e-learning technologies will be produced and be freely distributed that can be integrated in educational programmes at university level. The project will also contribute to capacity building through supporting scholarships and exchanges programmes of students at PhD and post-doctoral level.
- Close collaboration will be established with associations, such as the OE-A and in ETPs (NANOfutures, Photovoltaics, Photonics21, EuMaT, ENIAC, MANUFUTURE).
- Newsletters, press releases, flyers, for the public
- Dissemination Kit, which will contain suitable material (e.g. text and pictures including copyright clearance), which the EC may use for its own dissemination
- Participation and presentations in Conferences, Workshops and Summer schools. Participations of partners are scheduled in Scientific and Technical Conferences, as the E-MRS and mainly the possibility to organize a Symposium on “Organic Materials, Processes & Applications”, in relation with Conferences: Global Plastic Electronics, LOPE-C, Printed Electronics, Organic Semiconductors Conference, E-MRS, MRS, Drupa in Europe and USA.
- Participations in technical, commercial Fairs and Exhibitions. Dissemination of the results towards other industrial partners & potential target user groups will also be made by the consortium via participation in major technical exhibitions (ACHEMA, SolarExpo, International Converting Exhibition Trade Show, Intersolar etc).
- Actively participation in the EC activities organized at programme level relating to NMP in ICT or in Horizon2020, with the objective of providing input towards common activities and receiving feedback (e.g. from clusters), offering advice and guidance and receiving information relating to NMP and ICT programmes implementation, standards, policy and regulatory activities, national or international initiatives, etc.
- Participations in exhibitions, platforms, thematic networks. The partners have been participating for several years in such events for the promotion of their activities.

Participant 16

Organisation Name: INURU GmbH. (INURU)

Website: http://www.inuru.de/

Organisation Description:

Inuru Gmbh is a German-based printed light & displays and printed electronics company, positioned as an innovative R&D&I player in ultra-thin, low cost electronic devices industry. Inuru is the world leader of OLED containing paper products for advertisement purposes. Inuru has its own inks, printing process, battery and OLED production, which is done by ink-jet printing. The core of INURU's technology is a low cost OLED solution, based on paper-thin & bright organic light source, printed with standard printing equipment in ambient conditions. This cutting-edge eco-friendly technology finds use in a wide range of low-cost and flexible applications, in particular, full displays for smartphones and TVs. INURU builds on the expertise and strong research track-record of a team composed by high-calibre professionals in the field of organic and printed electronics. INURU run its own facility and printed flexible OLED manufacturing production line in Berlin.

Vision:

INURU is sumerian and means "in-light". It is composed out of the words "INA" (engl. in, into) and "NURU" (light). Sumerian thereby is the world's first written language. The word INURU fully reflects the vision of our company: FIRST we want to use light to make things visible and THEN we want to use the same light to enlighten people by bringing smart content through our printed display solutions.

Relevant Publications:

- Winner of industrial prize "german packaging award 2017" in collaboration with Coca Cola and Karl Knauer

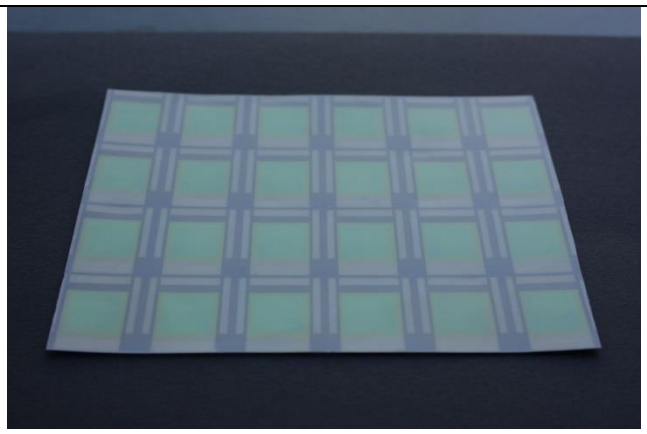
Relevant Projects:

- Project leader of BMBF project "ORIGAMI/IonT"

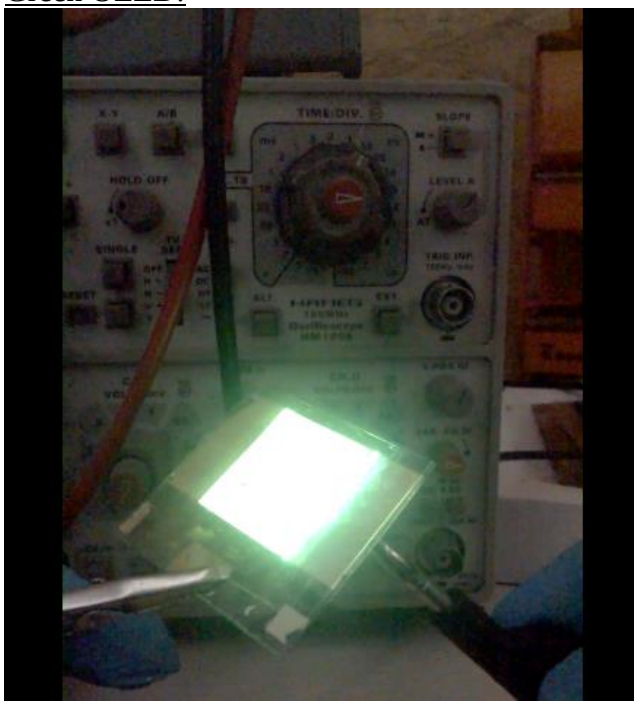
Infrastructure/Equipment:

- 4-point-probe, 3 Ink-jet-printers, blade-coater, atomic force microscope, fluorescence spectrometer, OTFT measurement set-up (Ossila), OLED characterization set-up (Ossila), evaporation chamber, spin-coater, flowbox systems, screen printer, R2R paper assembly, OLED ink-jet pilot line
- Printed flexible OLED manufacturing production line;
- Licensed software (process control and quality assurance);
- Characterization equipment;

OLED Sheet Flex:



Green OLED:



Key Personnel

Name: Marcin Ratajczak **Gender: Male**

Background:

During the last 10 years of being entrepreneur, Marcin has provided high quality customer solutions in many industries. Since 2012, he has been responsible for the overall development of INURU, as CEO and co-founder.

Education:

MSc degree from in Electronics Engineering from Freie Universität Berlin

Work Experience:

2012 – Present: CEO and Co-founder, Inuru Gmbh, Berlin, Germany

Product Development Experience:

- Go-to-market and technology development strategy;
- Business development & requirement analysis in advertising, packaging;
- Product development: supply chain; production process; customer experience.

Innovation Management:

Vice Chair Board of Directors, INAM – Innovation Network for Advanced Materials

Name: Patrick Barkowski **Gender: Male**

Background:

Patrick has been working with and for many different leading institutes in the field of organic and printed electronics (AG Koch - HU Berlin; Fraunhofer IAP; TU Chemnitz), and has collected a vast knowhow in physics, chemistry and processing. He has also done research on solution processed organic doping of organic semiconductors. At INURU, in his role as CTO & Co-founder, he leverages his background and previous professional experiences to develop printed light inks, which are unique in the industry.

Education:
MSc degree in Physics from Humboldt University of Berlin

Work Experience:
2014 – Present: CTO and Co-founder, Inuru Gmbh, Berlin, Germany

Product Development Experience:
- Developed full printed OLED stack
- Developed inks for printed OLEDs and OPVs
- Developed new low cost printed HTL material
- Developed barrier films for organic devices
- Developed printed circuits on paper
- Developed flexible paper based OLED

Intellectual Property Management:

- PCT/EP2017/073753
- PCT/EP2017/076172
- PCT/EP2017/074512
- PCT/EP2017/074514

Participant 17

Organisation Name: SMIT Thermal Solutions B.V. (SMIT)

Website: www.smitthermalsolutions.com

Organisation Description:

SMIT Thermal Solutions BV is a leading supplier of customized, innovative thermal and coatings equipment and processes, founded in 1936 in Nijmegen, The Netherlands. Currently the headquarters are located in Eindhoven, in an important hub of research and innovation in the south of Netherland and the company employs around 75 highly skilled engineers with competences covering all aspects required for equipment manufacturing, installation and commissioning.

Besides design and manufacture of thermal process solutions for high-volume manufacturing, the core competences of SMIT include deposition processes using thermal evaporation, (PE)CVD and ALD for both rigid and flexible substrates.

Current focus of SMIT is on thin film applications, acting as an expert partner to manufacturers aiming for cost-effective mass production.

The strategy of SMIT is to develop and commercialize state of the art industrial equipment that can serve a wide variety of applications. In particular, within this project, the aim of SMIT is to develop and test plasma enhanced (PE) spatial atomic layer deposition (sALD) compatible with an R2R system.

Our plan is to use the know-how developed within this project to offer added value to existing systems and as a building stone for next generation systems (R2R). A successful development will help market penetration of high quality flexible devices (OLED, OPV, CIGS) and will generate sales of R2R sALD systems and business expansion for SMIT and help create additional employment opportunities in the region.

Additionally, there are many other applications and markets that can as well benefit from this technology development to improve product quality and increase throughput. This will also contribute to increased revenue and can create additional expansion opportunities.

Relevant Publications:

5. FLEx ALD R2R and S2S. https://www.meyerburger.com/nl/en/technologies/specialized-technologies/spatial-atomic-layer-deposition/
6. TFE CONx (in particular JETx). https://www.meyerburger.com/nl/en/technologies/specialized-technologies/product-detail/product/conx-tfe/
7. Presentation for ALD (example). http://www.semi.org/en/dana-borsa-flexible-functional-devices-mass-production-level-flex-r2r-sald-platform
8. Presentation for barriers (example - Industrial scale manufacturing of transparent and flexible moisture barriers). https://www.fep.fraunhofer.de/en/events/rueckblick_2017/sidme17.html#tabpanel-2

Relevant Projects:

4. OPZuid. https://www.stimulus.nl/op-zuid/project/industriele-productie-dunne-film-barriers/
5. Enso. http://enso-ecsel.eu/
6. Collaboration with Holst: Development of a roll-to-roll PECVD deposition tool to be used for flexible OLED lighting and organic solar cells. https://www.oled-info.com/roth-rau-joins-holst-centre-research-barriers-flexible-electronics

Infrastructure/Equipment:

- Spectral ellipsometry
- Profilometer
- Low resolution SEM (>1µm)
- Climate chamber / ovens

Figure 1: Spectral ellipsometry

Figure 2: Profilometer

Figu (>1µm)

Figure 4: Climate chamber / ovens

Key Personnel

Name: Dana Borsa **Gender: Female**

Background:

Dr. Dana Borsa, is Business Development Manager at SMIT Thermal Solutions BV. Previously, as BU Manager Thin Film Coating and Projects at Meyer Burger (Netherlands) B.V, she was responsible for all business aspects including equipment, process and application developments. She has a PhD in physics from the Rijksuniversiteit Groningen focused on thin film coatings and later worked on switchable mirrors and hydrogen storage materials at the Vrije Universiteit Amsterdam and on high efficiency solar cell concepts at ECN before moving into a management role.

Education:

PhD – Centro de Microanalisis de Materiales, Universidad Autonoma de Madrid, Spain & Nuclear Solid State Physics Group, University of Groningen, NL

Work Experience:

2019 – present: SMIT Thermal Solutions B.V., Eindhoven, NL – Business Development Manager
2009 – 2019: Meyer Burger (Netherlands) B.V., Eindhoven, NL – BU Manager TFC and Projects, Manager Research, Project Leader
2007 – 2009: Energy Research Centre Nederland, Petten NL – Researcher Silicon PV
2004 – 2007: Vrije Universiteit Amsterdam, Amsterdam, NL – Researcher

Product Development Experience:

- Project manager customer project for the fabrication of a prototype system for battery applications; responsible for all phases
- Project leader for INREP H2020 project for the design, fabrication and commissioning of an injector head for sALD S2S tool
- Project leader 'SiOx for market implementation': budget and resource planning; technology scouting, process development and platform upgrade for 24/7 production
- Coordination CIP-SiNx program: definition and continuous topic update, resource planning (FTEs, tool time and consumables); progress monitoring and supervision of process development (>40 WP); field implementation and customer support
- Coordination customer sampling program: communication with customers and local sales team; define test program; realization and after care
- Sales support at HQ and on-site; regular customer visits and process and product marketing.

Name: Jos Bakema **Gender: Male**
Background:
Jos Bakema is Product Manager at SMIT Thermal Solutions BV and involved in various aspects of product development of thermal equipment. With a strong technical background from Rijksuniversiteit Groningen and Technical University Eindhoven, and many years of industry experience, he is an experienced manager who combines hands-on feel with sharp technological and business thinking.

Work Experience:
2016 – present: SMIT Thermal Solutions B.V., Eindhoven, NL – Product Manager
2015 – 2016: Solaytec., Eindhoven, NL – Program Manager
2006 – 2015: Smit Ovens, Son, NL – Manager Engineering
1997 – 2006: Gemco, NL – Project Manager, Process Engineer

4.2. Third parties involved in the project (third party resources)

15. Advent

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	No
Does the participant envisage that part of its work is performed by linked third parties	No
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	No
Does the participant envisage that part of the work is performed by International Partners (Article 14a of the General Model Grant Agreement)?	No

Section 5: Ethics and Security

5.1 Ethics

There any no ethical issues related to this proposal.

5.2 Security

Please indicate if your project will involve:

- activities or results raising security issues: NO
- 'EU-classified information' as background or results: NO